

SA-AI (Spalart–Allmaras with Autogenous Inception): Technical Summary

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1 Model

1.1 Transport equation and blended sources

$$\frac{D\tilde{\nu}}{Dt} = P - D + \frac{1}{\sigma} \nabla \cdot [(c_{\nu, \text{ai}} \nu + \tilde{\nu}) \nabla \tilde{\nu}] + \frac{c_{b2}}{\sigma} |\nabla \tilde{\nu}|^2, \quad (1)$$

$$P = \max[(1 - \sigma_P) a S \omega \tilde{\nu}, \sigma_P P_{\text{SA}}], \quad D = \sigma_D D_{\text{SA}}, \quad (2)$$

$\sigma_P \equiv 1$ (and with it $\sigma_D = 1$), $c_{\nu, \text{ai}} = 1$, and $b \equiv 0$ recover baseline SA exactly.

$$\begin{aligned} P_{\text{SA}} &= c_{b1} \tilde{S} \tilde{\nu}, & D_{\text{SA}} &= c_{w1} f_w \left(\frac{\tilde{\nu}}{d} \right)^2, & \tilde{S} &= \omega + \frac{\tilde{\nu}}{\kappa^2 d^2} f_{v2}, & f_{v2} &= 1 - \frac{\chi}{1 + \chi f_{v1}}, \\ f_{v1} &= \frac{\chi^3}{\chi^3 + c_{v1}^3}, & f_w &= g \left(\frac{1 + c_{w3}^6}{g^6 + c_{w3}^6} \right)^{1/6}, & g &= r + c_{w2}(r^6 - r), & r &= \min\left(\frac{\tilde{\nu}}{\tilde{S} \kappa^2 d^2}, 10\right), \end{aligned} \quad (3)$$

The non-negative $\tilde{S}$ modification and the negative- $\tilde{\nu}$ safeguards of Allmaras et al. (2012) are retained, with one consistency requirement: the negative-branch diffusion factor f_n is formed with the scaled coefficient $c_{\nu, \text{ai}} \nu$, which keeps the total diffusivity positive, $c_{\nu, \text{ai}} \nu + f_n \tilde{\nu} > 0$.

$$\sigma_P = \sigma_t(\chi; \tau), \quad \sigma_t(\chi; \tau) = \max\left[1 - e^{-(\chi-1)/\tau}, 0\right], \quad \sigma_D = 1 - \frac{c_{b1}}{\kappa^2 c_{w1}} (1 - \sigma_P). \quad (4)$$

1.2 Local indicators

In a thin quasi-two-dimensional layer the profile is captured to second order about any wall-normal point by three velocity scales formed from the wall distance d : the velocity u , the shear du' , and the curvature $\frac{1}{2}d^2u''$. Normalized by their common magnitude $R = \sqrt{u^2 + (du')^2 + (\frac{1}{2}d^2u'')^2}$, they place the local state on the unit sphere of directions

$$(X, Y, Z) = (u, du', \frac{1}{2}d^2u'')/R, \quad (5)$$

$$\hat{\Omega} = \frac{Y}{\sqrt{X^2 + Y^2}} \quad (\text{vorticity fraction}), \quad \hat{I} = Y - X - Z \quad (\text{accumulated inflection}). \quad (6)$$

A near-wall parabola $u = By + Cy^2$ satisfies $\hat{I} = 0$ identically.

$$I(d) \equiv R \hat{I} = d u' - u - \frac{1}{2} d^2 u'' = - \int_0^d \frac{1}{2} s^2 u'''(s) ds, \quad (7)$$

Every profile leaves the wall with $\hat{I}$ third-order small ($u_w''' = 0$ on a steady wall). The vorticity Reynolds number $Re_\Omega = d^2 \omega / \nu$ is reserved for the onset gate. Fig. 1.

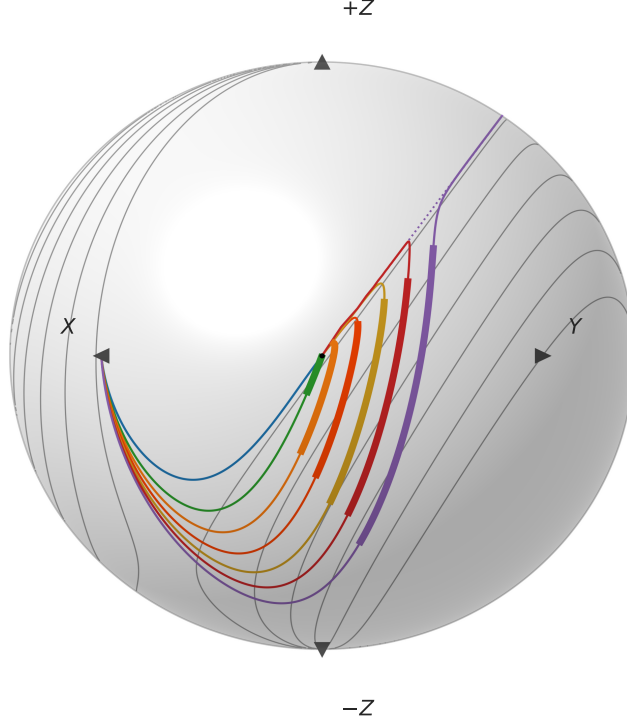


Figure 1: Indicator sphere. Falkner–Skan profiles $\beta = +0.30, 0, -0.10, -0.16, -0.19, -0.1988$; separated Stewartson $\beta = -0.19, H \approx 4.9$. Attached profiles at $Re_\theta = 500$, reversed-flow at $Re_\theta = 200$; $\hat{\Omega}\hat{I}$ contours $\{0.025, 0.2, 0.4, 0.6, 0.8, 1.0\}$.

In the solver the triple is assembled from four local vector-calculus objects: the velocity $\mathbf{u}$, the vorticity $\boldsymbol{\omega}$, the wall-distance vector $\mathbf{d} = d \nabla d$, and the curvature vector $\boldsymbol{\ell} = \frac{1}{2} \langle \mathbf{d}, \mathbf{d} \rangle \nabla^2 \mathbf{u}$, with $\nabla^2 \mathbf{u} = -\nabla \times \boldsymbol{\omega}$ supplying the second derivative.

$$(X, Y, Z) = \left(\langle \mathbf{u}, \mathbf{u} \rangle, \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle \langle \mathbf{d}, \mathbf{d} \rangle \langle \boldsymbol{\omega}, \boldsymbol{\omega} \rangle}, \langle \boldsymbol{\ell}, \mathbf{u} \rangle \right), \quad \hat{\Omega} = \frac{Y}{\sqrt{X^2 + Y^2}}, \quad \hat{I} = \frac{Y - X - Z}{\sqrt{X^2 + Y^2 + Z^2}}, \quad (8)$$

The solver’s equivalent ratio realization ($|\mathbf{u}|, d|\boldsymbol{\omega}|, \frac{1}{2}d^2(\nabla^2 \mathbf{u}) \cdot \hat{\mathbf{u}}$) coincides with it wherever $\mathbf{u} \neq 0$.

1.3 Amplification rate and onset threshold

The complete amplification source is

$$P_{\text{AI}} = a S \omega \tilde{\nu}, \quad a = a_{\text{max}} \text{clip} \langle \hat{\Omega} \hat{I} \rangle_0^1, \quad S = S(Re_\Omega / Re_\Omega^c(\hat{\Omega} \hat{I})), \quad S(\zeta) = \frac{1}{2} [1 + \tanh((\zeta - 1)/w)], \quad (9)$$

$$Re_\Omega^c(\hat{\Omega} \hat{I}) = k \text{softmin}_2 \left(C, A + B \langle \hat{\Omega} \hat{I} \rangle_+^{-2} \right), \quad \text{softmin}_2(x_1, x_2) = \frac{x_1 x_2}{\sqrt{x_1^2 + x_2^2}}, \quad \langle x \rangle_+ = \max(x, 0). \quad (10)$$

Fig. 2: graze ratios 0.93–1.14, root-mean-square $\sim 6\%$, from incipient separation through $\beta = +0.15$. The marched Blasius $N=9$ crossing lands at $Re_\theta = 1181$, 7% past Drela’s 1108; across the attached family the $N=1$ stations land on the Drela–Giles stations to 6% root-mean-square (adverse wedges within $\pm 4\%$, the favorable pair 11–12% early). By $\beta = +0.35$ the rate is $\approx 0.1 \times$ Drela and the onset $\approx 3 \times$ its critical Re_θ ; by $\beta \approx 0.45$ transition has retreated beyond any station of practical relevance, and stronger favorable wedges are suppressed outright (Figs. 3–4).

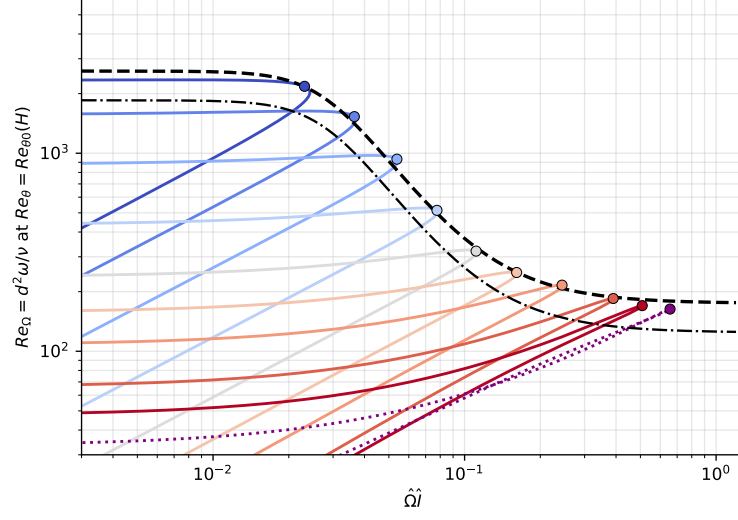


Figure 2: Onset-threshold graze. Attached family $\beta = +0.15, +0.10, +0.05, 0, -0.05, -0.10, -0.15, -0.19, -0.1988$, each at its Drela-Giles $Re_{\theta 0}(H)$; separated Stewartson $\beta = -0.19$ ($H \approx 4.9$, $Re_{\theta 0} \approx 26$). $\max_y \hat{\Omega} \hat{I} = 0.162/0.078/0.037$ at $\beta = -0.10/0/+0.10$. Envelope $k = 1$; model threshold $k = 0.712$.

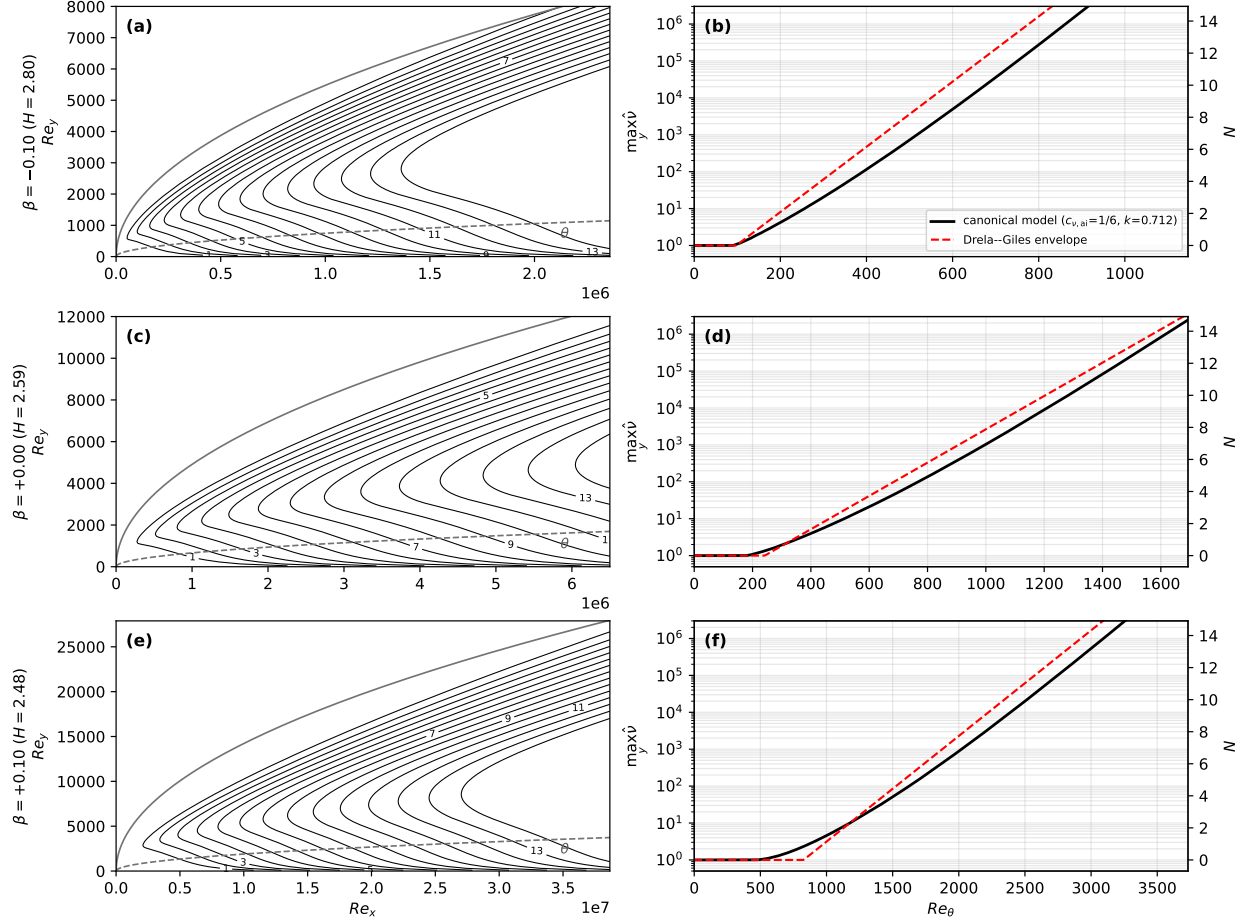


Figure 3: Disturbance transport, three wedges. Adverse $\beta = -0.10$ ($\max_y \hat{\Omega} \hat{I} = 0.162$), Blasius $\beta = 0$ (0.078), favorable $\beta = +0.10$ (0.037); seeded $\hat{\nu}_\infty = 1$, $a_{\max} = 0.19$, $(c_{\nu, ai}, k) = (\frac{1}{6}, 0.712)$. Anchor $N = 9$ crossing 7% past Drela; favorable $N = 1$ station $\sim 11\%$ early.

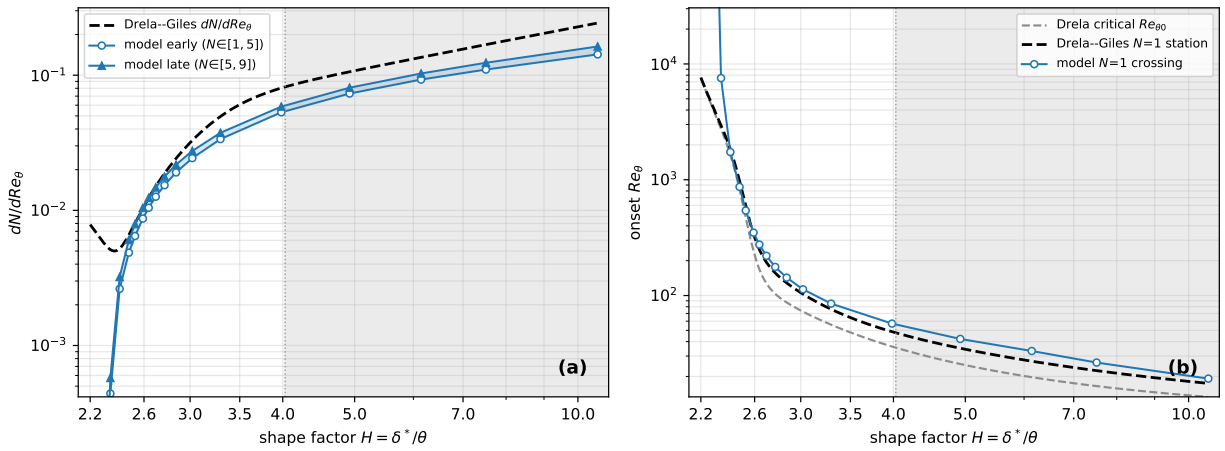


Figure 4: Calibrated model vs Drela-Giles. Stagnation $H \approx 2.2$, Blasius $H \approx 2.59$, separation limit $H = 3.98$, reversed-flow Stewartson $H \approx 10.6$; $a_{\max} = 0.19$, $c_{\nu, ai} = 1/6$. (b) onset tracks the Drela-Giles $N = 1$ stations within $\sim \pm 12\%$ from $H = 2.44$ ($\beta = +0.15$) to $H = 3.98$.

1.4 Reduced laminar diffusion

The molecular coefficient in the diffusion term of Eq. (1) only is scaled by $c_{\nu,\text{ai}} < 1$, leaving the mean-flow viscosity, the SA production and destruction, and $\chi = \tilde{\nu}/\nu$ (formed with the true ν) untouched. The frozen-profile growth eigenvalue on Blasius at $Re_\theta = 400$ delivers only 0.49 of the Drela–Giles rate, and the marched inception fails outright (the Blasius $N = 1$ crossing sits at $Re_\theta = 392$ even with the threshold removed entirely). Down the ladder $c_{\nu,\text{ai}} = 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{6}, \frac{1}{12}$ that station’s eigenvalue ratio climbs 0.49, 0.75, 0.87, 1.02, 1.13: the value

$$c_{\nu,\text{ai}} = \frac{1}{6}$$

puts the just-past-critical stations on the correlation (the separation branch holds 0.82–0.91). The equilibrium disturbance-band thickness scales as $\sqrt{c_{\nu,\text{ai}}}$, so every halving thins the band the grid must resolve by $\sqrt{2}$.

1.5 Handover blend and destruction tie

The destruction is not gated off but tied: σ_D tracks σ_P with slope $c_{b1}/(\kappa^2 c_{w1}) \approx 0.249$ from the floor $(1 + c_{b2})/(\sigma c_{w1}) \approx 0.751$, the two summing to one by the definition of c_{w1} . The tie is derived from the constant-stress wall layer, on which SA’s solution is the linear profile $\tilde{\nu} = \kappa u_\tau y$ with height-independent terms $P_{\text{SA}} = c_{b1} u_\tau^2$, $D_{\text{SA}} = c_{w1} \kappa^2 u_\tau^2$, and self-diffusion $(1 + c_{b2}) \kappa^2 u_\tau^2 / \sigma$; requiring the gated balance to hold pointwise gives Eq. (4), under which the linear profile is an exact solution of the gated equation at every height and for any τ —continuously and discretely (verified to below 10^{-10} in profile and log-law intercept with two unrelated discretizations). In the laminar range the destruction floor is inert at the disturbance’s quadratic scale: its ratio to the amplification production stays below 10^{-3} up to the $\chi = 1$ crossing for every wedge and seed tested, and at $\chi = c_{v1}$ is at most 1.4% of the σ_P -weighted SA production it is tied against; marching the calibration instrument with the floor included moves the anchor and prediction stations by less than 1%.

1.6 Eddy-viscosity assembly in lifted transitional layers

The near-wall damping f_{v1} enforces the attached-wall-layer identity $\chi = \kappa y^+$. Writing $y^+ := \chi/\kappa$, the reconstruction

$$q = \frac{|\mathbf{u}| d/\nu}{y^+ U^+(y^+)}, \quad U^+(y^+) = \frac{1}{\kappa} \ln(1 + \kappa y^+) + 7.8(1 - e^{-y^+/11} - \frac{y^+}{11} e^{-y^+/3}) \quad (\text{Reichardt}), \quad (11)$$

The assembly is blended,

$$\mu_t = [(1 - b) f_{v1}(\chi) + b] \rho \tilde{\nu}, \quad b = s(\chi) G(q), \quad s = \text{clip}(\chi - 1, 0, 1), \quad G = \text{clip}((q - 2)/2, 0, 1), \quad (12)$$

inert for $\chi < 1$ (negligible μ_t either way), for $\chi \gtrsim 30$ ($f_{v1} \rightarrow 1$), and in the attached inner layer ($q = 1$, G shut). Controlled on/off batteries bound the net effect on attached-layer outer edges below 1.5 drag counts (flat plate and NLF at their cold-budget protocol).

1.7 Freestream seed

A case seeded with χ_∞ reaches the handover where its envelope crosses $N_{\text{crit}} = \ln(c_{v1}/\chi_\infty)$.

$$N_{\text{crit}}(Tu) = -8.43 - 2.4 \ln(Tu_{\text{frac}}), \quad \chi_\infty = c_{v1} e^{-N_{\text{crit}}}, \quad (13)$$

One e -fold in χ_∞ is one unit of N_{crit} , which on the Blasius envelope ($dN/dRe_\theta \approx 10^{-2}$) shifts onset by $\Delta Re_\theta \approx 100$. Strongly bypass-dominated conditions ($Tu \gtrsim 1\%$) are out of scope: at $Tu = 1\%$ the map leaves only $N_{\text{crit}} \approx 2.6$, and by $Tu \approx 3\%$ the seed exceeds c_{v1} outright.

1.8 Constants

Baseline SA, unchanged:

$$c_{b1} = 0.1355, \quad \sigma = \frac{2}{3}, \quad c_{b2} = 0.622, \quad \kappa = 0.41, \quad c_{w1} = \frac{c_{b1}}{\kappa^2} + \frac{1 + c_{b2}}{\sigma}, \quad c_{w2} = 0.3, \quad c_{w3} = 2, \quad c_{v1} = 7.1.$$

SA-AI:

constant	value	determination
$a_{\max}$	0.19	tanh-layer Kelvin–Helmholtz temporal eigenvalue, $\omega_{i,\max}/\omega_{\text{peak}} = 0.1897$
(C, A, B)	(2600, 175, 2)	graze envelope of the attached Falkner–Skan family at its critical Re_θ (rms $\sim 6\%$)
k	0.712	Blasius march crosses $N = 1$ at the Drela–Giles station $Re_\theta = 338$ at $c_{\nu,\text{ai}} = \frac{1}{6}$
w	0.35	onset ramp width; $N = 1$ stations move $< 1\%$ halved, $< 2.5\%$ doubled
$c_{\nu,\text{ai}}$	1/6	frozen-profile growth eigenvalue reaches the Drela–Giles rate just past critical (1.02 at $Re_\theta = 400$)
τ	4	numerics bracket: handover 78% complete at $\chi = c_{v1}$, 97% at $\chi \approx 14.8$
σ_D tie	Eq. (4)	derived: constant-stress wall-layer balance; linear wall solution exact
bypass (s, G)	Eq. (12)	parameter-free law-of-the-wall discriminator; G ramps over $q \in (2, 4)$
$(A, B)_{\text{Mack}}$	(−8.43, 2.4)	Mack 1977 e^N correlation, wholesale

1.9 Running the model: branch selection and seed handling

On the frozen Hiemenz stagnation field with the freestream seed exactly zero, standard SA sustains the wedge above a critical nose Reynolds number bisected to $Re_r = 4.66\text{--}4.74 \times 10^5$ (Fig. 5).

Table 1: Attachment-anchored branch: steady $\max \chi$ versus nose Reynolds number $Re_r = L^2$ on the frozen Hiemenz wedge, standard SA, $\chi_\infty = 0$. Below the bisected critical band the layer collapses to $\chi = 0$.

$L = \sqrt{Re_r}$	Re_r	steady $\max \chi$
≤ 683	$\leq 4.66 \times 10^5$	$< 10^{-3}$ (collapses to 0)
683–689	$4.66\text{--}4.74 \times 10^5$	critical band (bisected)
689	4.74×10^5	2.19
700	4.90×10^5	6.15
1000	1.00×10^6	24.1
3000	9.00×10^6	75.7

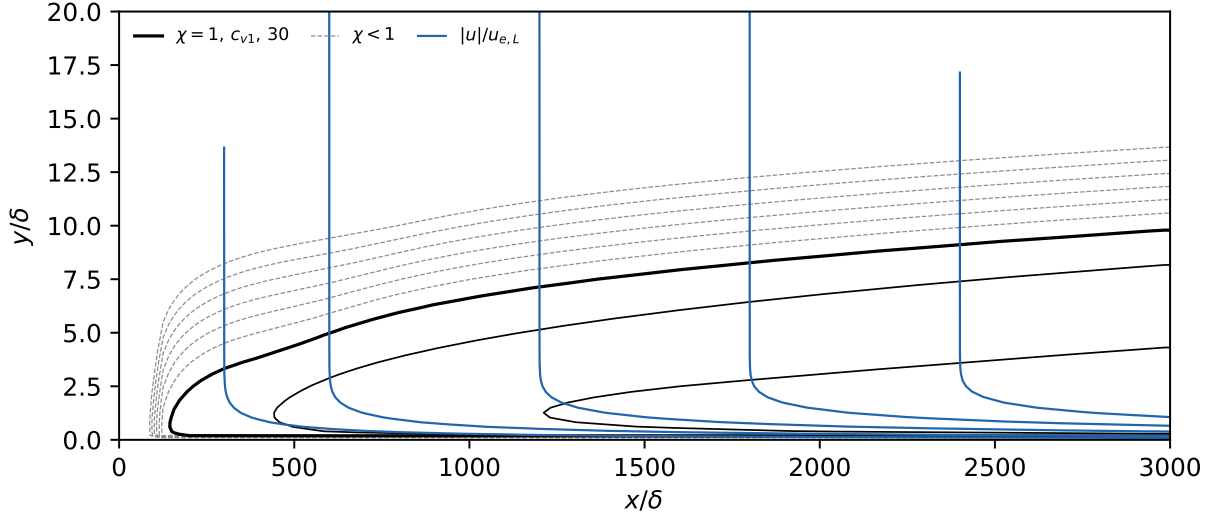


Figure 5: Attachment-anchored turbulent branch, $L = 3000$ ($Re_r = 9 \times 10^6$), $\max \chi = 76$. Standard SA, frozen Hiemenz, $\chi_\infty = 0$. χ contours dashed per decade for $\chi < 1$ (10^{-6} – 10^{-1}), solid at 1, c_{v1} , 30; velocity magnitude $|u|/u_{e,L}$ (blue, normalized by the $x = L$ edge speed) at 0.1, 0.2, 0.4, 0.6, 0.8. Branch data in Table 1.

The protocol used for every result in this paper: the $\tilde{\nu}$ -equation residual is scaled by f^{1-g} , where g rises smoothly from 0 (laminar) to 1 (turbulent) with χ and $f = 10^{-2}$ (the solver uses $g = 1 - e^{-(\chi - c_{v1})/4}$ above $\chi = c_{v1}$, zero below). Because the freestream boundary condition flows through the same scaling, the freestream viscosity ratio supplied to the solver is $\chi_\infty f$; the seeds quoted throughout are the physical χ_∞ , recovered at the boundary-layer edge. (In the reference implementation the scaling is `AI_LAMINAR_SLOWDOWN` = f , and the case files carry the pre-multiplied boundary seed $\chi_\infty f$; running a committed case at $f = 1$ without rescaling the seed mis-seeds it by $1/f$.)

2 Results

Coupled RANS: Flow360, unstructured node-centered second-order finite volume, Roe flux, $M = 0.1$. Airfoil cases: $\chi_\infty = c_{v1} e^{-9} \approx 8.76 \times 10^{-4}$ (the $N_{\text{crit}} = 9$ anchor; via Mack’s map, $Tu \approx 0.07\%$); two independently generated mesh families—all-triangular unstructured cavity meshes, constrained-Delaunay through the boundary layer, and all-quadrilateral structured O-grids—at three refinement levels, 1.5×10^4 to 2.6×10^5 nodes. Transition locations are the solver’s near-wall $\chi = 1$ crossing (the $\chi = c_{v1}$ crossing sits up to $0.05c$ aft where the front grows slowly); bubble stations are signed- C_f zero crossings. The cross-solver replication is a cell-centered, pressure-based, incompressible OpenFOAM implementation of the model on the same flat-plate grid specification and the same structured airfoil grids (the port omits the f_{v1} bypass); thirty airfoil cases plus the flat-plate sweep.

2.1 Flat plate, $Tu = 0.04$ – 0.60%

Structured 320×80 grid, $Re_x \leq 6 \times 10^6$, $y^+ \lesssim 0.3$; five seeds from Mack’s map ($\chi_\infty \approx 2.3 \times 10^{-4}$, 1.2×10^{-3} , 6.3×10^{-3} , 2.9×10^{-2} , 1.5×10^{-1}). Figure 6: at the $\chi = c_{v1}$ crossing the onset brackets the Abu-Ghannam & Shaw correlation within 10% across the range. Figure 7: $\chi = 1$ crossings within 7% in Re_θ . Table 2: both solvers’ crossings.

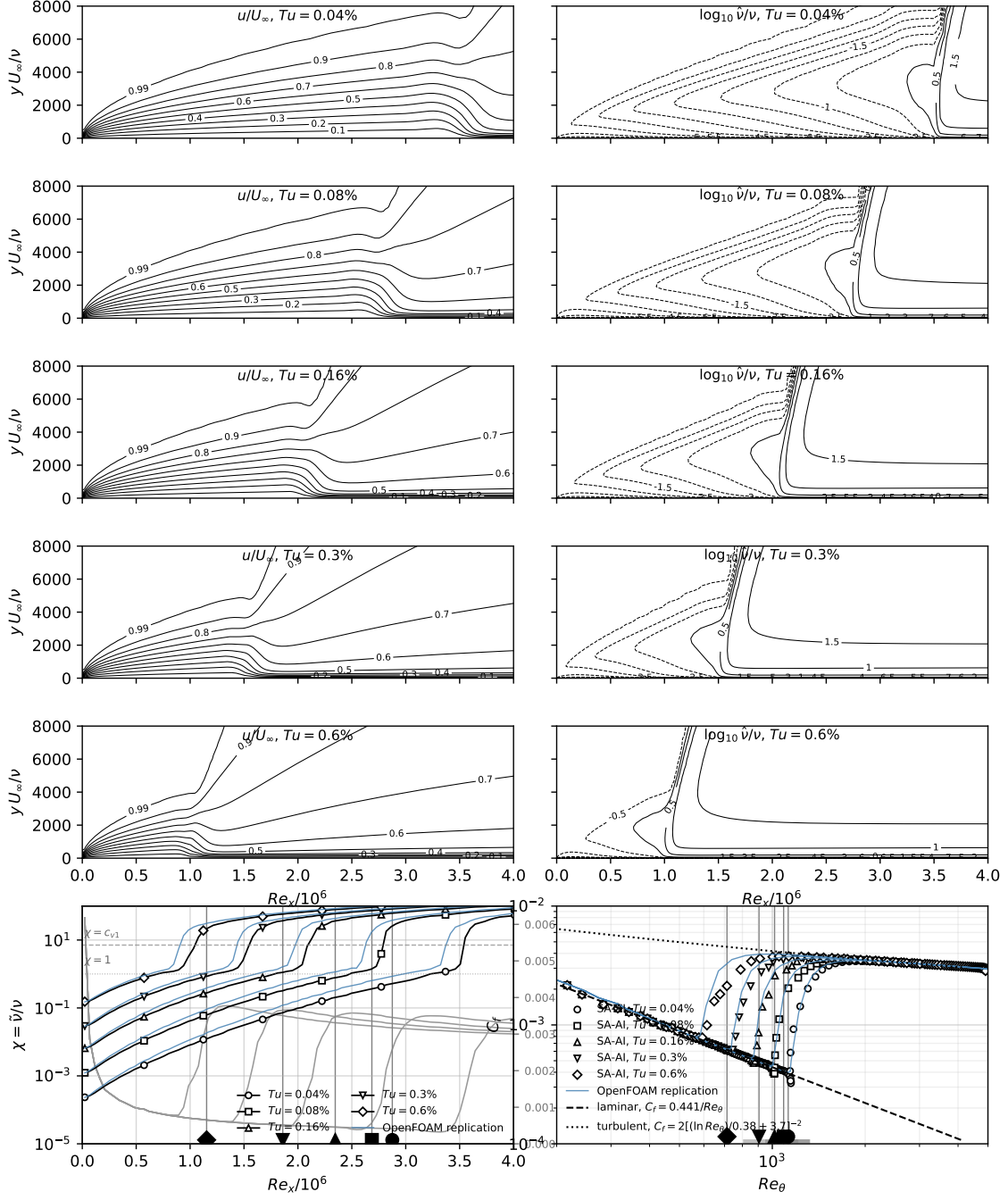


Figure 6: Flat plate, Flow360. Five levels $Tu = 0.04\text{--}0.60\%$; references $\chi = 1$ and $\chi = c_{v1} = 7.1$; laminar $C_f = 0.441/Re_\theta$, turbulent $C_f = 2[(\ln Re_\theta)/0.38 + 3.7]^{-2}$; $\chi = c_{v1}$ onset brackets Abu-Ghannam & Shaw within 10%.

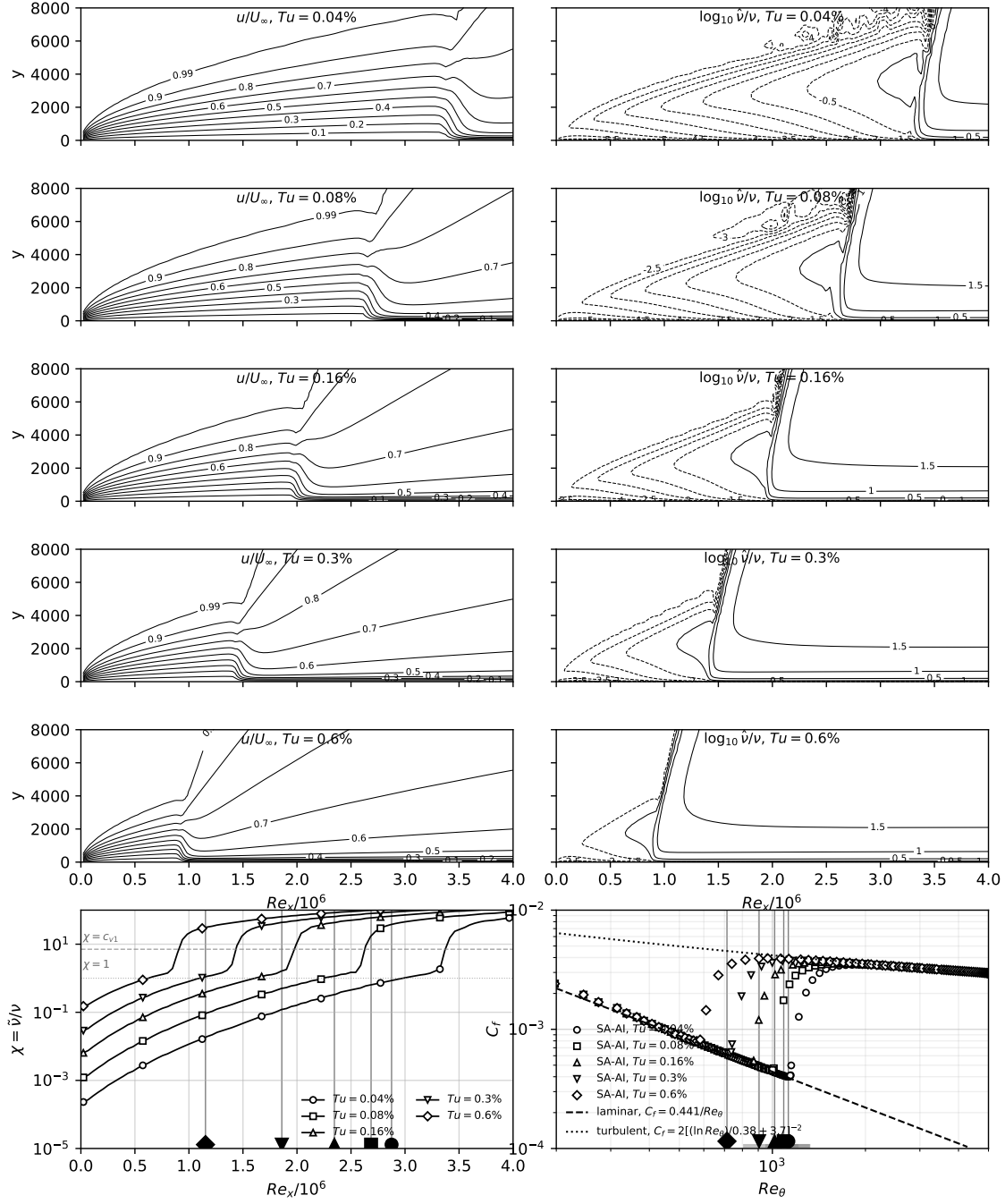


Figure 7: Flat plate, OpenFOAM. Same sweep as Fig. 6, five levels $Tu = 0.04\text{--}0.60\%$.

Table 2: Flat-plate transition-onset Re_θ (integrated from the computed profiles) at the $\chi = 1$ and $\chi = c_{v1}$ crossings, Flow360 and the OpenFOAM replication (OF), against the Abu-Ghannam & Shaw correlation.

Tu [%]	AGS Re_θ	$\chi = 1$	$\chi = c_{v1}$	OF $\chi = 1$	OF $\chi = c_{v1}$
0.04	1126	1149	1203	1098	1154
0.08	1088	1002	1069	949	1004
0.16	1017	845	934	798	887
0.3	905	700	815	656	739
0.6	713	524	685	487	589

2.2 NLF(1)-0416 at $Re = 4 \times 10^6$

$\alpha \in \{0^\circ, 4^\circ, 9^\circ, 15^\circ\}$ on both families at L0–L2 plus $\alpha = -8^\circ, -4^\circ$ on all six grids (thirty-six cases). Figure 8: transition fronts against the LTPT microphone measurements, Coder’s AFT (OVERFLOW) at $N_{crit} = 10.07$ and recalibrated 7.18, the fourteen First AIAA Transition Modeling Workshop submittals, SA-LM2015, Langtry–Menter, and XFOIL. Figure 9: drag polar against Somers TP-1861, the e^9 reference, and fully turbulent SA. Figures 10–12: surface distributions. Table 3: forces and fronts. Table 4: at L2 every front within $0.016c$ and the lift within 0.9% of the SA-AI values.

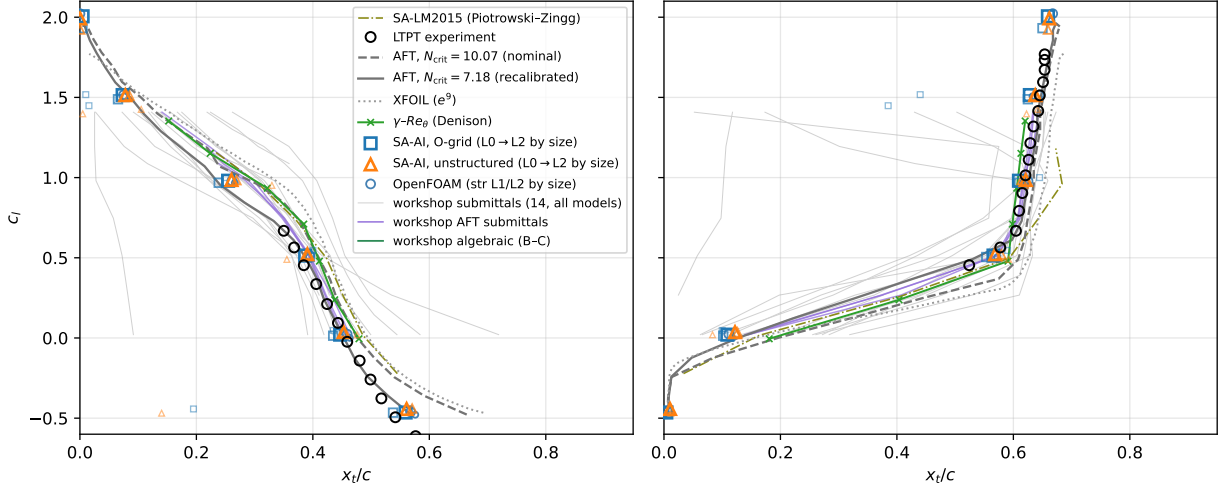


Figure 8: NLF(1)-0416 transition fronts, $Re = 4 \times 10^6$; near-wall $\chi = 1$ convention. SA-AI at $\alpha = 0^\circ, 4^\circ, 9^\circ, 15^\circ, -8^\circ, -4^\circ$; workshop sweep $\alpha = -4^\circ - 8^\circ$.

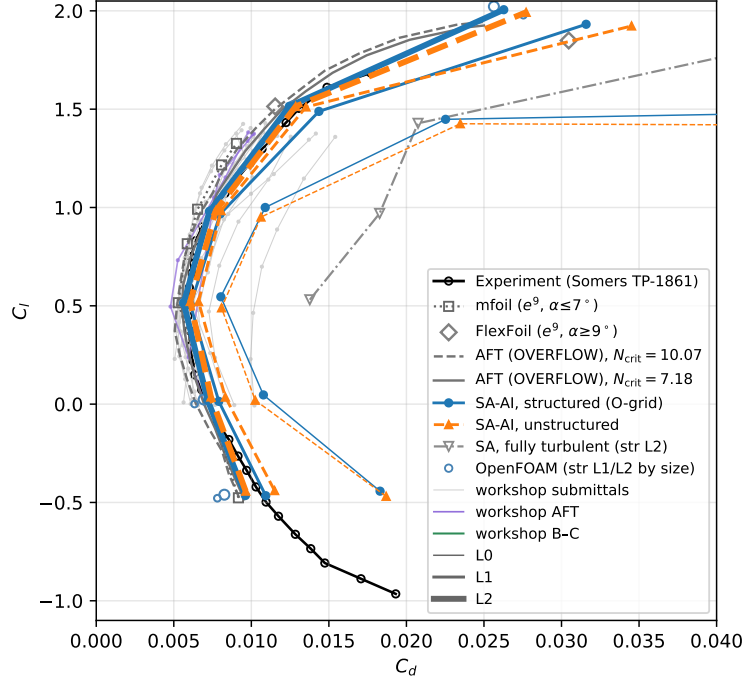


Figure 9: NLF(1)-0416 drag polar, $Re = 4 \times 10^6$. Fully turbulent SA ($\chi_\infty = 3$): 1.6–2.5 $\times$ above the SA-AI drag, +144/+150% in the bucket at $\alpha = 0^\circ/4^\circ$, +67/+74% at $9^\circ/15^\circ$.

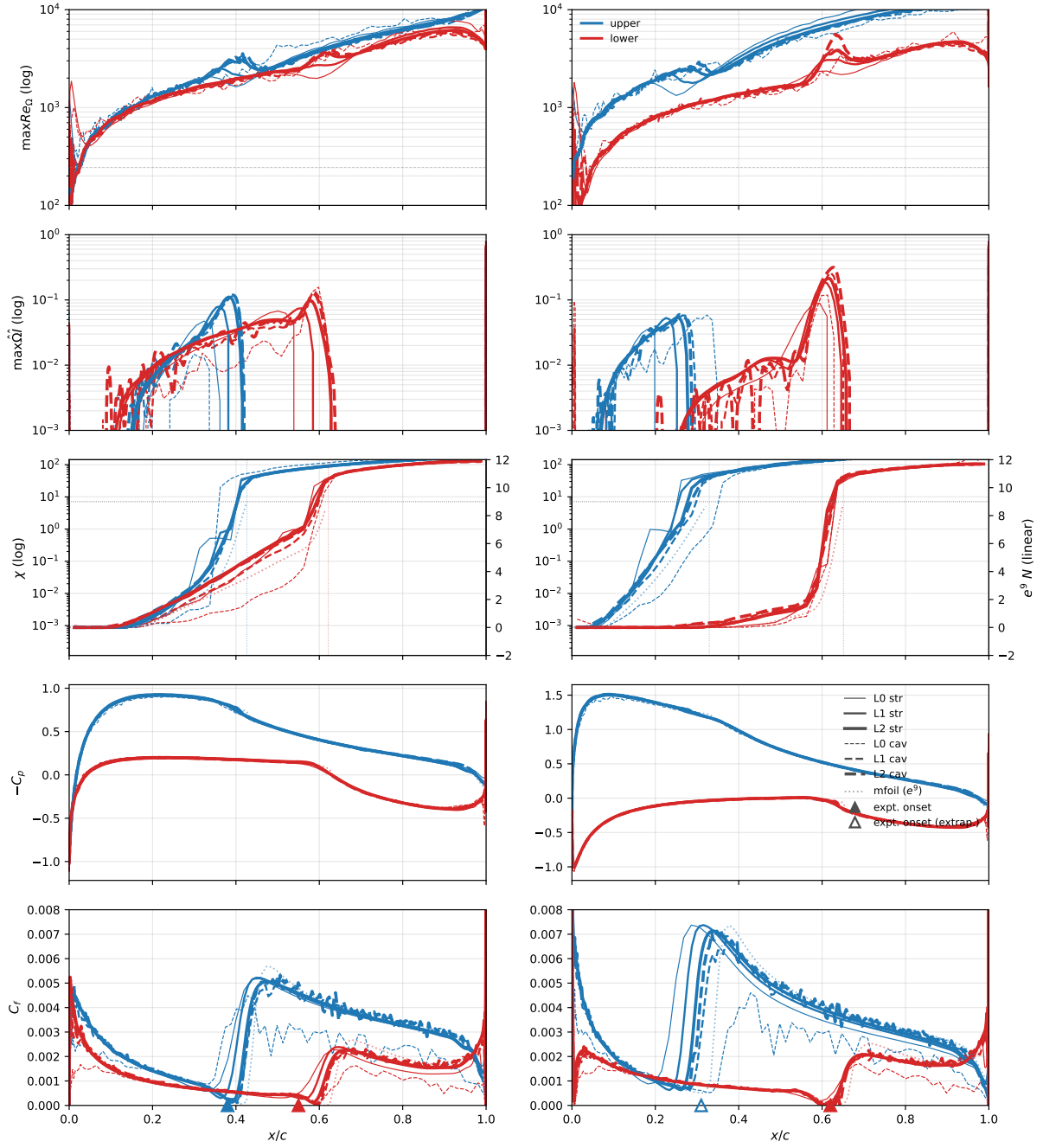


Figure 10: NLF(1)-0416, $\alpha = 0^\circ, 4^\circ$, $Re = 4 \times 10^6$.

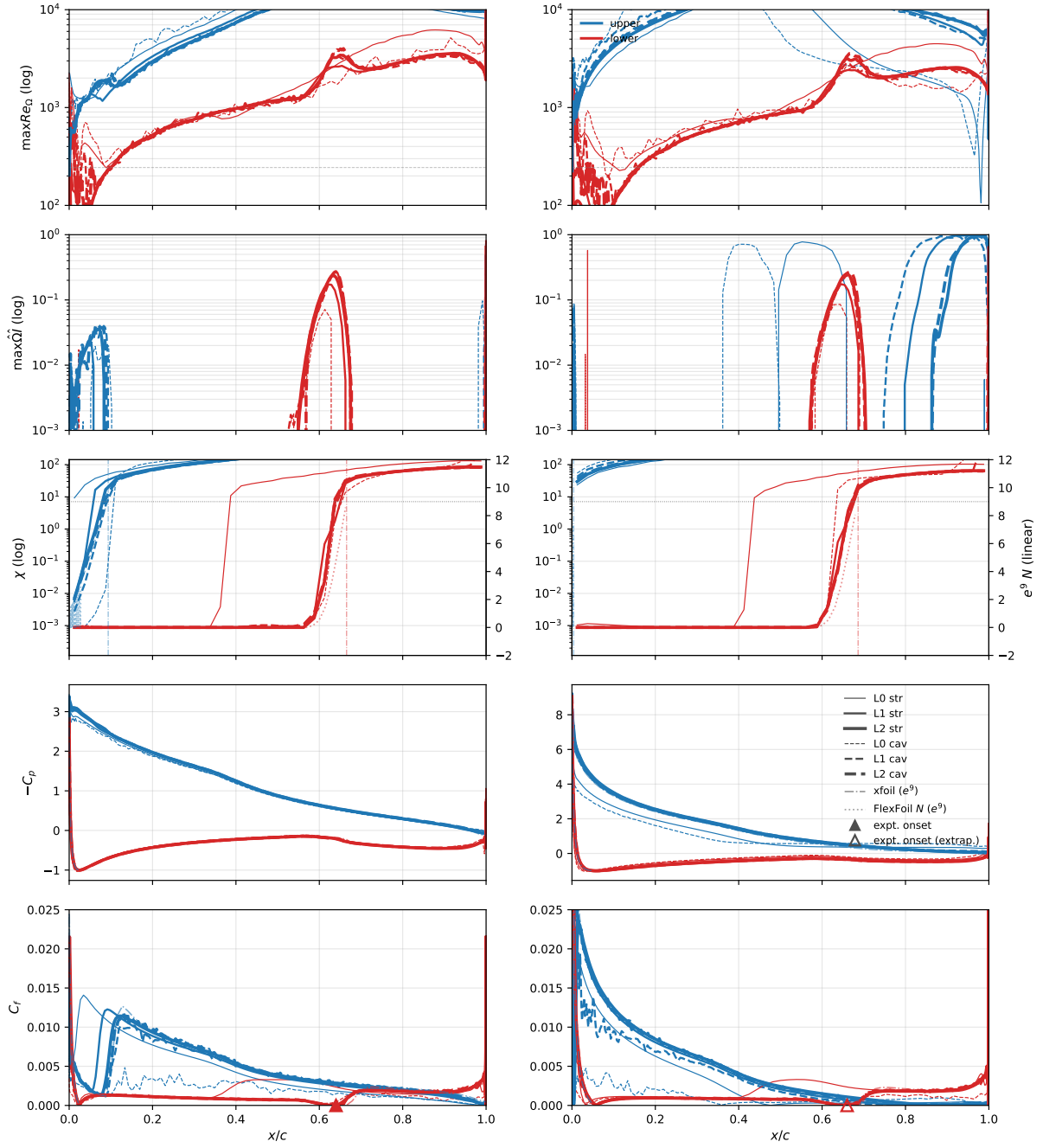


Figure 11: NLF(1)-0416, $\alpha = 9^\circ, 15^\circ$, $Re = 4 \times 10^6$.

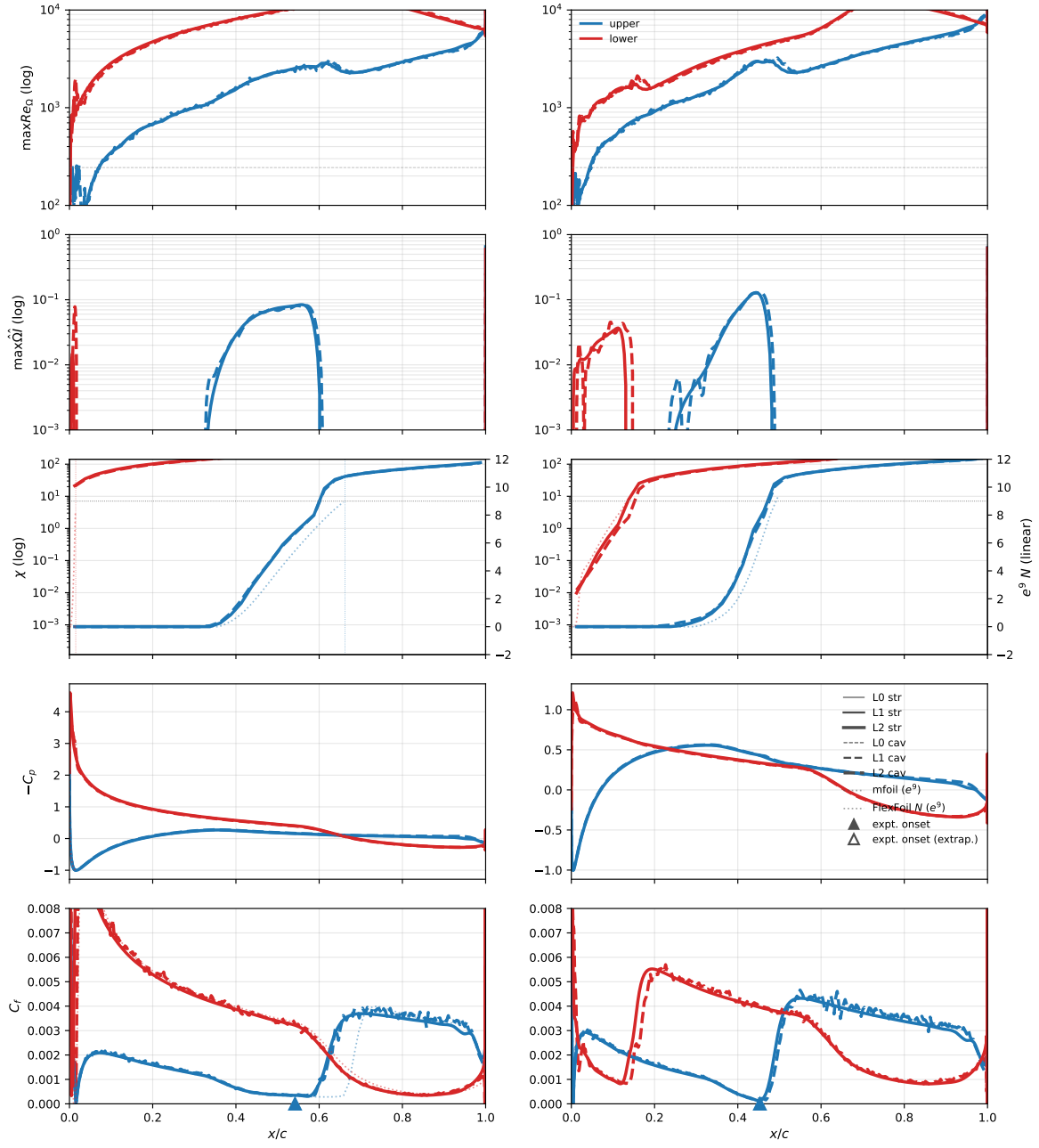


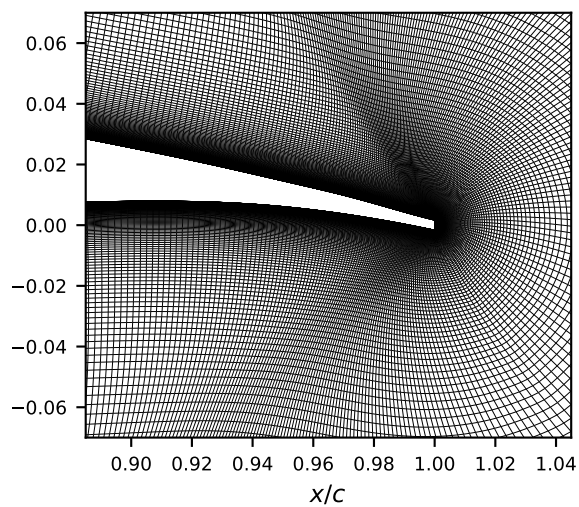
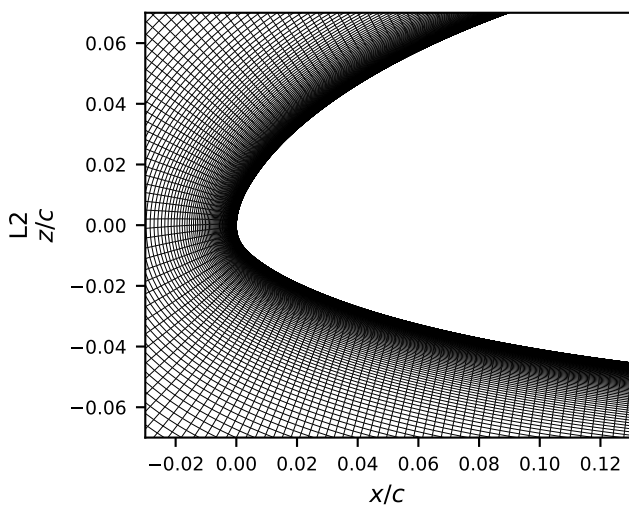
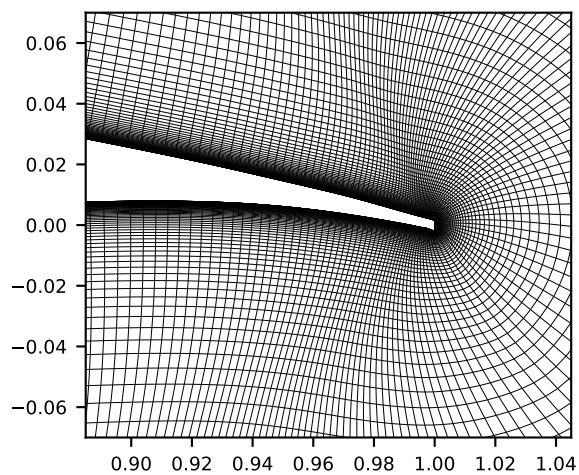
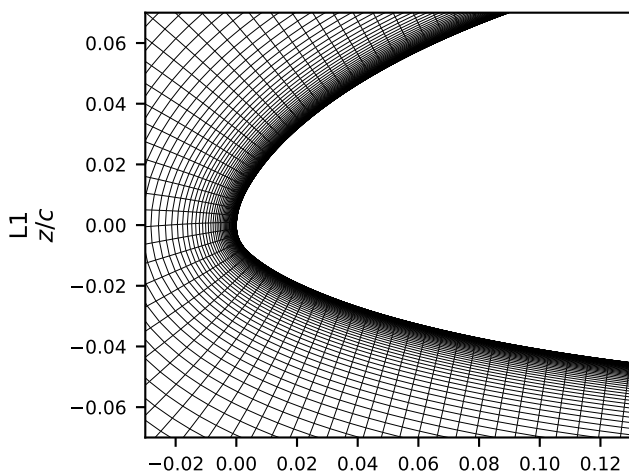
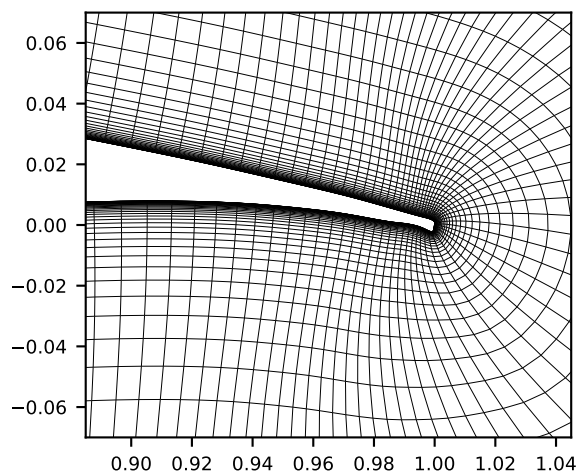
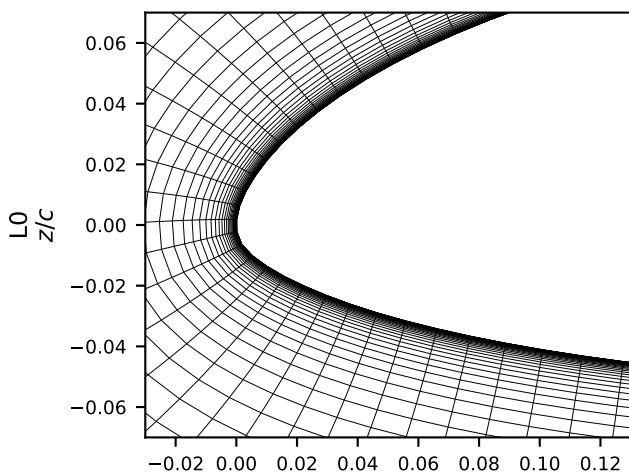
Figure 12: NLF(1)-0416, $\alpha = -8^\circ, -4^\circ$, $Re = 4 \times 10^6$.

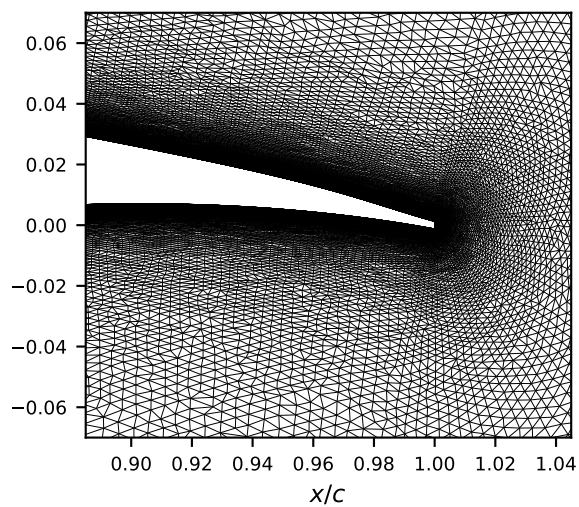
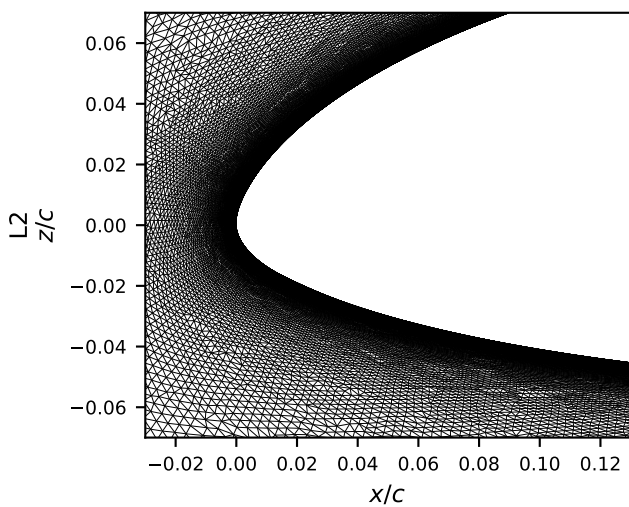
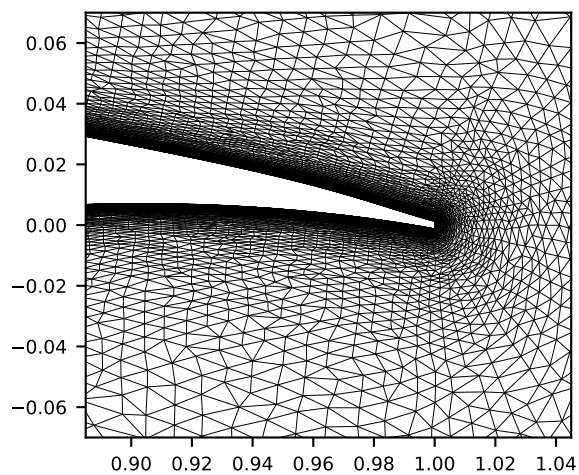
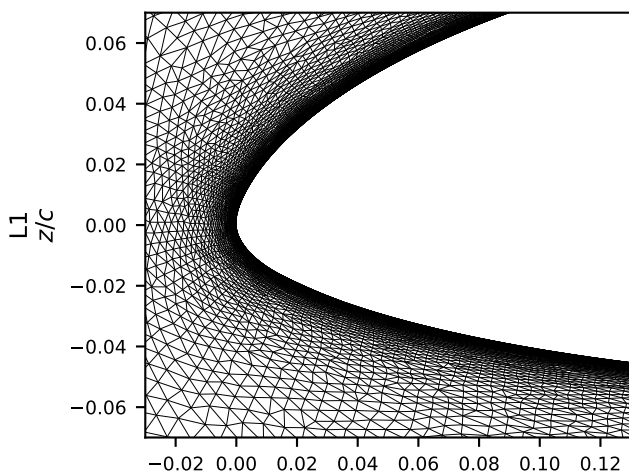
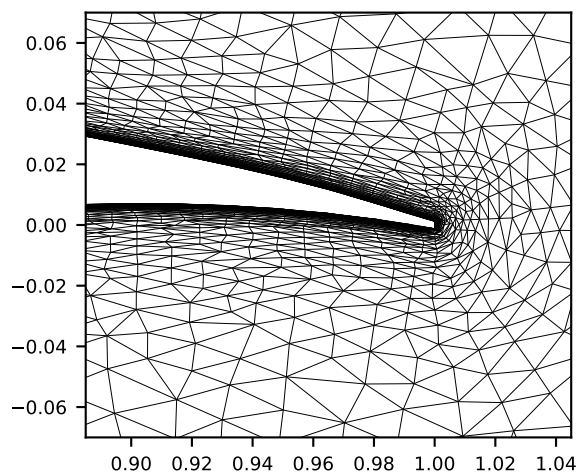
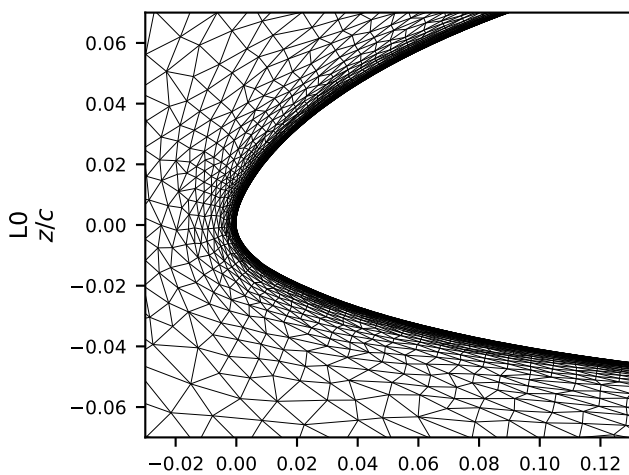
Table 3: NLF(1)-0416, $Re=4\times 10^6$: computed forces and transition locations (near-wall $\chi=1$ front stations).

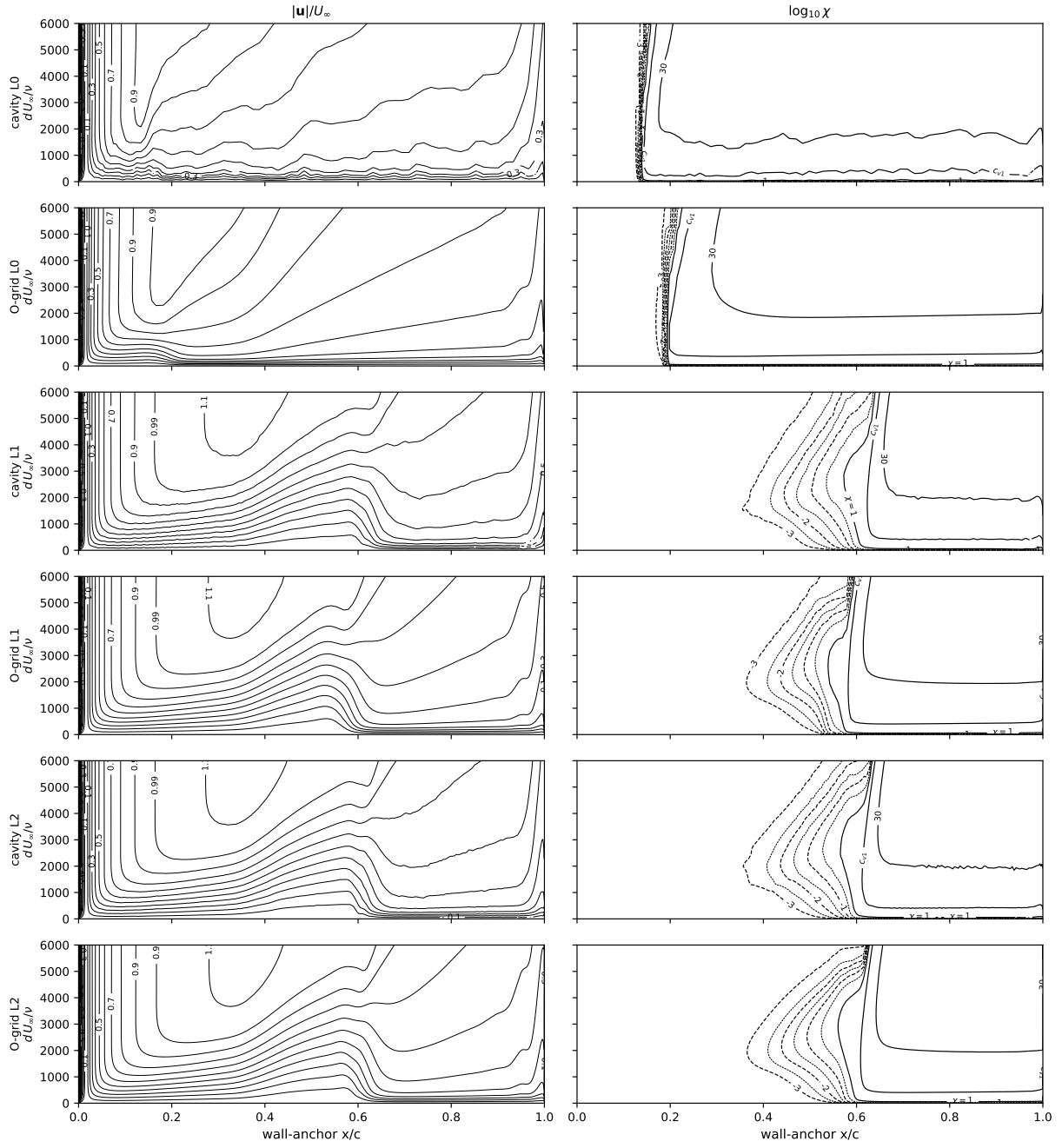
α	grid	structured O-grid				unstructured cavity			
		C_L	C_D	x_{tr}^{up}	x_{tr}^{lo}	C_L	C_D	x_{tr}^{up}	x_{tr}^{lo}
-8	L0	-0.4426	0.01830	0.195	0.011	-0.4689	0.01869	0.140	0.012
-8	L1	-0.4656	0.01093	0.538	0.006	-0.4397	0.01150	0.571	0.011
-8	L2	-0.4646	0.00962	0.559	0.004	-0.4426	0.00964	0.561	0.010
-4	L0	0.0469	0.01078	0.435	0.100	0.0197	0.01025	0.476	0.083
-4	L1	0.0142	0.00790	0.435	0.102	0.0357	0.00832	0.452	0.126
-4	L2	0.0208	0.00718	0.447	0.109	0.0363	0.00739	0.453	0.122
0	L0	0.5458	0.00803	0.400	0.590	0.4896	0.00808	0.355	0.592
0	L1	0.5048	0.00607	0.395	0.553	0.5209	0.00660	0.392	0.579
0	L2	0.5139	0.00564	0.387	0.566	0.5219	0.00606	0.391	0.569
4	L0	1.0001	0.01090	0.250	0.645	0.9513	0.01062	0.330	0.621
4	L1	0.9678	0.00800	0.238	0.610	0.9839	0.00805	0.269	0.614
4	L2	0.9805	0.00730	0.255	0.610	0.9848	0.00772	0.260	0.621
9	L0	1.4483	0.02252	0.015	0.385	1.4251	0.02345	0.105	0.641
9	L1	1.4886	0.01436	0.065	0.625	1.5112	0.01353	0.085	0.635
9	L2	1.5146	0.01245	0.073	0.628	1.5139	0.01286	0.078	0.638
15	L0	1.5172	0.07109	0.010	0.440	1.3976	0.11602	0.005	0.623
15	L1	1.9317	0.03159	0.004	0.650	1.9232	0.03454	0.001	0.659
15	L2	2.0057	0.02628	0.005	0.659	1.9924	0.02772	0.000	0.662

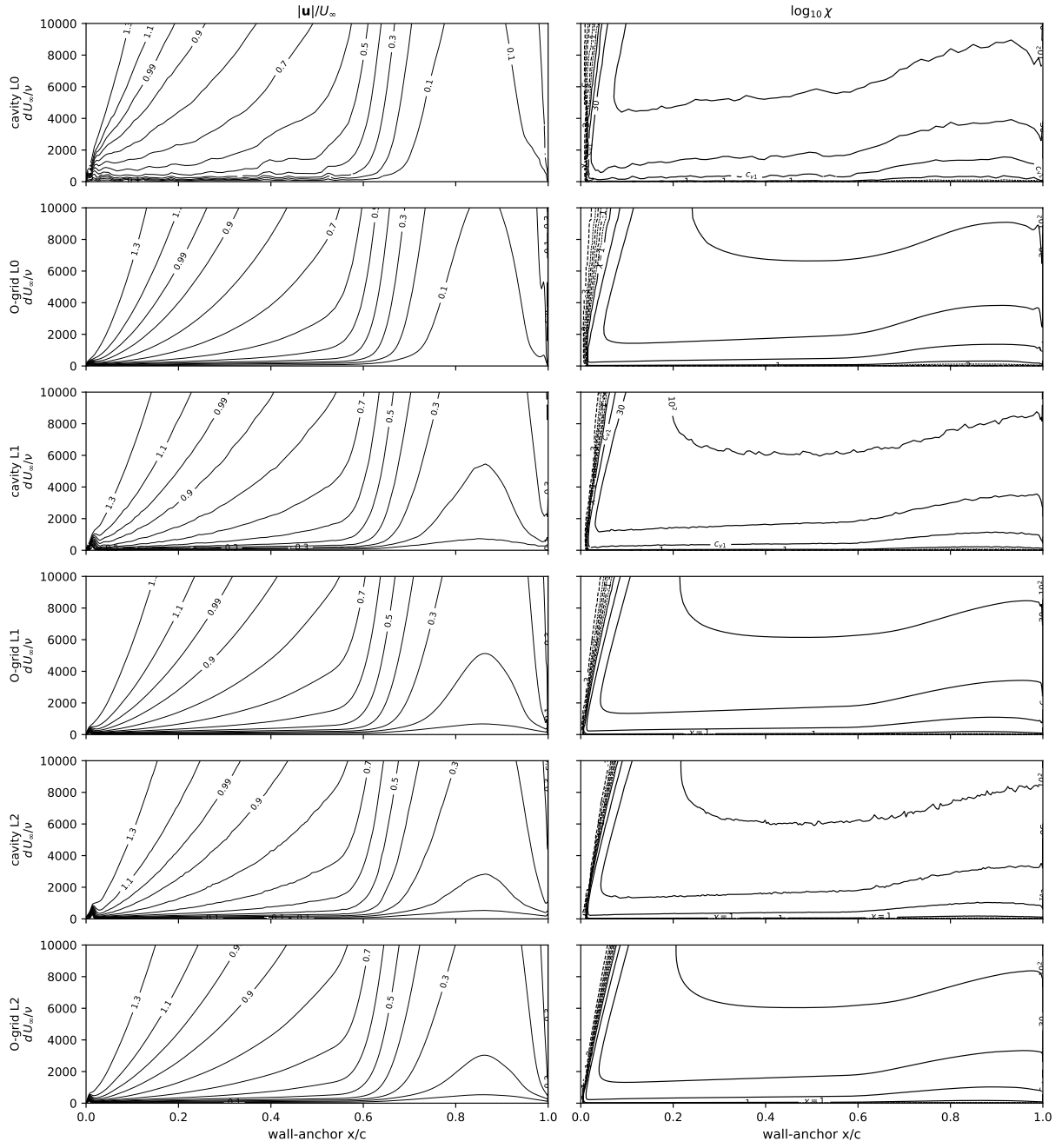
Table 4: NLF(1)-0416: the independent OpenFOAM implementation on the structured L0, L1 and L2 grids, fronts at the same near-wall $\chi=1$ convention (the port omits the f_{v1} bypass).

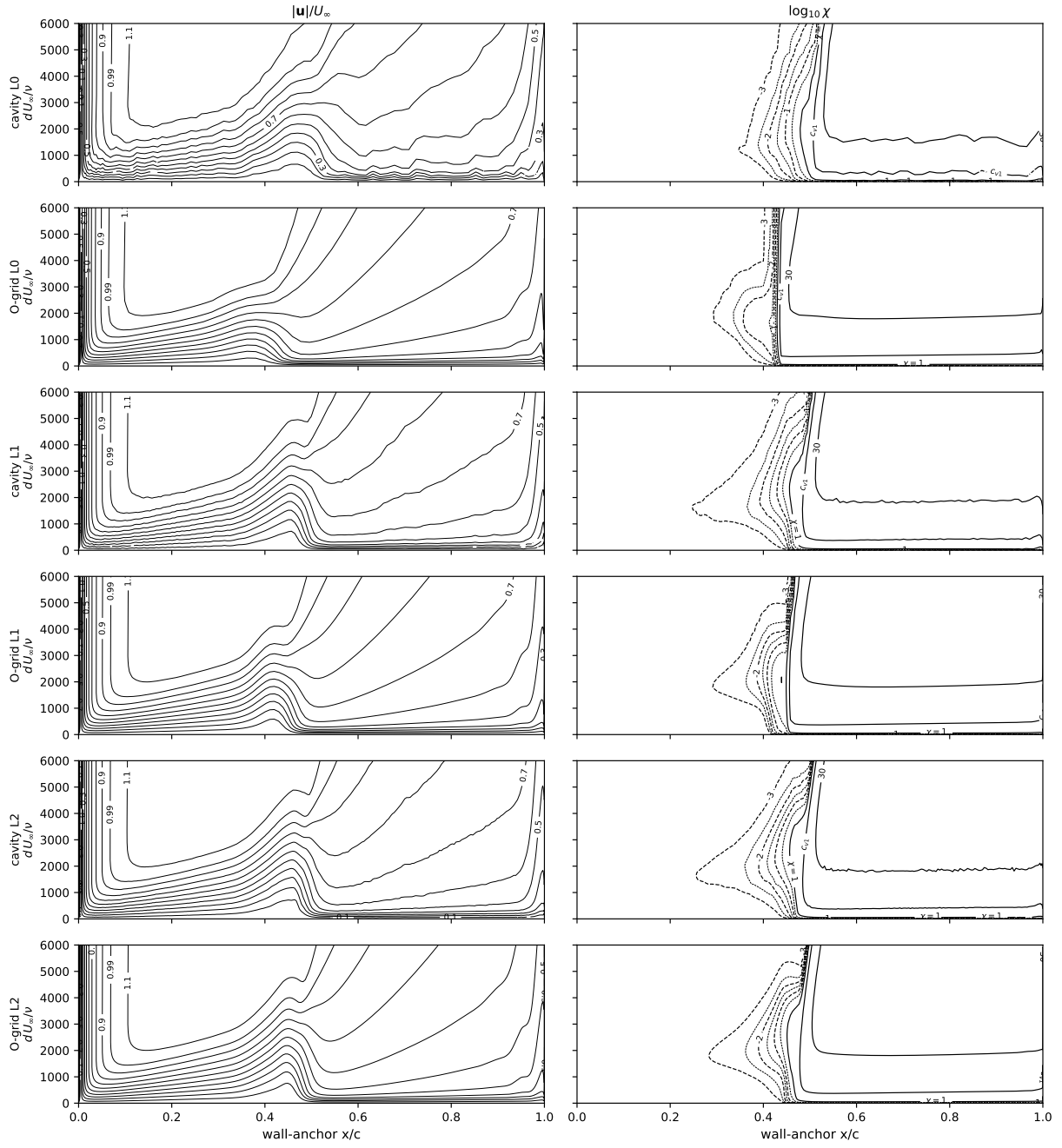
α	grid	C_L	C_D	x_{tr}^{up}	x_{tr}^{lo}
-8	L0	-0.4460	0.01241	0.556	0.007
-8	L1	-0.4785	0.00781	0.576	0.000
-8	L2	-0.4606	0.00826	0.568	0.000
-4	L0	0.0388	0.00748	0.424	0.018
-4	L1	0.0005	0.00634	0.464	0.117
-4	L2	0.0239	0.00693	0.459	0.111
0	L0	0.5473	0.00669	0.387	0.564
0	L1	0.5034	0.00603	0.393	0.572
0	L2	0.5153	0.00573	0.395	0.568
4	L0	1.0114	0.00955	0.271	0.618
4	L1	0.9670	0.00778	0.266	0.616
4	L2	0.9793	0.00749	0.261	0.625
9	L0	1.5277	0.01496	0.063	0.618
9	L1	1.5066	0.01273	0.076	0.633
9	L2	1.5151	0.01270	0.075	0.644
15	L0	1.7210	0.05409	0.000	0.634
15	L1	1.9780	0.02754	0.000	0.665
15	L2	2.0221	0.02564	0.000	0.667

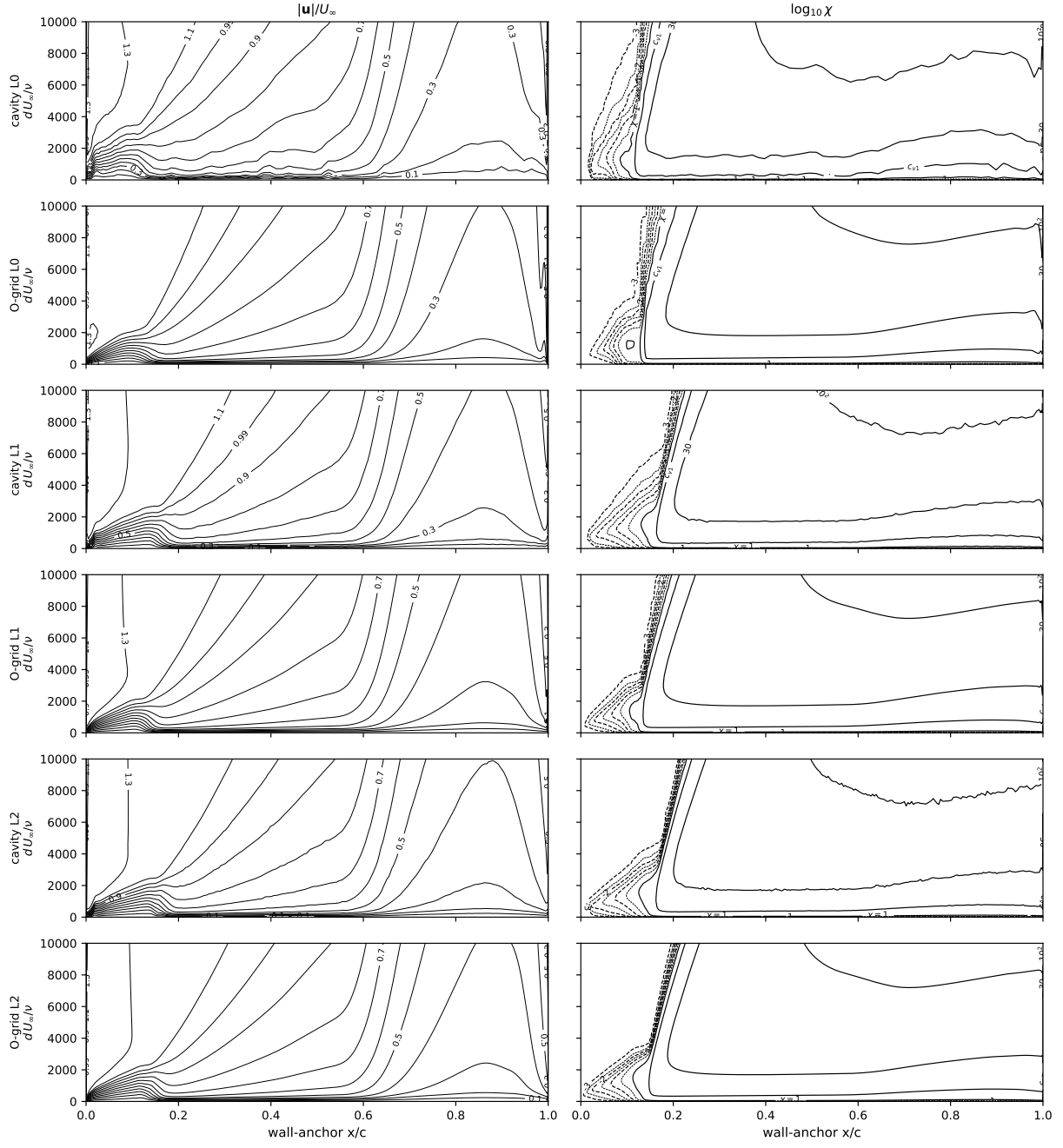


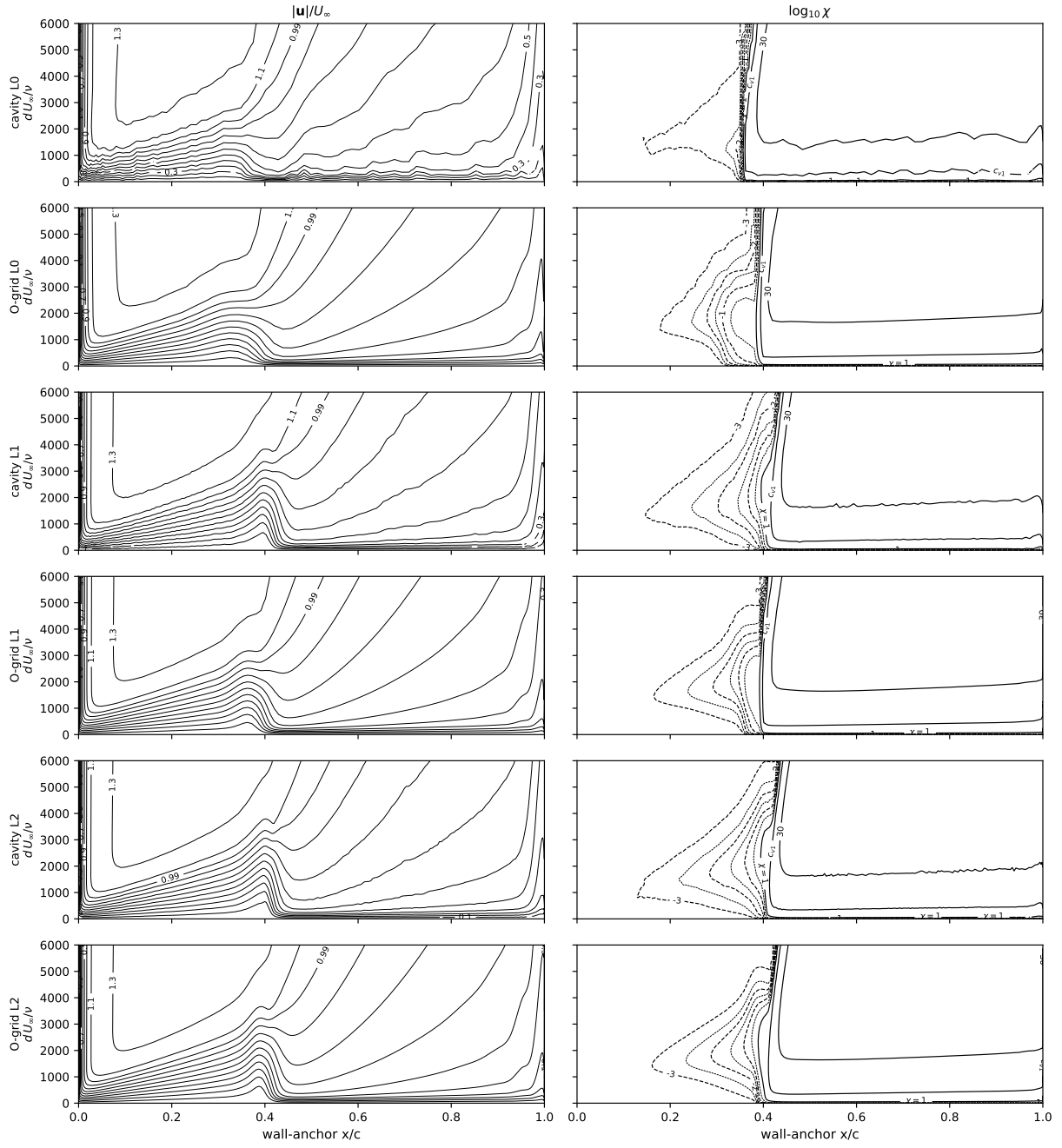


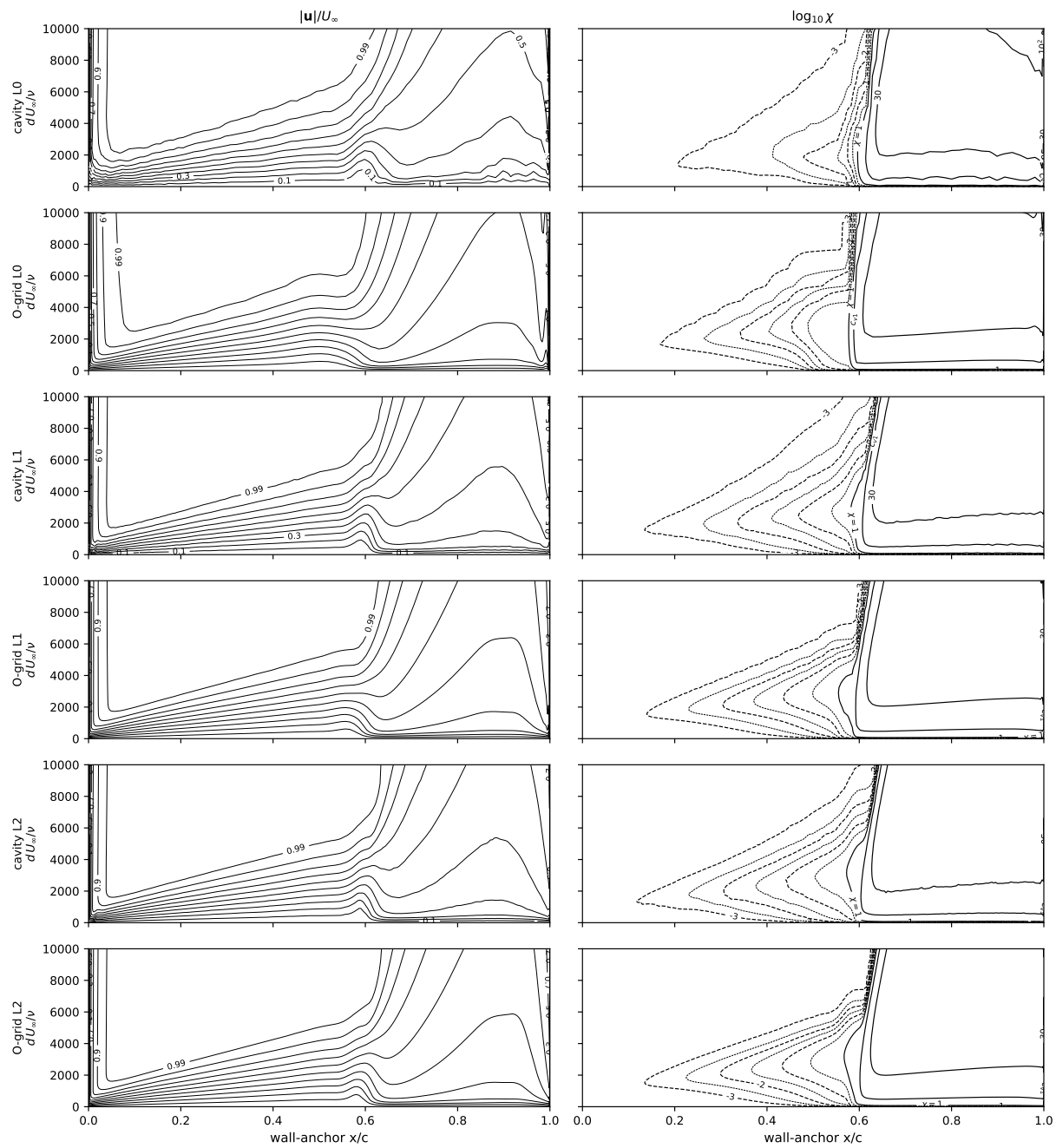


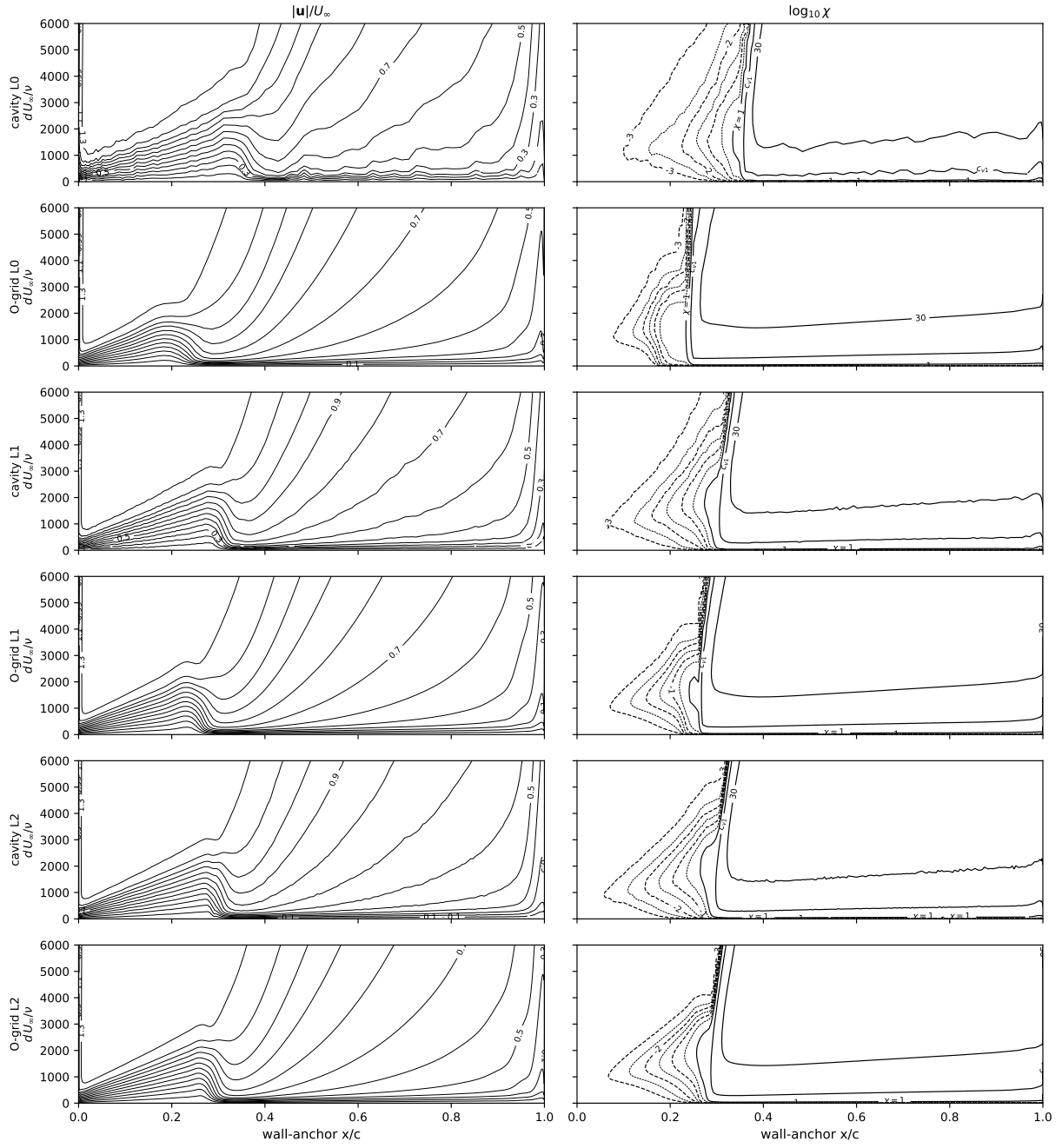


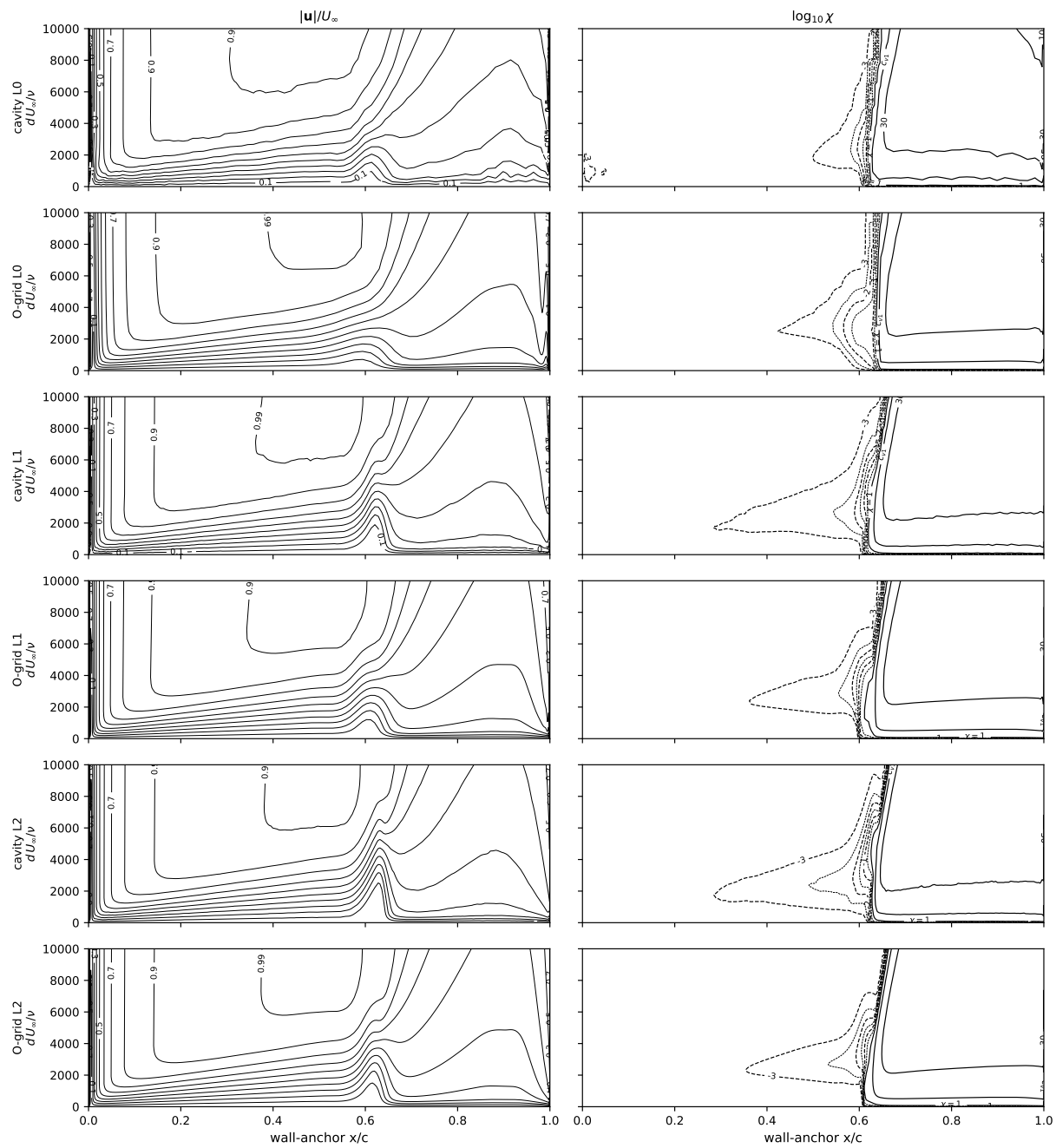


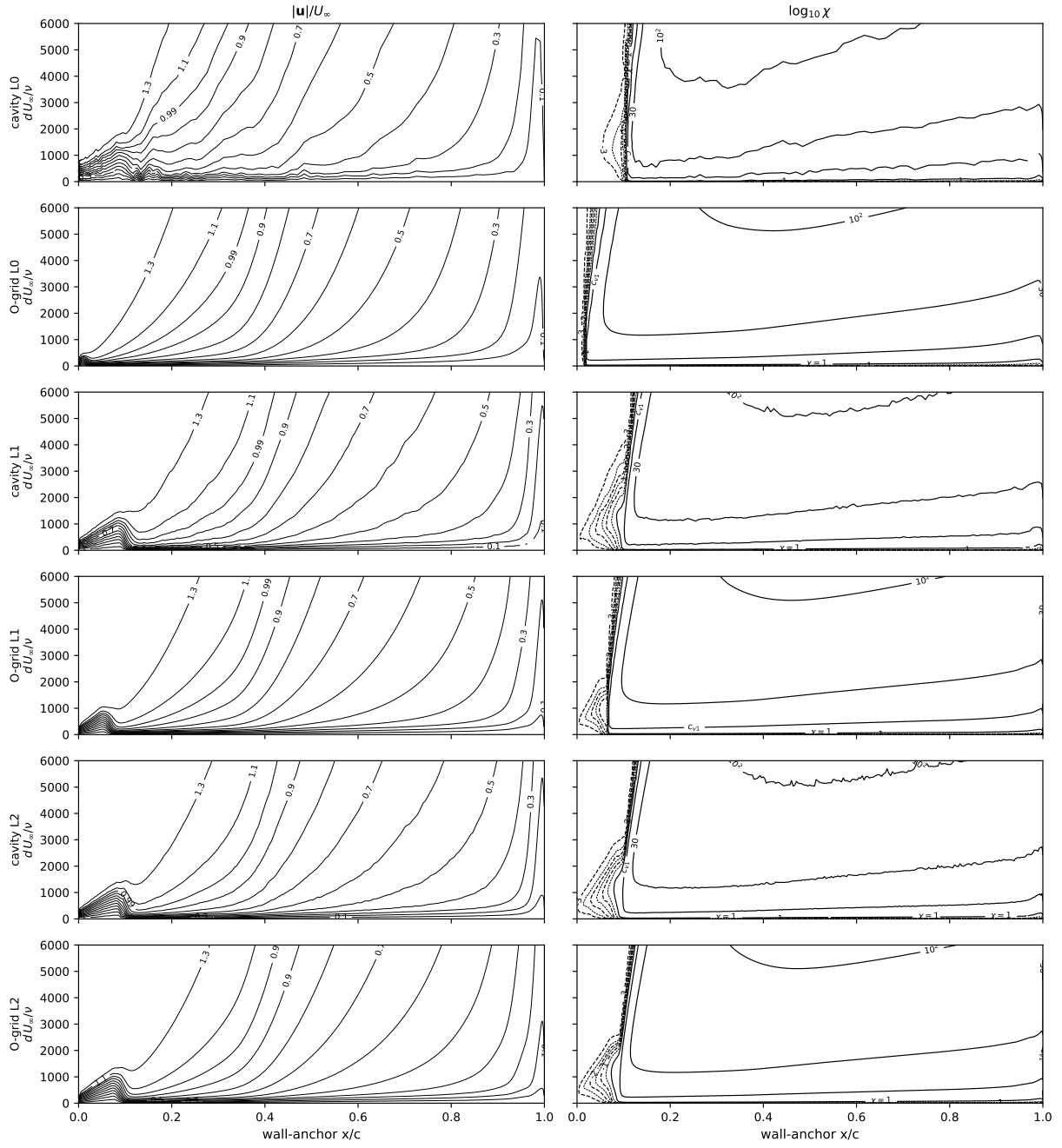


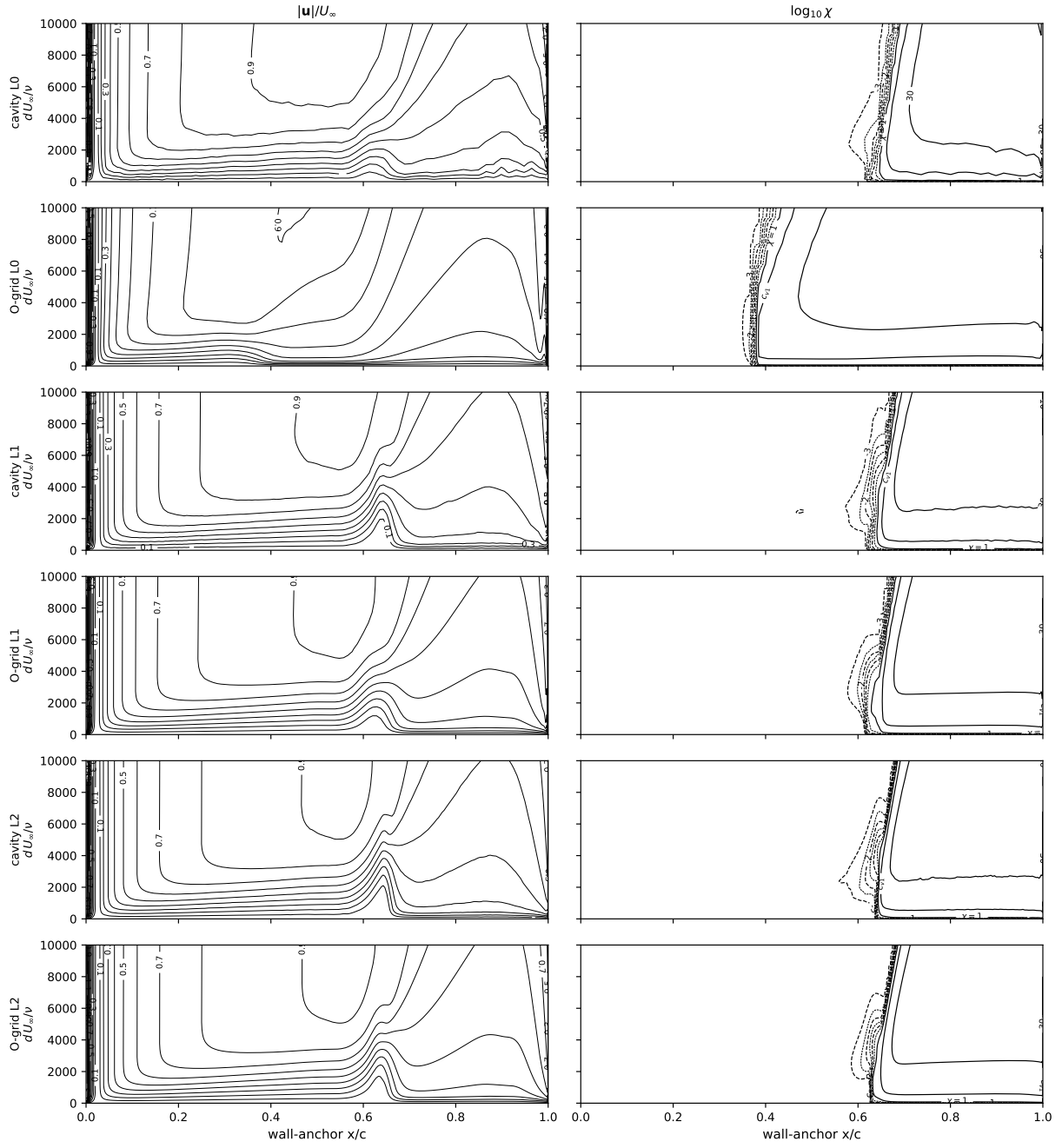


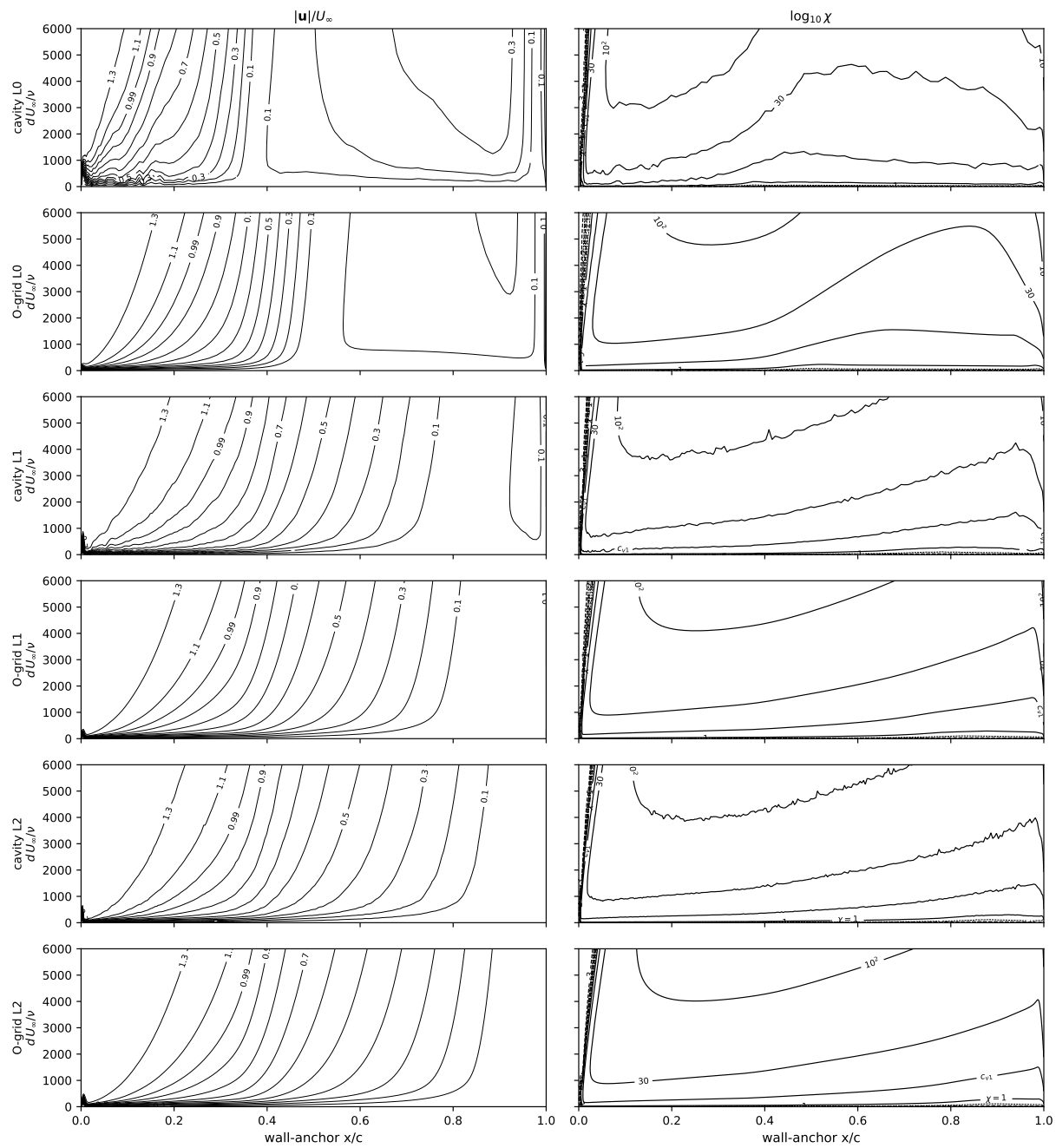


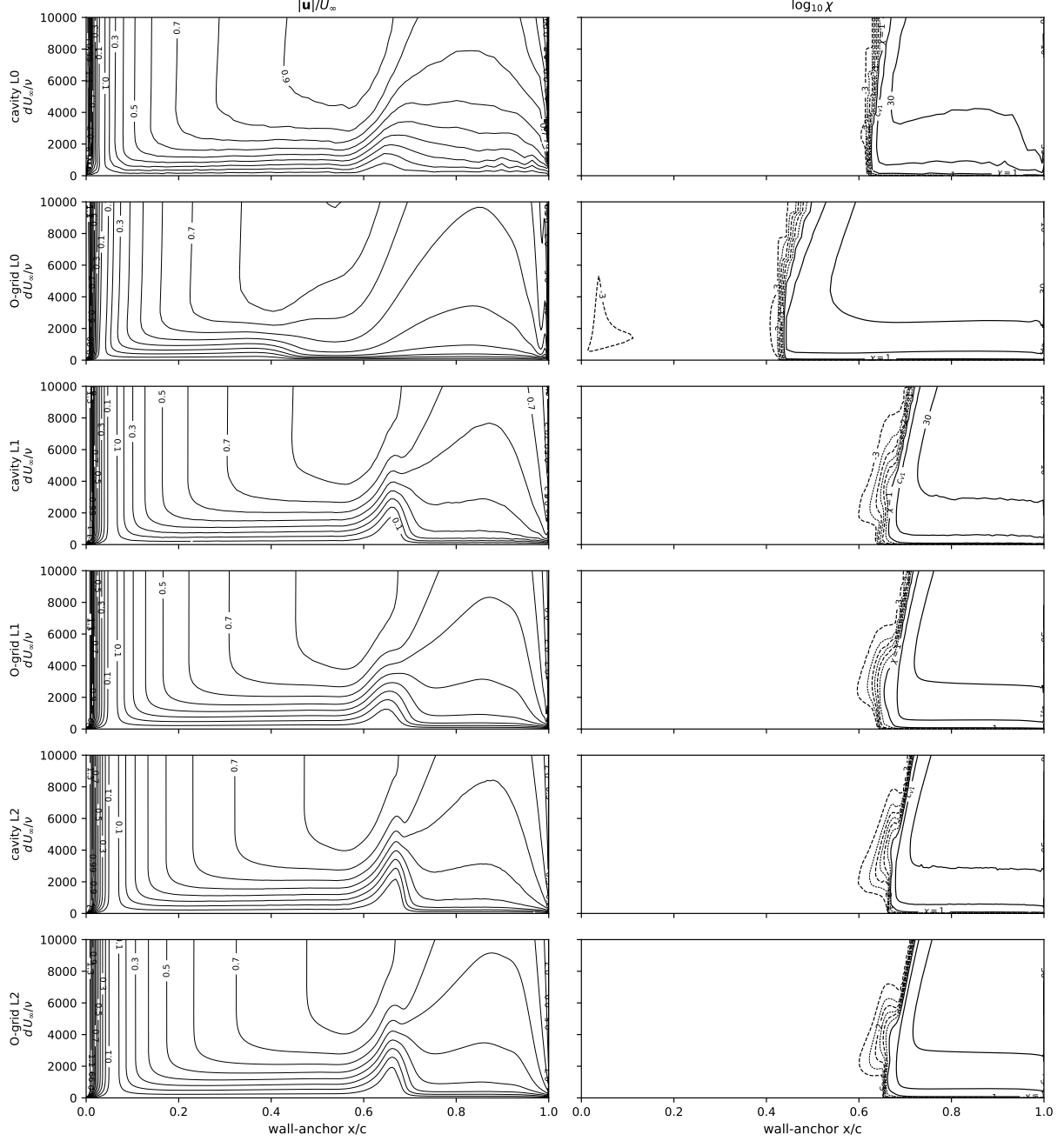












2.3 Eppler 387 at $Re = 2 \times 10^5$

$\alpha \in \{0^\circ, 2^\circ, 5^\circ, 7^\circ\}$ on both families at L0–L2 plus six extension incidences ($-2^\circ, 1^\circ, 3^\circ, 4^\circ, 6^\circ, 8.5^\circ$) on the finest pair; same seed. Figure 13: drag polar against the LTPT measurement (McGhee et al., TM-4062), the e^9 reference, fully turbulent SA, and the digitized $\gamma-Re_\theta$ and SA-BC curves (-6.4% to $+3.7\%$ over $\alpha = 0^\circ$ – 7°). Figure 14: bubble stations against the oil flow, Selig et al., Cole–Mueller, and the e^9 references—reattachment 0.02 – $0.04c$ late at $\alpha \leq 5^\circ$, at the edge of bubble collapse at 7° . Figures 15–16: surface suites. Table 5: forces. Table 6: bubble stations. Table 7: at L2 lift within 0.5% , separation 0.030 – $0.037c$ ahead of and reattachment 0.033 – $0.045c$ behind the SA-AI stations.

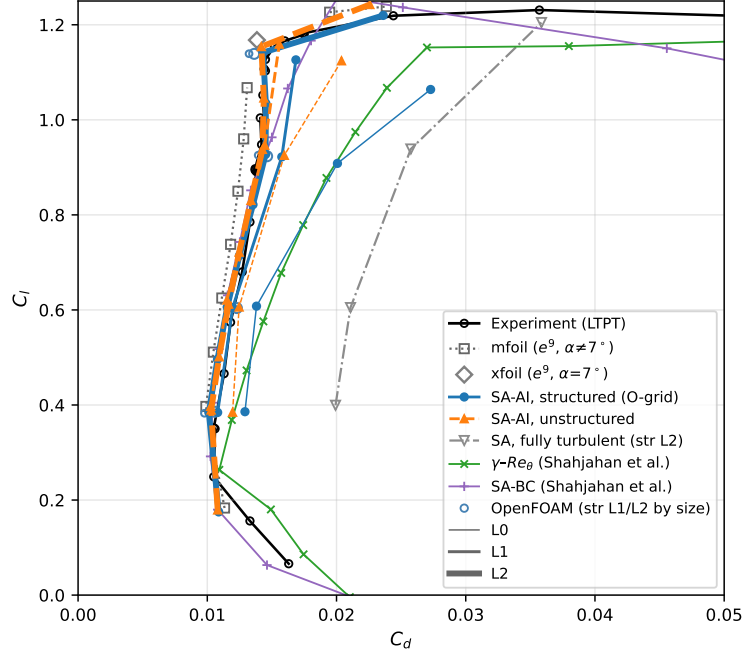


Figure 13: Eppler 387 drag polar, $Re = 2 \times 10^5$. Fully turbulent SA ($\chi_\infty = 3$) 77–97% above the SA-AI drag in the bucket, +151% at $\alpha = 7^\circ$; mesh families agree to $\Delta C_d \leq 2.4 \times 10^{-4}$ over $\alpha = -2^\circ$ – 7° and 1.0×10^{-3} at 8.5° ; OpenFOAM port omits the bypass, worth -6 to -10 counts at $\alpha = 0$ – 5° .

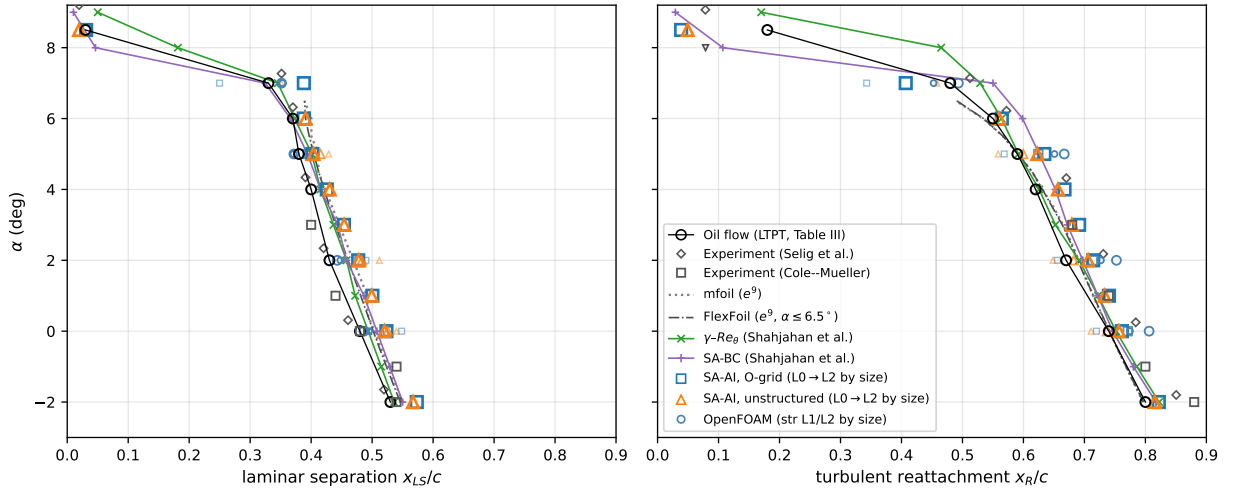


Figure 14: Eppler 387 bubble stations, $Re = 2 \times 10^5$; signed- C_f zero crossings against oil flow (TM-4062 Table III), Selig, Cole–Mueller, e^9 . At $\alpha = 7^\circ$ XFOIL $C_l = 1.17$, mfoil $C_l = 0.928$.

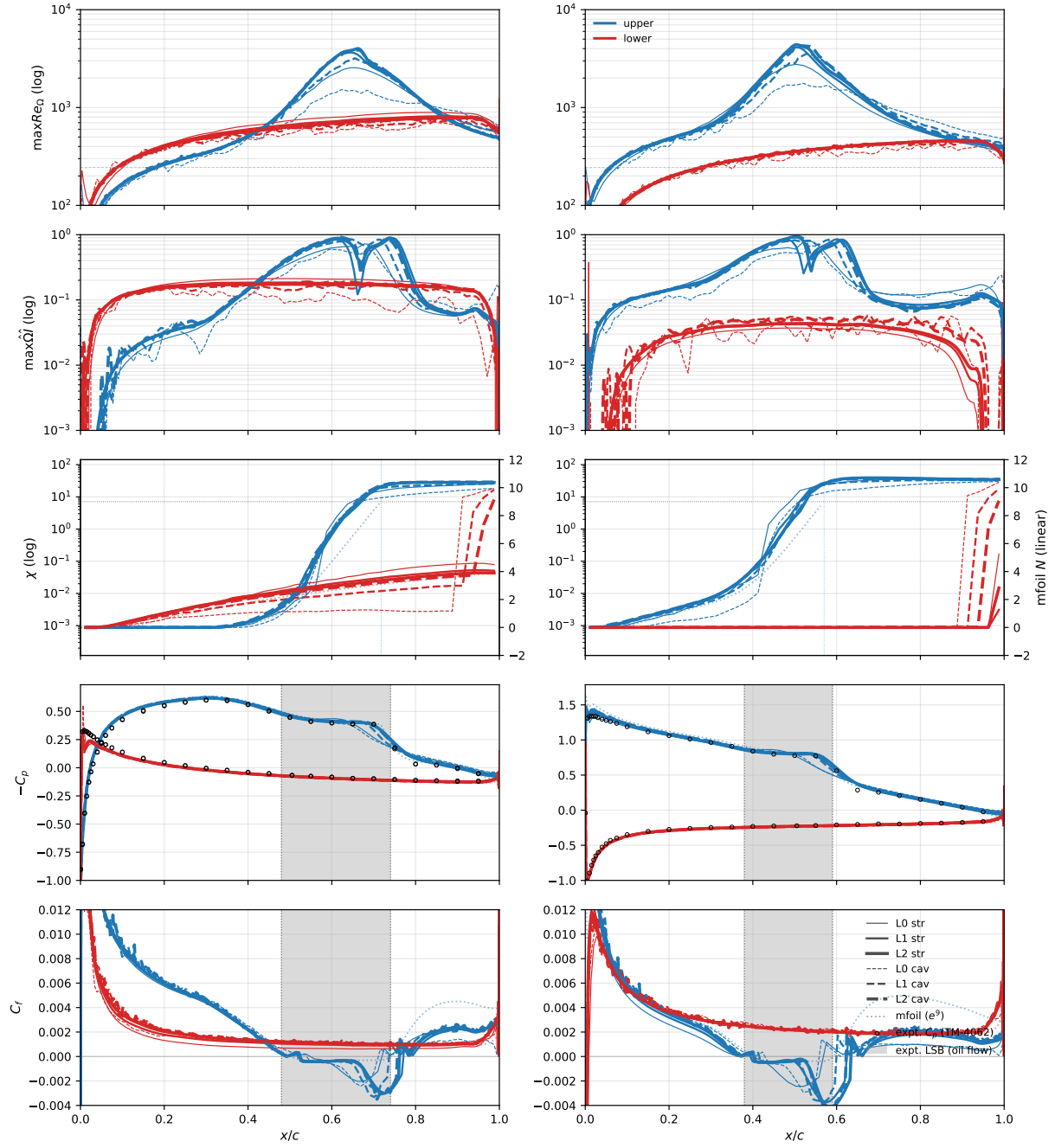


Figure 15: Eppler 387, $\alpha = 0^\circ, 5^\circ$, $Re = 2 \times 10^5$.

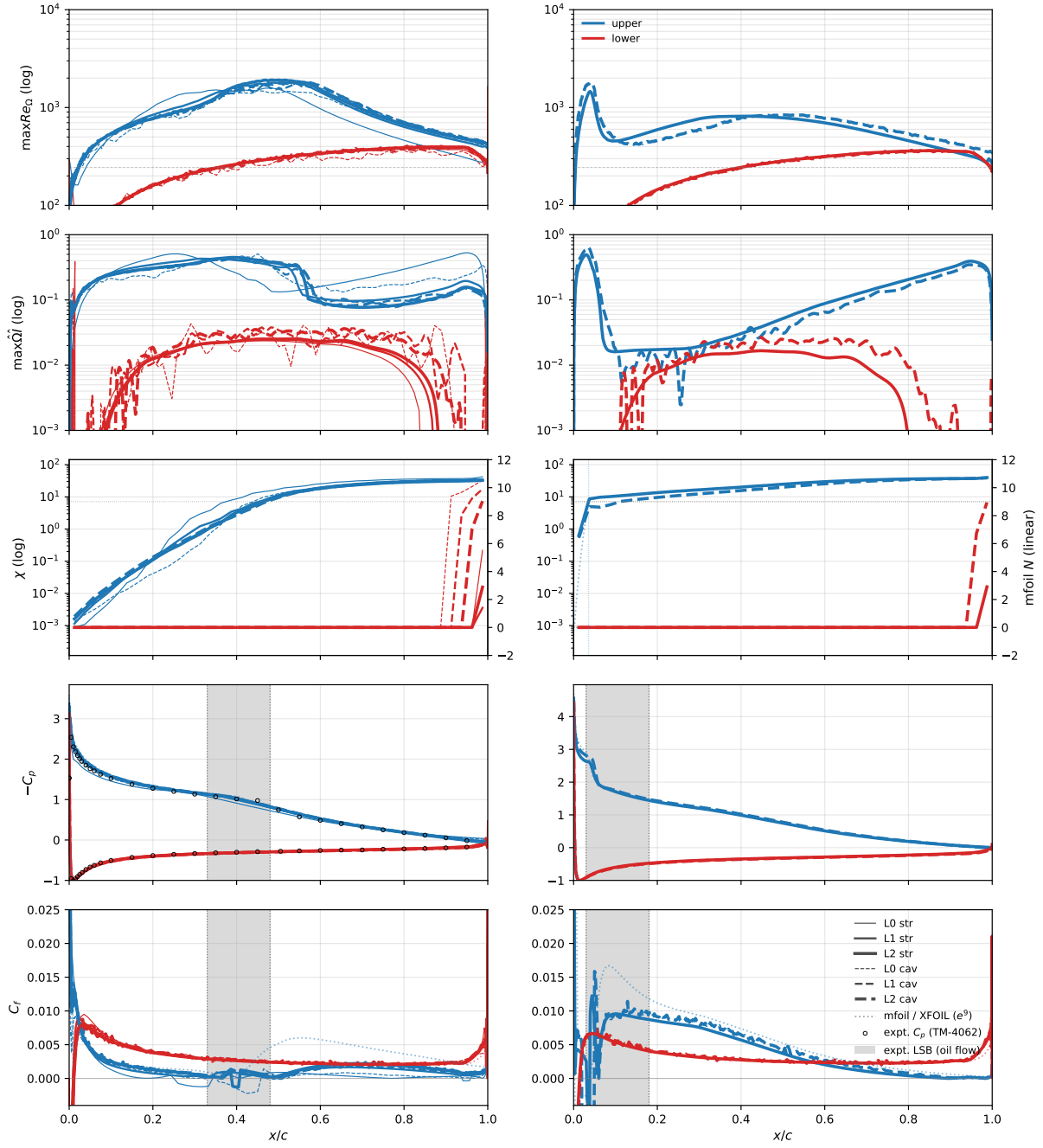


Figure 16: Eppler 387, $\alpha = 7^\circ, 8.5^\circ$, $Re = 2 \times 10^5$.

Table 5: Eppler 387, $Re = 2 \times 10^5$: computed forces (the -2° – 8.5° extension incidences exist on the finest pair only).

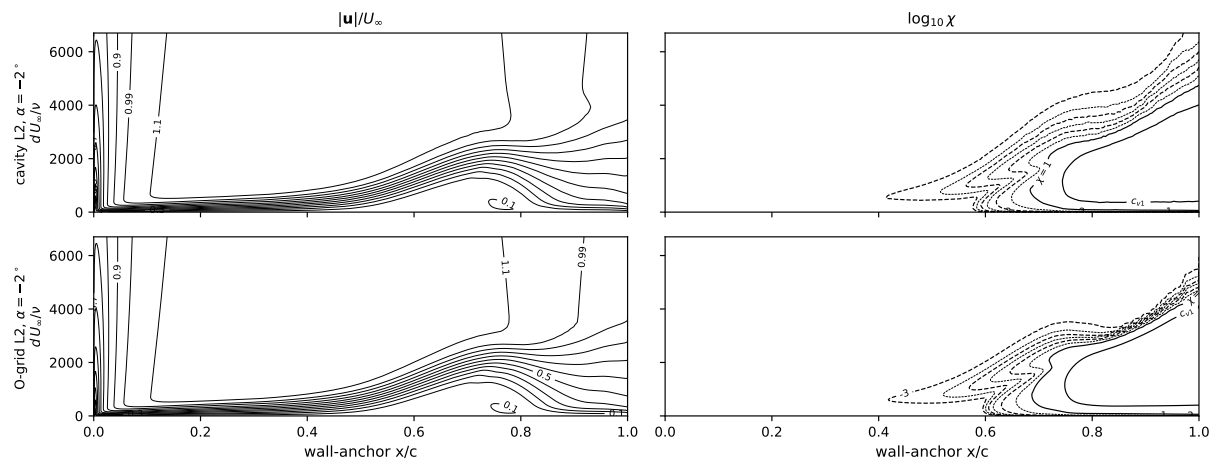
α	grid	structured C_L	O-grid C_D	unstructured C_L	cavity C_D
-2	L2	0.1752	0.01089	0.1791	0.01080
0	L0	0.3859	0.01292	0.3847	0.01196
0	L1	0.3843	0.01080	0.3969	0.01026
0	L2	0.3844	0.01010	0.3894	0.01030
1	L2	0.4957	0.01078	0.5012	0.01092
2	L0	0.6079	0.01381	0.6061	0.01246
2	L1	0.6048	0.01196	0.6207	0.01152
2	L2	0.6059	0.01154	0.6124	0.01167
3	L2	0.7146	0.01247	0.7218	0.01245
4	L2	0.8218	0.01354	0.8304	0.01341
5	L0	0.9083	0.02007	0.9253	0.01596
5	L1	0.9218	0.01575	0.9465	0.01444
5	L2	0.9274	0.01451	0.9375	0.01428
6	L2	1.0346	0.01453	1.0456	0.01439
7	L0	1.0639	0.02728	1.1241	0.02038
7	L1	1.1262	0.01686	1.1593	0.01557
7	L2	1.1415	0.01431	1.1535	0.01419
8.5	L2	1.2190	0.02310	1.2611	0.02150

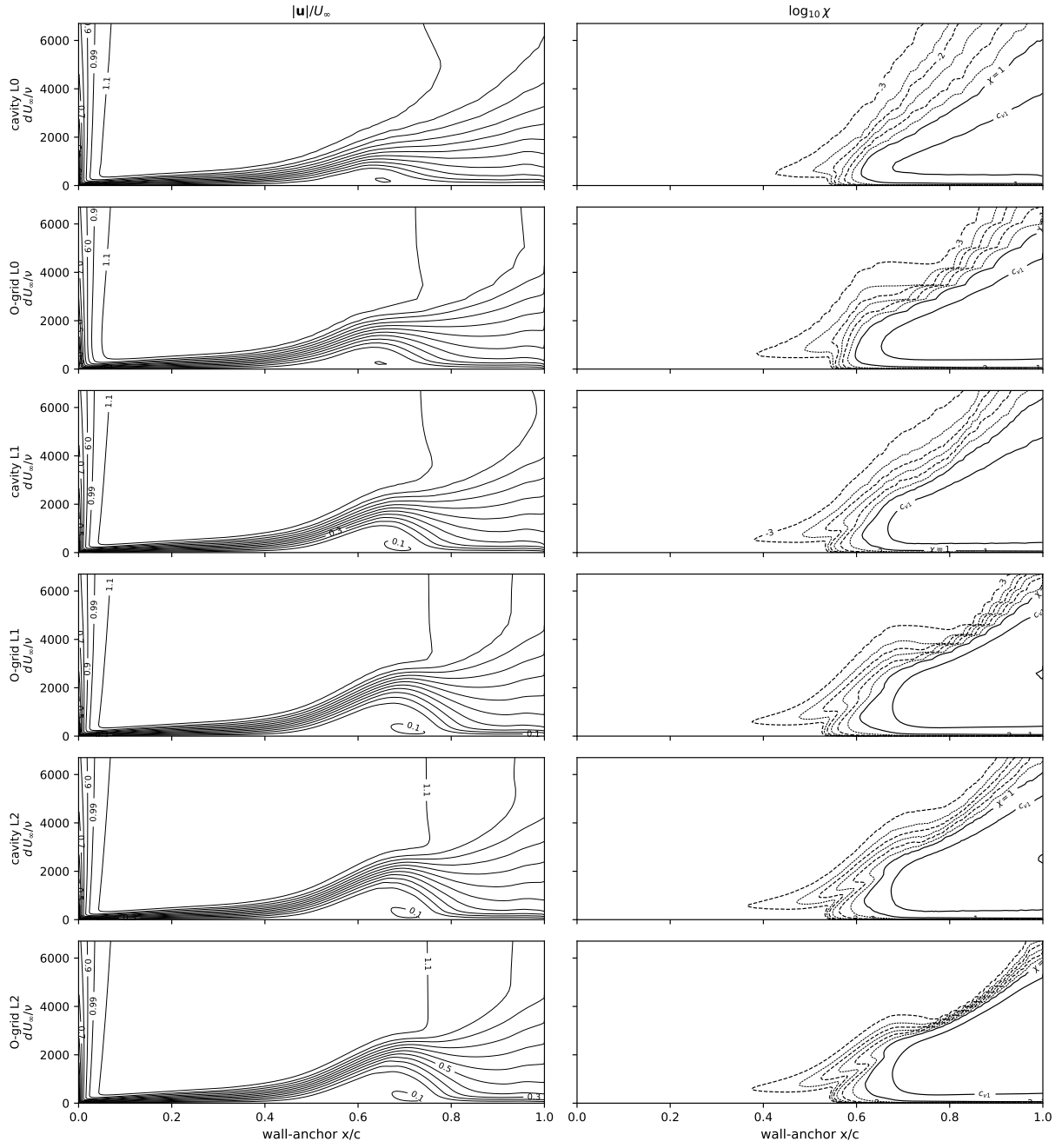
Table 6: Eppler 387, $Re = 2 \times 10^5$: computed upper-surface laminar-separation and turbulent-reattachment stations (signed- C_f zero crossings; missing entries transition ahead of separation and have no bubble). The measured oil-flow reattachments (TM-4062 Table III) at $\alpha = 0^\circ/2^\circ/5^\circ$ are 0.74/0.67/0.59 c .

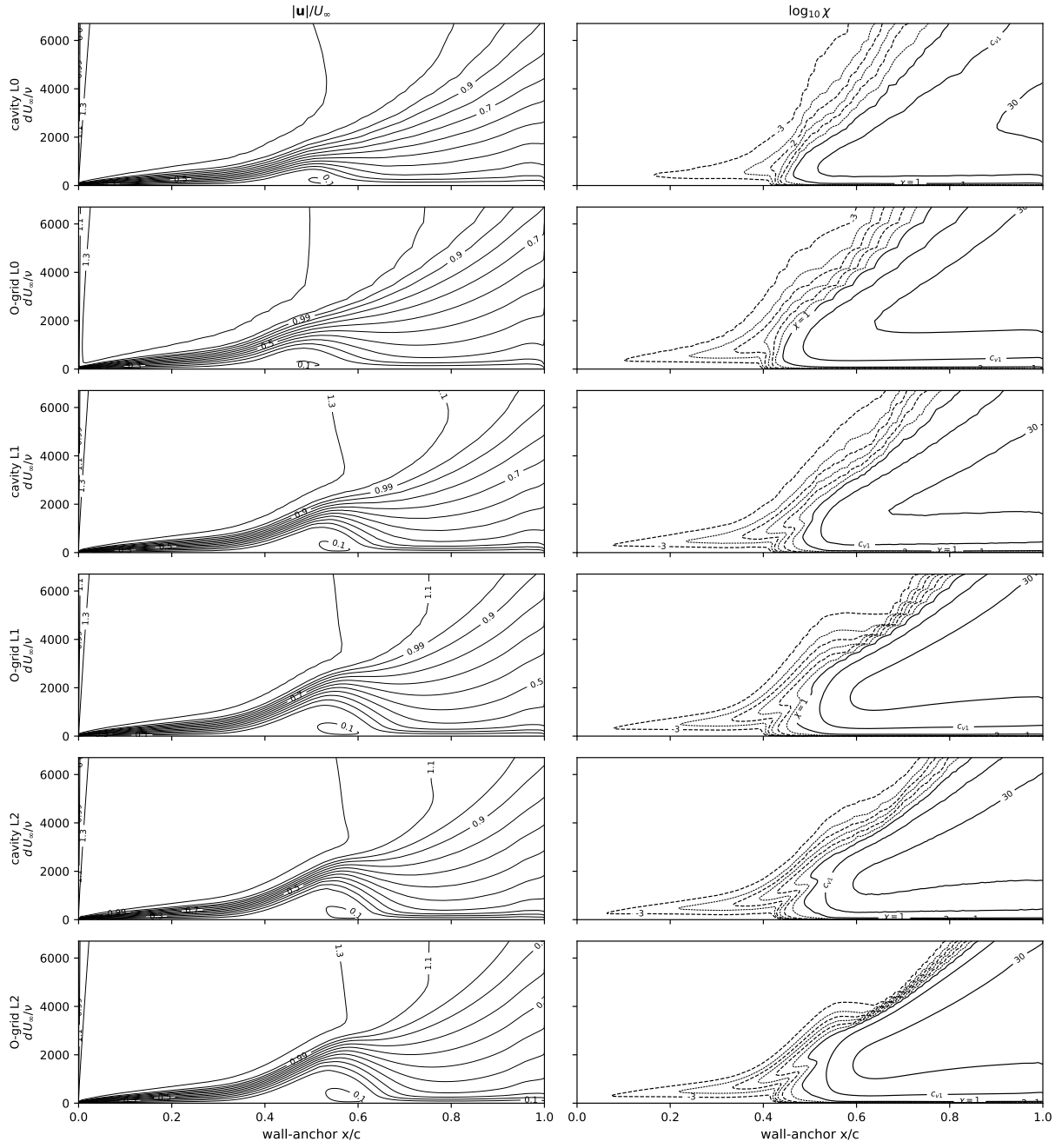
α	grid	structured x_{LS}/c	O-grid x_R/c	unstructured x_{LS}/c	cavity x_R/c
-2	L2	0.573	0.822	0.567	0.815
0	L0	0.549	0.720	0.540	0.711
0	L1	0.526	0.771	0.526	0.737
0	L2	0.523	0.761	0.520	0.757
1	L2	0.500	0.740	0.499	0.733
2	L0	0.490	0.656	0.512	0.649
2	L1	0.479	0.713	0.483	0.682
2	L2	0.477	0.714	0.478	0.706
3	L2	0.454	0.691	0.454	0.679
4	L2	0.426	0.667	0.430	0.657
5	L0	0.393	0.568	0.429	0.558
5	L1	0.394	0.624	0.415	0.600
5	L2	0.402	0.634	0.404	0.623
6	L2	0.388	0.565	0.391	0.560
7	L0	0.250	0.343	0.351	0.458
7	L1	—	—	—	—
7	L2	0.388	0.407	—	—
8.5	L2	0.031	0.039	0.019	0.049

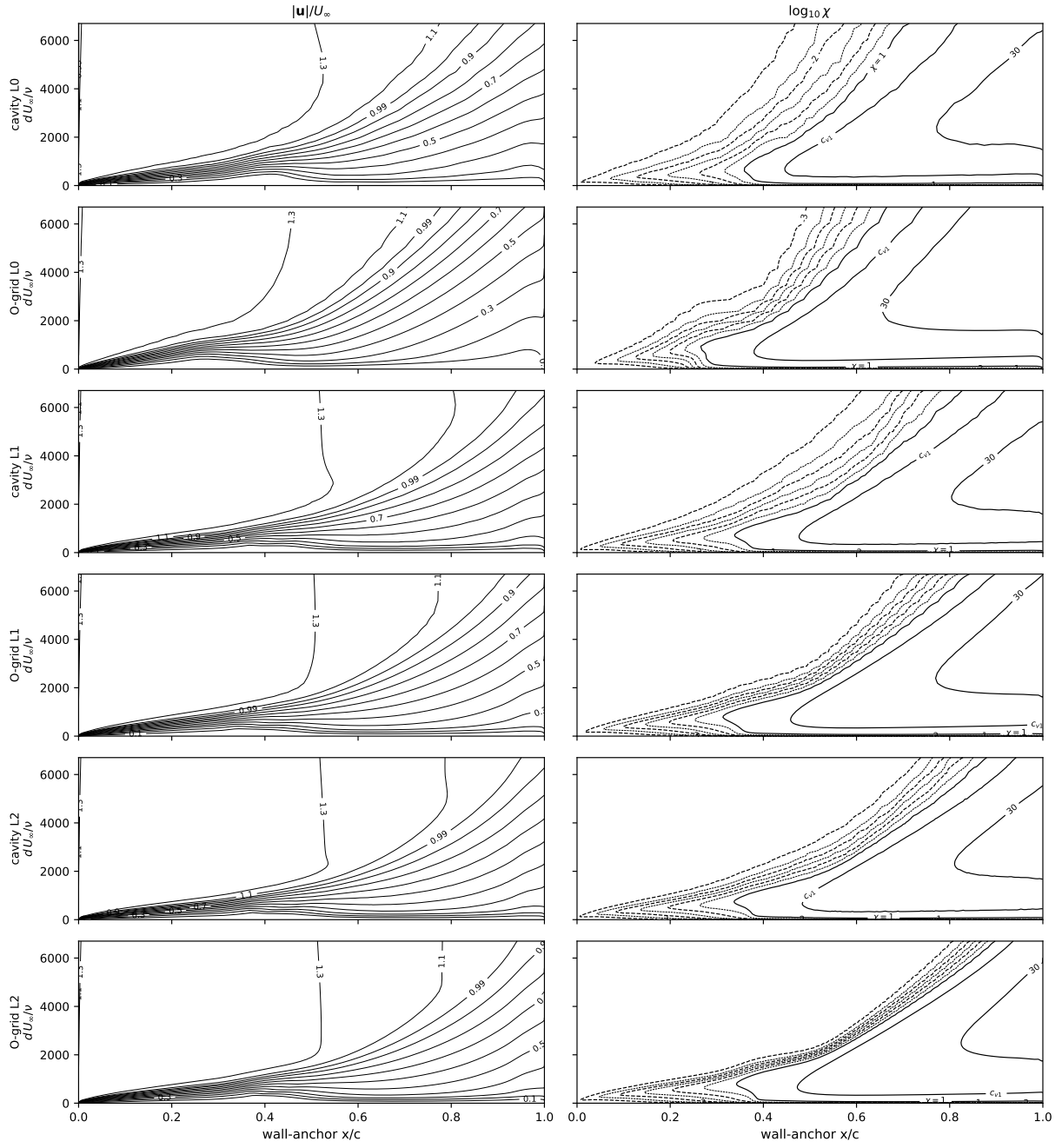
Table 7: Eppler 387: the independent OpenFOAM implementation on the structured L0, L1 and L2 grids (fronts at the near-wall $\chi = 1$ convention, stations at the signed- C_f zero crossings; the port omits the f_{v1} bypass).

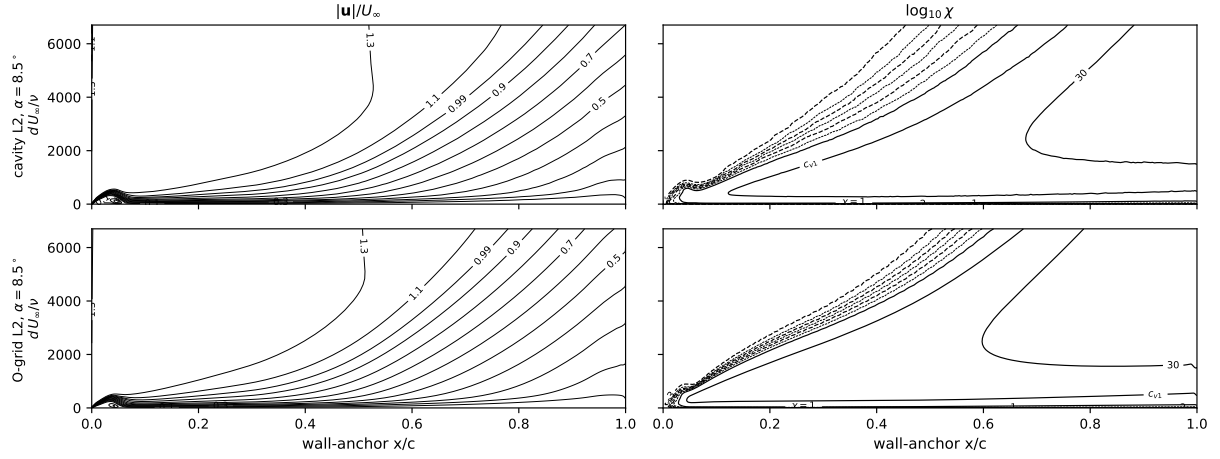
α	grid	C_L	C_D	x_{tr}^{up}	x_{LS}	x_R
0	L0	0.3887	0.01167	0.638	0.493	0.784
0	L1	0.3826	0.00978	0.605	0.495	0.775
0	L2	0.3857	0.01058	0.621	0.486	0.806
2	L0	0.6133	0.01406	0.568	0.447	0.739
2	L1	0.6037	0.01174	0.543	0.448	0.727
2	L2	0.6039	0.01229	0.564	0.443	0.752
5	L0	0.9292	0.01665	0.457	0.344	0.654
5	L1	0.9256	0.01394	0.468	0.371	0.651
5	L2	0.9234	0.01463	0.487	0.372	0.667
7	L0	1.1173	0.01817	0.139	—	0.201
7	L1	1.1391	0.01323	0.329	0.353	0.453
7	L2	1.1388	0.01363	0.379	0.351	0.493











2.4 Eppler 387 Reynolds sweep at $\alpha = 5^\circ$

$Re \in \{6 \times 10^4, 10^5, 3 \times 10^5, 4.6 \times 10^5\}$ on the same grids (24 further solutions). Figure 17: the refinement fan closes onto the measurement at $2\text{--}4.6 \times 10^5$ and opens at 10^5 , the model's bursting boundary, one Reynolds step above the measurement's (6×10^4). Figures 18–19: surface suites. Table 8: L2 coefficients.

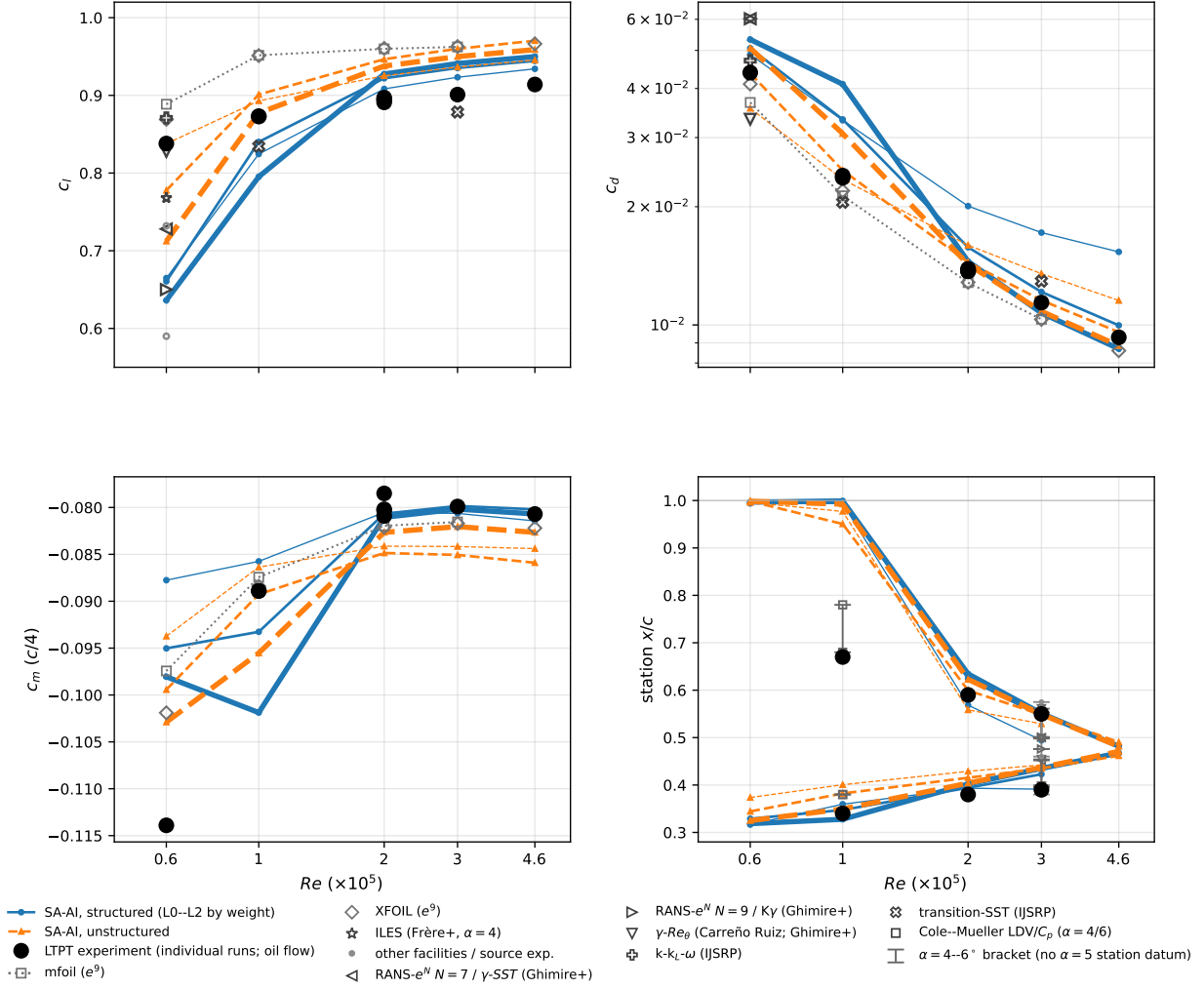


Figure 17: Reynolds-sweep forces and stations, $\alpha = 5^\circ$, $Re = 6 \times 10^4$ – 4.6×10^5 . LTPT runs 1/2/4/1/1 at $0.6/1/2/3/4.6 \times 10^5$.

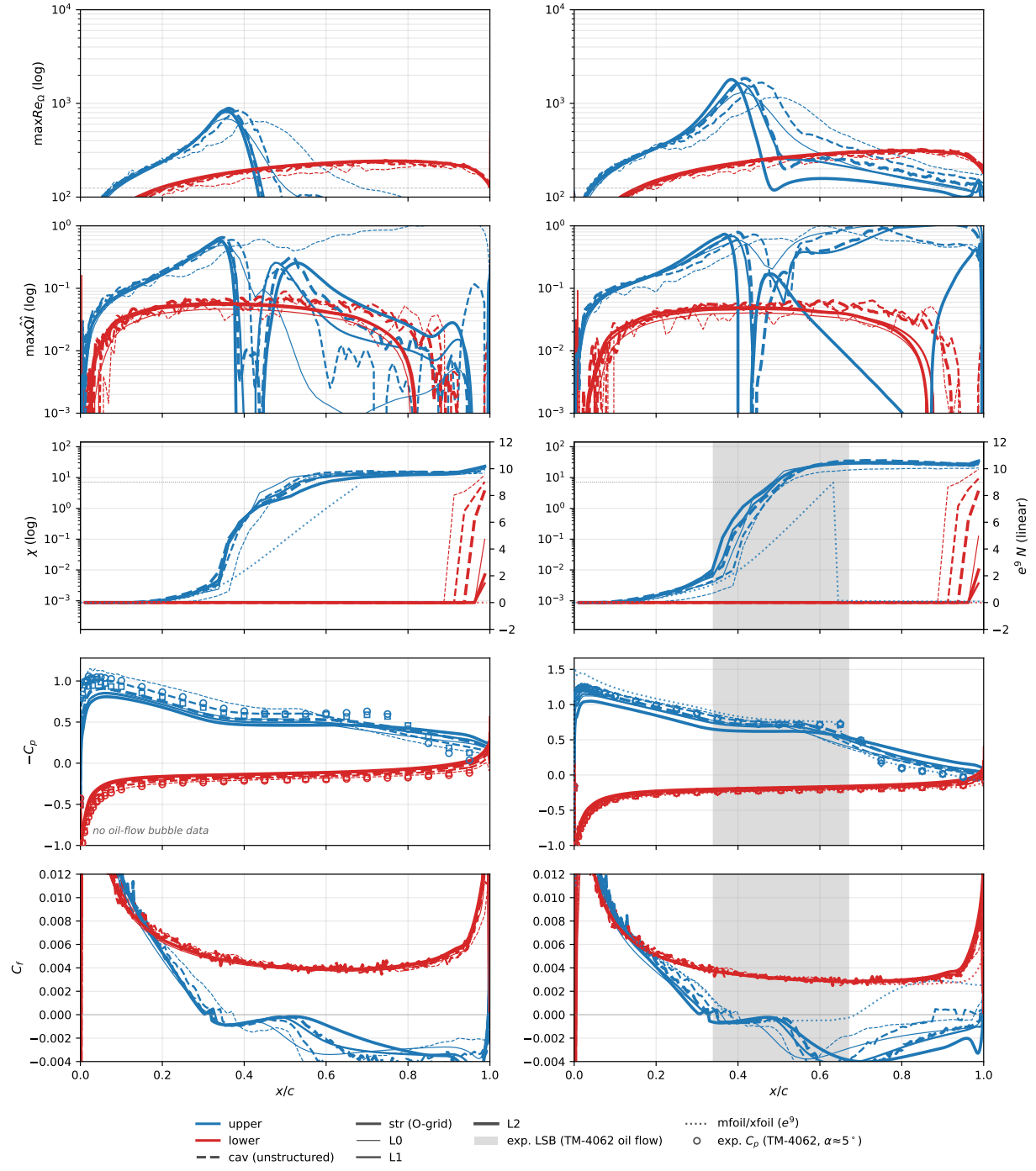


Figure 18: Reynolds sweep, $Re = 6 \times 10^4, 10^5$.

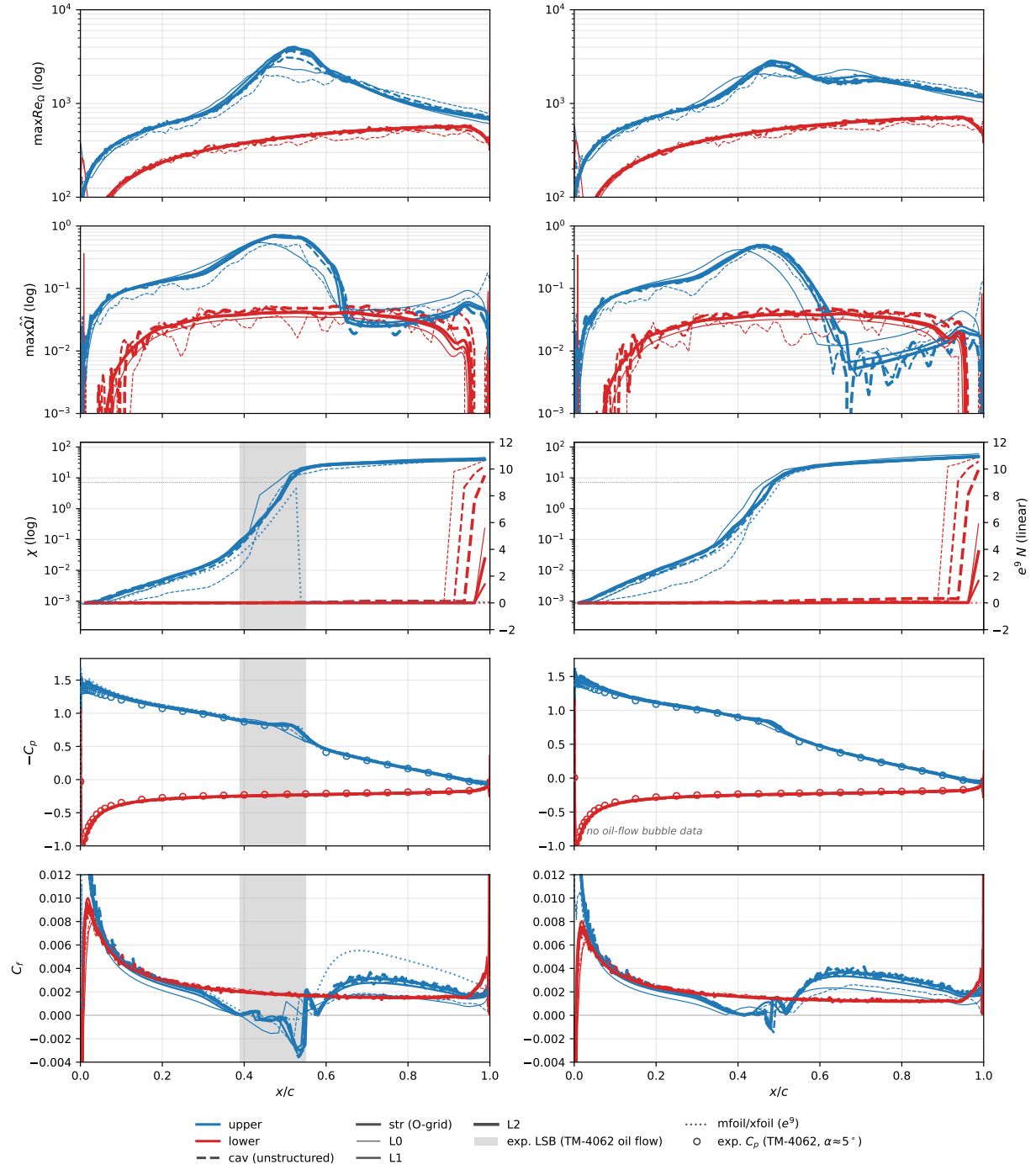
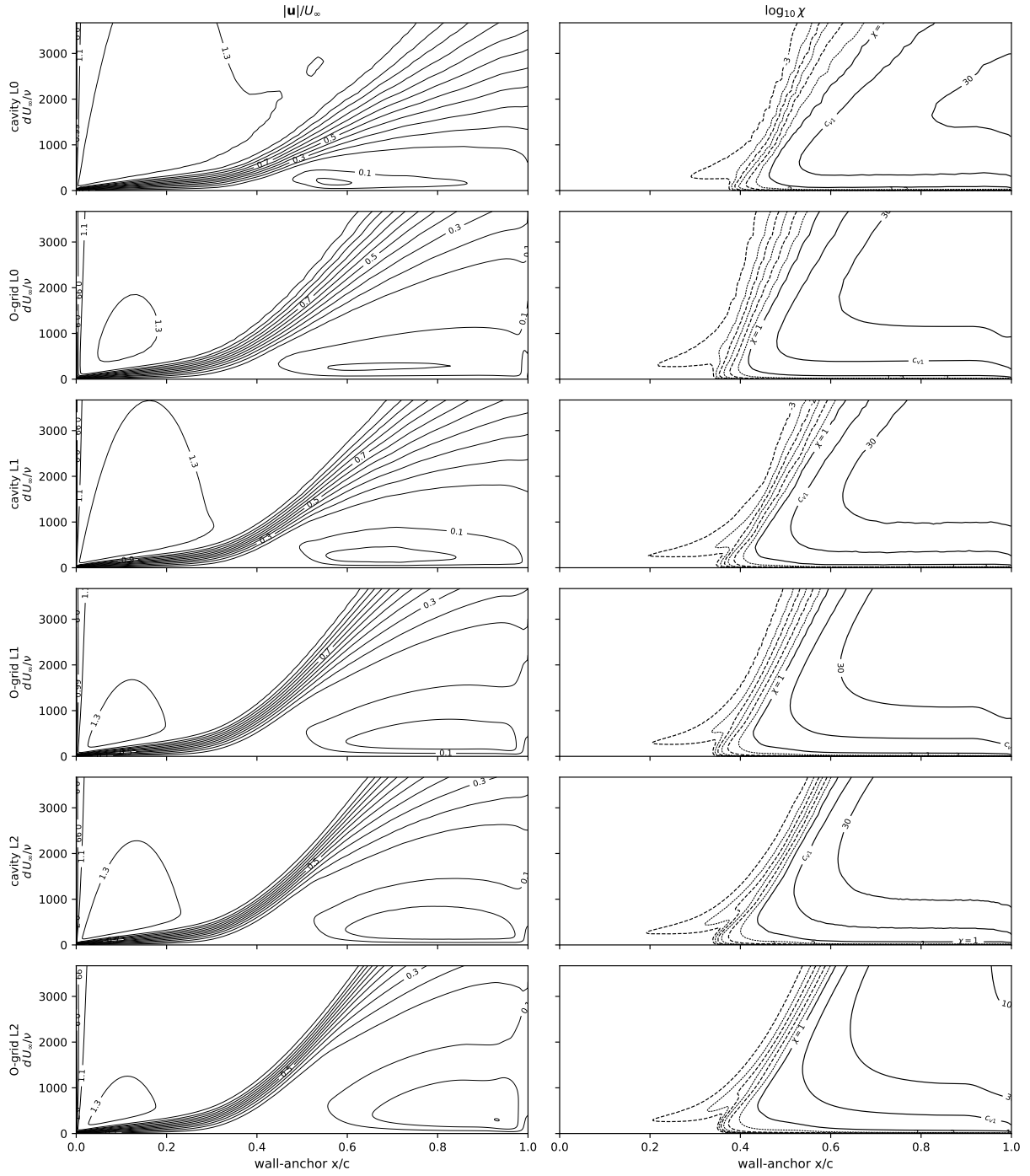


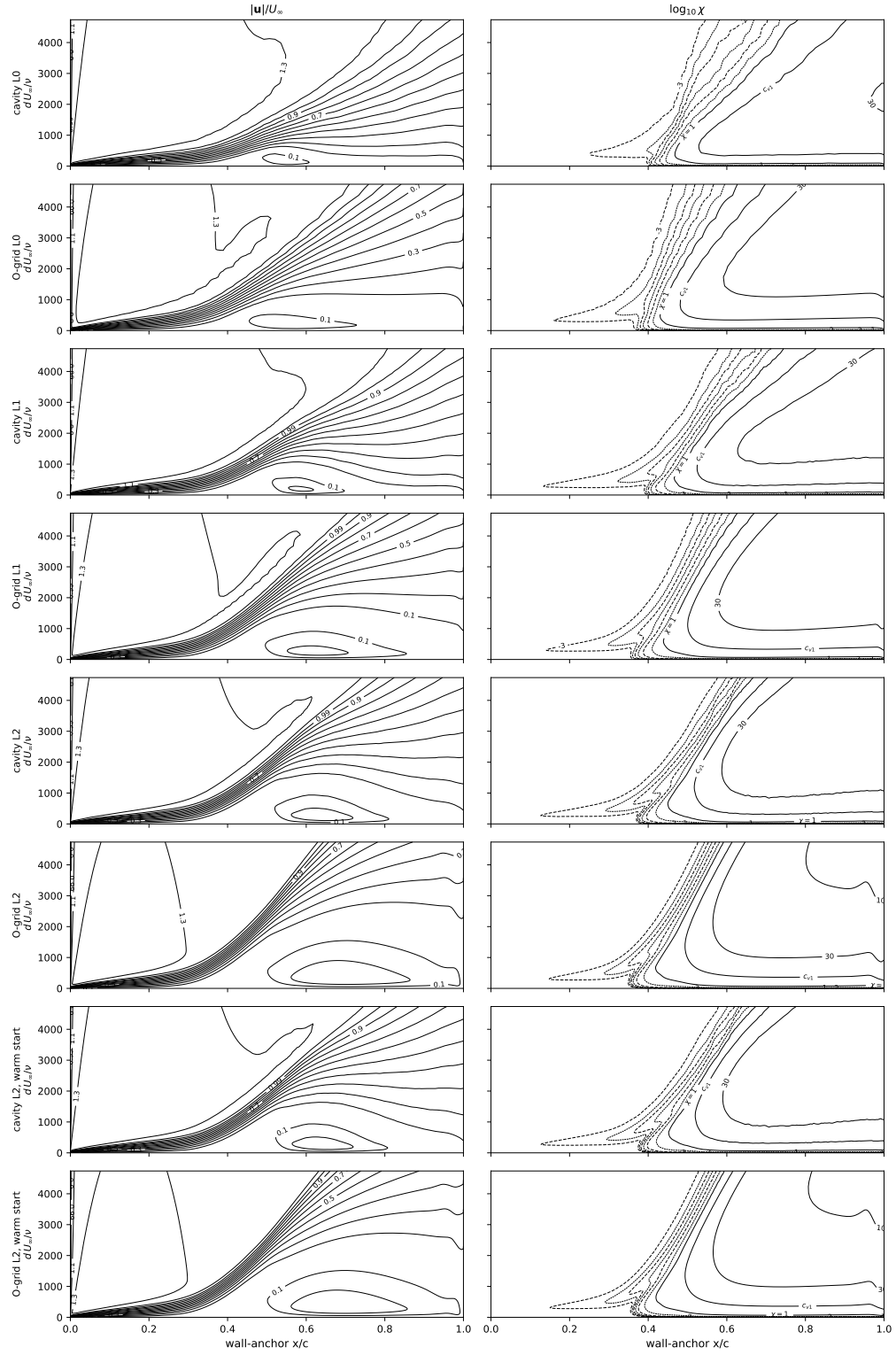
Figure 19: Reynolds sweep, $Re = 3, 4.6 \times 10^5$.

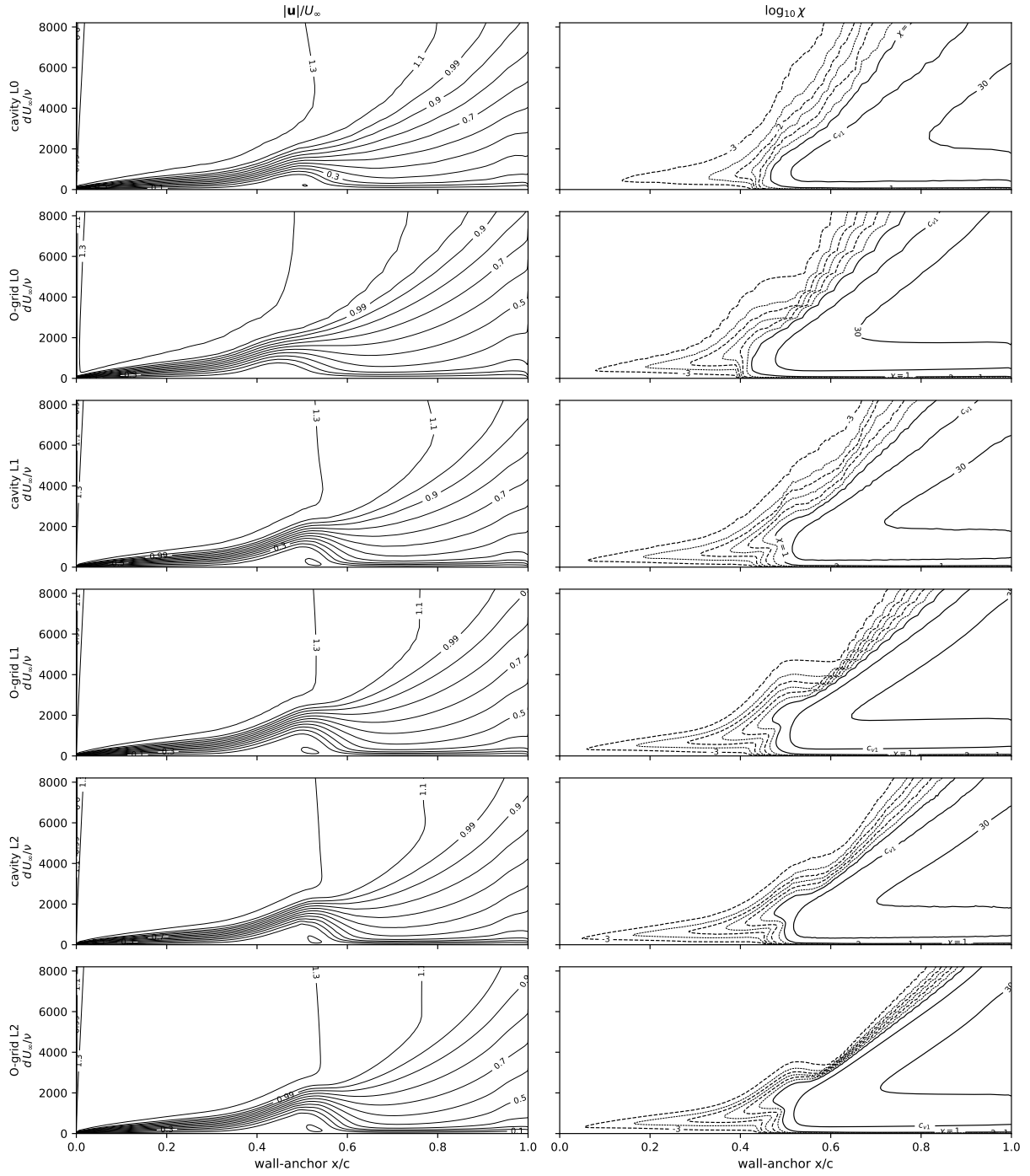
Table 8: Eppler 387 at $\alpha = 5^\circ$ across the Reynolds sweep: L2 section coefficients against the e^9 panel reference and the LTPT measurement (read at the tabulated angle nearest 5°). Computed moments transferred to the quarter chord.

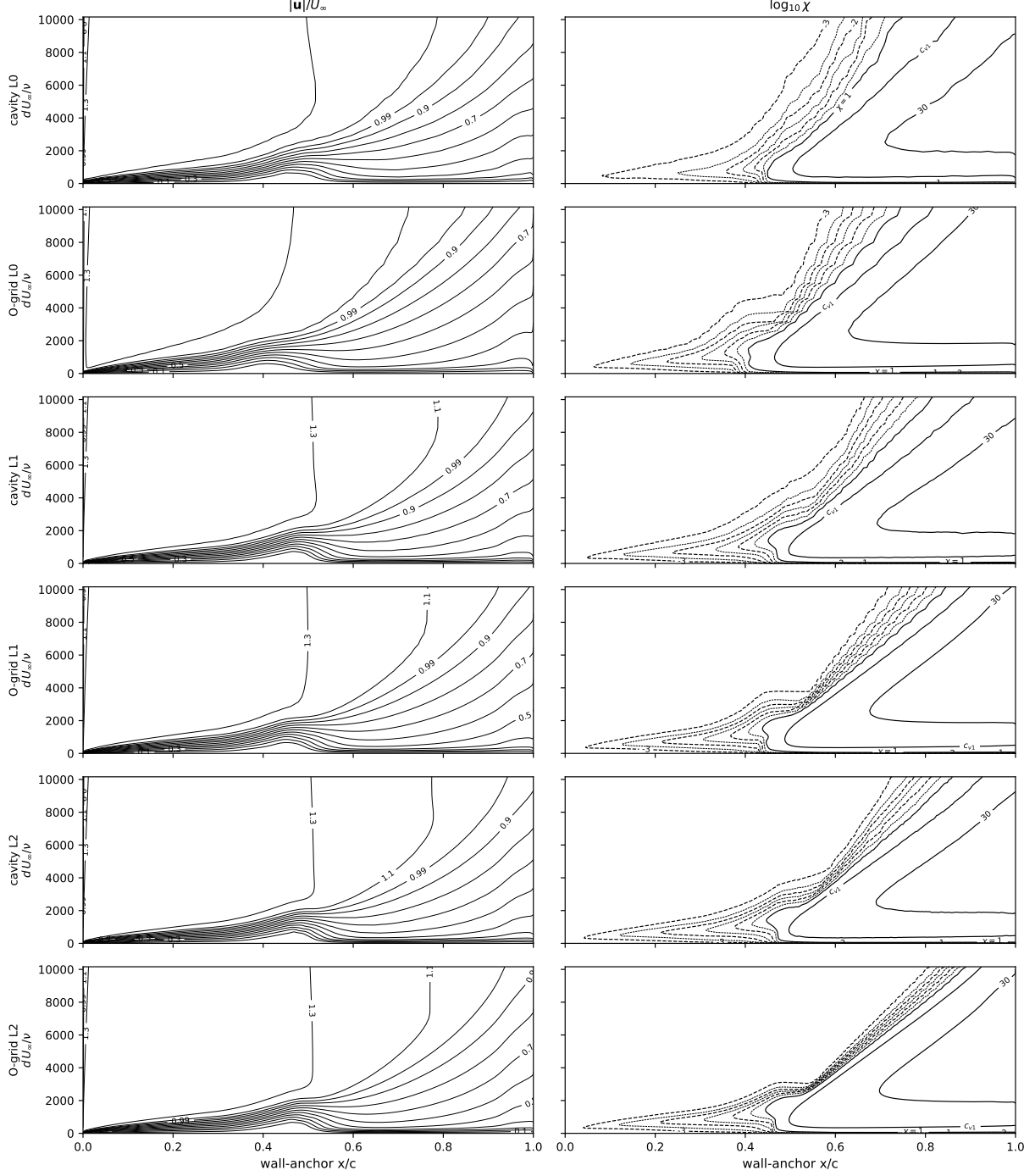
Re	SA-AI, structured L2			SA-AI, unstructured L2			e^9 panel			Exp (LTPT)		
	c_l	c_d	c_m	c_l	c_d	c_m	c_l	c_d	c_m	c_l	c_d	c_m
6×10^4	0.636	0.0533	-0.098	0.712	0.0506	-0.103	0.869 [‡]	0.0411 [‡]	-0.102 [‡]	0.838	0.0439	-0.114
1×10^5	0.795	0.0410	-0.102	0.877	0.0307	-0.096	0.952	0.0213	-0.087	0.873	0.0237	-0.089
2×10^5	0.927	0.0145	-0.081	0.938	0.0143	-0.083	0.960	0.0128	-0.082	0.891	0.0138	-0.081
3×10^5	0.941	0.0107	-0.080	0.950	0.0108	-0.082	0.962	0.0103	-0.082	0.901	0.0114	-0.080
4.6×10^5	0.950	0.0087	-0.081	0.959	0.0089	-0.083	0.966	0.0086	-0.082	0.914	0.0093	-0.081

[‡]XFOIL, whose forces sit closest to the measurement, is the reference at this bistable condition; mfoil gives $c_l = 0.889$, $c_d = 0.0368$, $c_m = -0.097$, and FlexFoil sides with XFOIL ($c_l = 0.868$, $c_d = 0.0406$, $c_m = -0.102$).









2.5 Cylinder drag crisis, $Re_D = 1\text{--}2 \times 10^7$

A steady-branch traverse over 7.3 decades. Twelve Reynolds numbers, $Re_D = 6 \times 10^4\text{--}2 \times 10^6$, at three seeds ($\chi_\infty = 3.9 \times 10^{-4}$, 1.1×10^{-2} , 0.22 ; Mack's map at $Tu = 0.05, 0.2, 0.7\%$) on one quasi-two-dimensional O-grid (1200×148 , first spacing $8 \times 10^{-6} D$, outer boundary $100 D$; $y^+ \leq 0.64$ at 2×10^6), each seed swept as an upward and a downward continuation ladder plus twelve cold starts at the middle seed—84 steady cases; a large-domain laminar grid (600×121 , outer boundary $1000 D$) carries $Re_D = 1\text{--}10^3$ and a fine supercritical grid (1600×170 , first spacing $10^{-6} D$; $y^+ \leq 0.71$ at 10^7 , 1.4 at the 2×10^7 endpoint, quoted with that caveat)

carries 4×10^6 – 2×10^7 —129 cases in all. At the grid seams C_d agrees to 0.5–0.6% at $Re_D = 300$ and 10^3 and to 2.9% between cold-started pairs at 2×10^6 ; 37 of 129 cases are flagged steady limit cycles; every case lands on the symmetric branch, $|C_L| < 2 \times 10^{-7}$.

Figure 20: the crisis moves from $Re_D \approx 5$ – 7×10^5 at the two quiet seeds to ≈ 3 – 4×10^5 at $Tu = 0.7\%$, and the supercritical floor ($C_d = 0.217$ – 0.233 at 2×10^6) rises with the seed. Figure 21: the transition and separation angles. Figure 22: fields at three stations of the traverse. Table 9: the matrix C_d . Below the shedding onset ($Re_D \approx 46$) the steady solution is the physical flow: $C_d = 10.67$ at $Re_D = 1$ and 2.797 at 10, -1.7% against the steady benchmark of Dennis & Chang (+6.4% at 100). Seed independence holds to 0.2% from 10^4 to 4×10^4 , and χ first crosses unity at 4×10^4 .

Six concessions:

1. Through the crisis band the steady branch is non-unique and weakly unsteady: 27 of the 84 matrix cases end in a limit cycle (tail peak-to-peak median 0.011, maximum 0.199 in C_d), the three protocols separate by up to 0.055 in C_d there (against 0.003 elsewhere), and the downward ladder lands at higher drag.
2. $C_L = 0$ in all 84 matrix cases.
3. The subcritical plateau, $C_d \approx 0.84$.
4. The supercritical turbulent separation angles (106 – 118°).
5. The computed floors sit flat at 0.185/0.199/0.215 (by seed) through 2×10^7 while measurement rebuilds toward $C_d \approx 0.7$; the $\chi = 1$ front marches from 101° to 82° over 2×10^6 – 2×10^7 , so the failure is the turbulent separation, pinned near 120° .
6. On the fine grid the downward ladder finds a second supercritical steady state, 13–22% lower in C_d (separation 123.5° against 120.4° , base pressure -0.18 against -0.23 at 10^7).

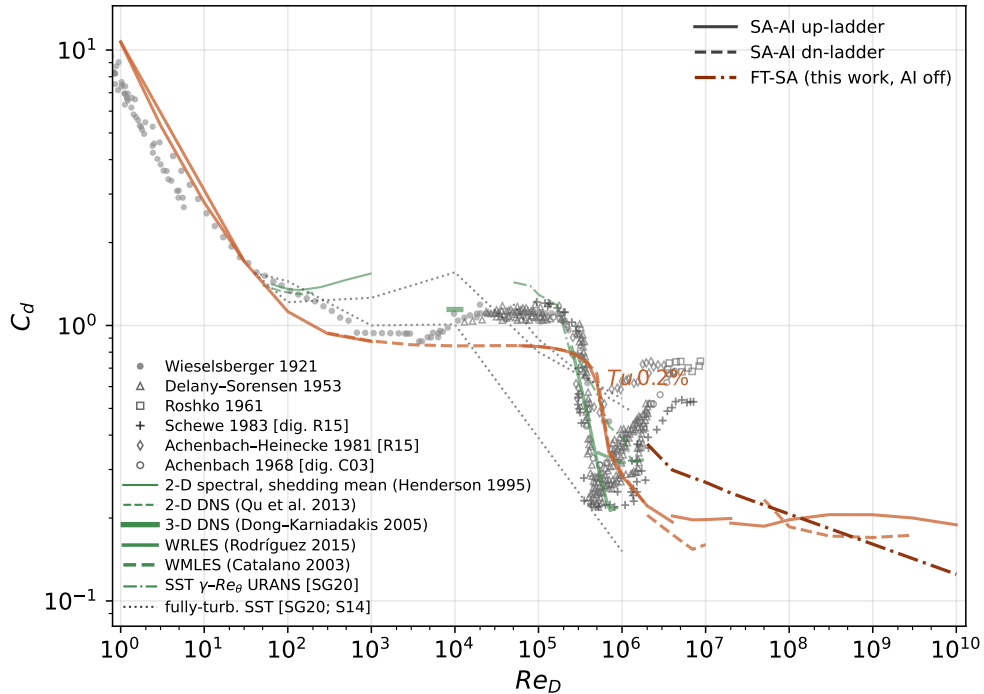


Figure 20: Cylinder drag crisis, $Re_D = 1$ – 10^{10} ; middle seed $Tu = 0.2\%$, up (solid)/down (dashed) ladders.

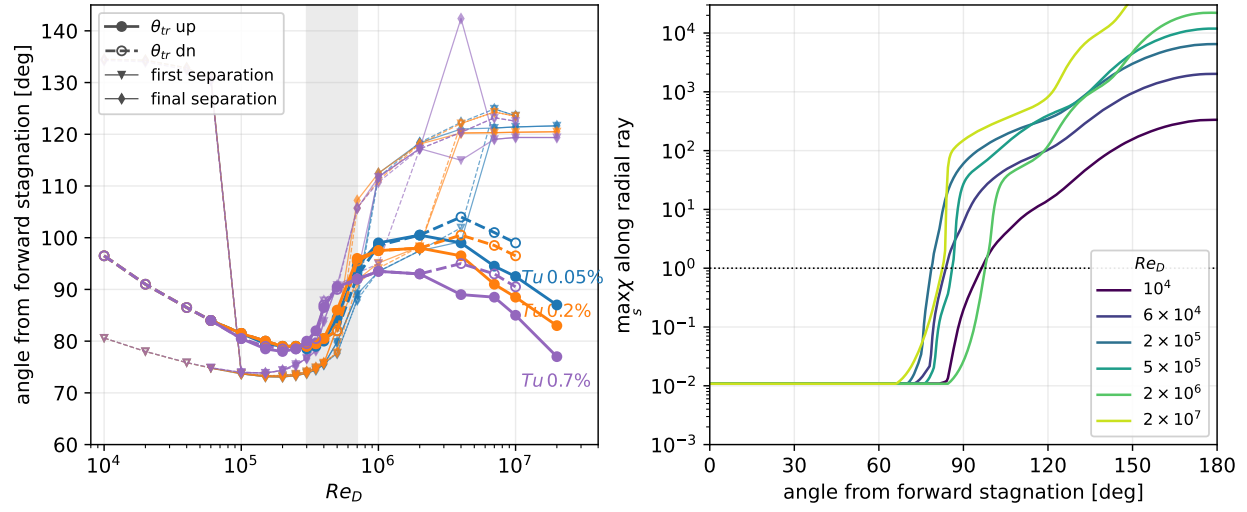


Figure 21: Transition and separation angles; three seeds $Tu = 0.05, 0.2, 0.7\%$.

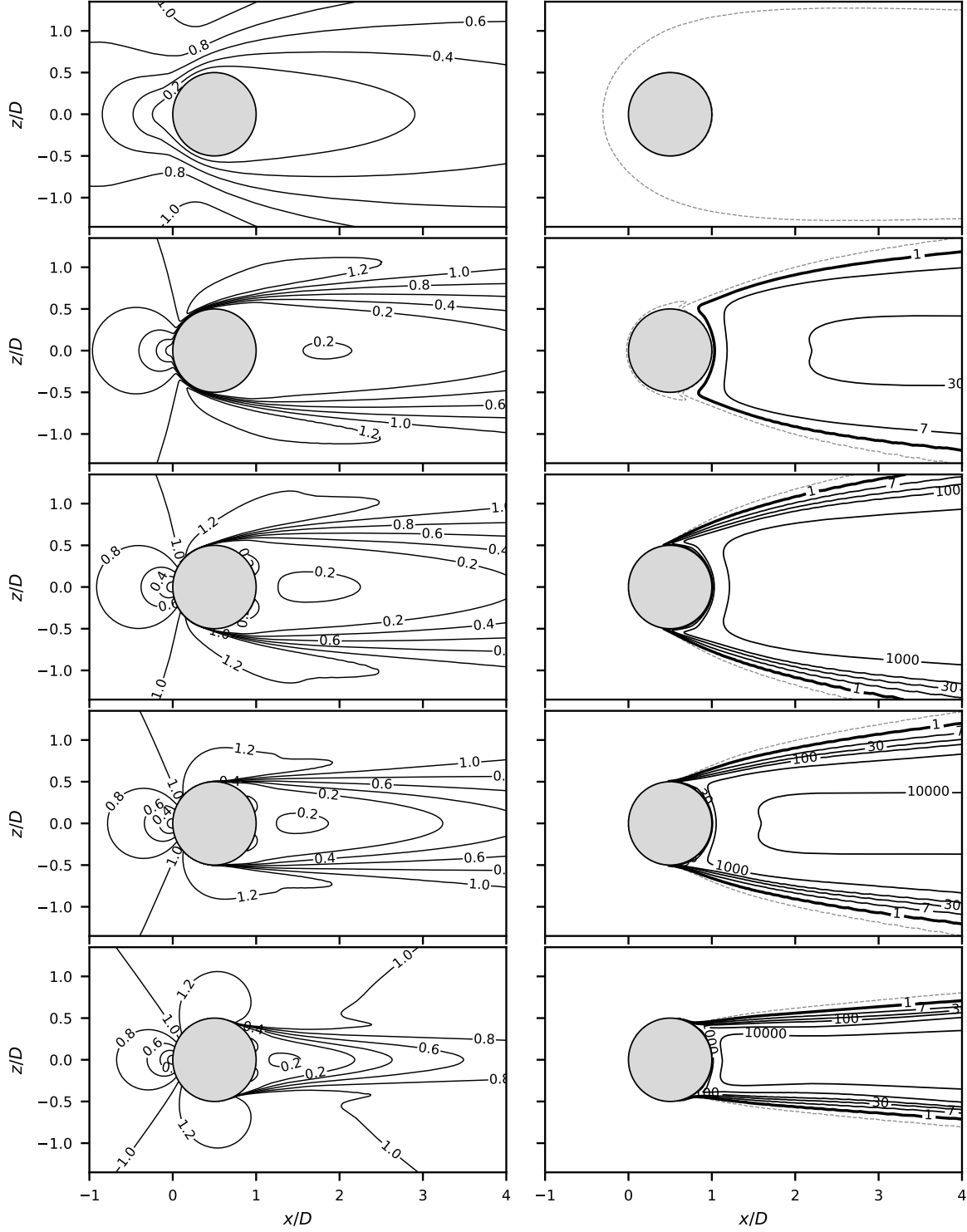


Figure 22: Drag-crisis fields, $Re_D = 10, 10^3, 10^5, 5 \times 10^5, 2 \times 10^6$ (rows, up-ladder $Tu = 0.2\%$); columns $|\mathbf{u}|/U_\infty$ and χ (dashed $\chi < 1$, solid $\chi = 1$, c_{v1} , 30, 10^2 – 10^4). $C_d = 2.80, 0.87, 0.84, 0.62, 0.22$.

Table 9: Cylinder steady drag-crisis matrix: C_d (median over the final-stage tail window) for the 84 matrix cases. Bracketed values: tail-window C_d peak-to-peak for cases flagged as steady limit cycles; $|C_L| < 2 \times 10^{-7}$ in all 84 cases.

Re_D ($\times 10^5$)	Tu 0.05%		Tu 0.2%			Tu 0.7%	
	up	dn	up	dn	cold	up	dn
0.6	0.845	0.845	0.845	0.845	0.845	0.842	0.842
1	0.840	0.839	0.839	0.839	0.839	0.831	0.832
1.5	0.828	0.827	0.828	0.827	0.827	0.809	0.812
2	0.814	0.812	0.813	0.811	0.811	0.781	0.786
2.5	0.796	0.794	0.795	0.793	0.793	0.745	0.755
3	0.776	0.774	0.774	0.771	0.772	0.677 [0.075]	0.709
3.5	0.753	0.750	0.749	0.748	0.749	0.616 [0.199]	0.671 [0.030]
4	0.725	0.726	0.719	0.722	0.724	0.495 [0.122]	0.547 [0.086]
5	0.644 [0.076]	0.670 [0.036]	0.620 [0.129]	0.668 [0.047]	0.650 [0.136]	0.430 [0.008]	0.439
7	0.409 [0.051]	0.407 [0.024]	0.344 [0.002]	0.355	0.347 [0.003]	0.365 [0.006]	0.356
10	0.286 [0.006]	0.285	0.284 [0.005]	0.290	0.282 [0.004]	0.294 [0.007]	0.303
20	0.219 [0.013]	0.217 [0.008]	0.221 [0.011]	0.220 [0.008]	0.220 [0.008]	0.233 [0.010]	0.232 [0.005]

2.6 Daedalus wing (three-dimensional deployment)

The Daedalus human-powered-aircraft wing: half-span 17.07 m, root chord 0.917 m, aspect ratio 37.8, DAE-11/21/31 sections; $Re_{root} = 5 \times 10^5$, $\alpha \in \{4^\circ, 5^\circ, 6^\circ\}$ ($C_L \approx 1.02$ – 1.23), flight-quiet seed $\chi_\infty = 8.76 \times 10^{-6}$ ($N_{crit} \approx 13.6$); two mesh families (spanwise-stacked structured O-grid, 0.72–36.7M nodes; unstructured prism/tet/hex, 1.70–54.9M nodes) at three levels, eighteen solutions; the reference is AVL with strip-wise XFOIL section polars at matched local c_l and N_{crit} , trimmed to the RANS lift. Figure 23: the families agree at L2 to 0.35–0.63% in C_L and 0.7–1.7 counts in C_D , and the finest-grid drag runs 15.6/10.3/5.6 counts (7.3/4.4/2.2%) below the matched-lift reference at $\alpha = 4^\circ/5^\circ/6^\circ$. Figure 24: surface maps at $\alpha = 4^\circ$ against the e^N strips—at $\eta = 0.31$, separation within $0.017c$ and reattachment within $0.009c$ at every incidence, the $\chi = 1$ front sits $\sim 0.09c$ ahead of the strips’ N_{crit} contour, and the tip vortex trips transition to the leading edge for $\eta \gtrsim 0.992$. Table 10: totals by level ($\alpha = 5^\circ, 6^\circ$ surface maps and section sheets below).

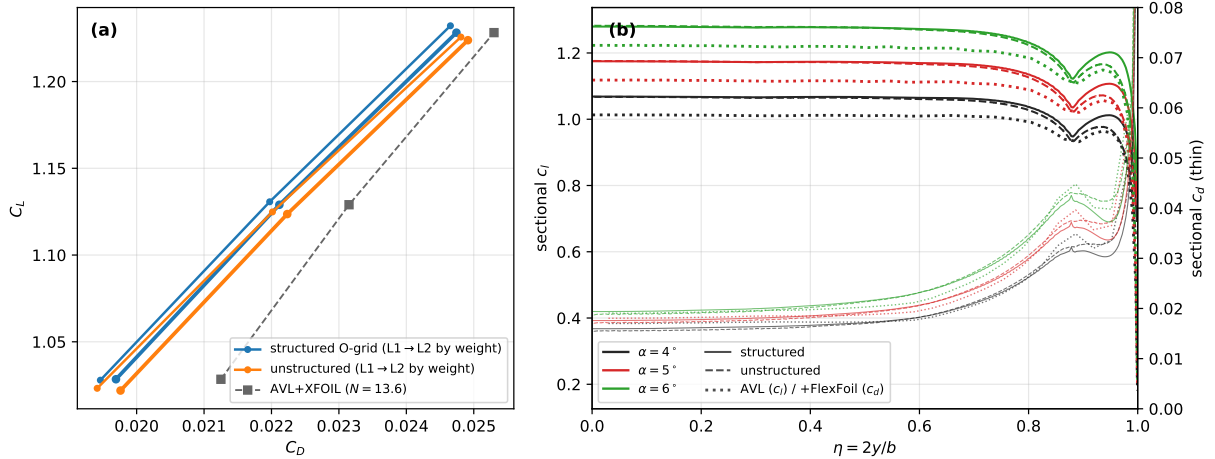


Figure 23: Daedalus polar and sections. Families agree at L1 to 0.46–0.53% in C_L and 0.4–1.6 counts in C_D , at L0 to 0.74–0.97% and 18–21 counts; taper break at $\eta = 0.88$.

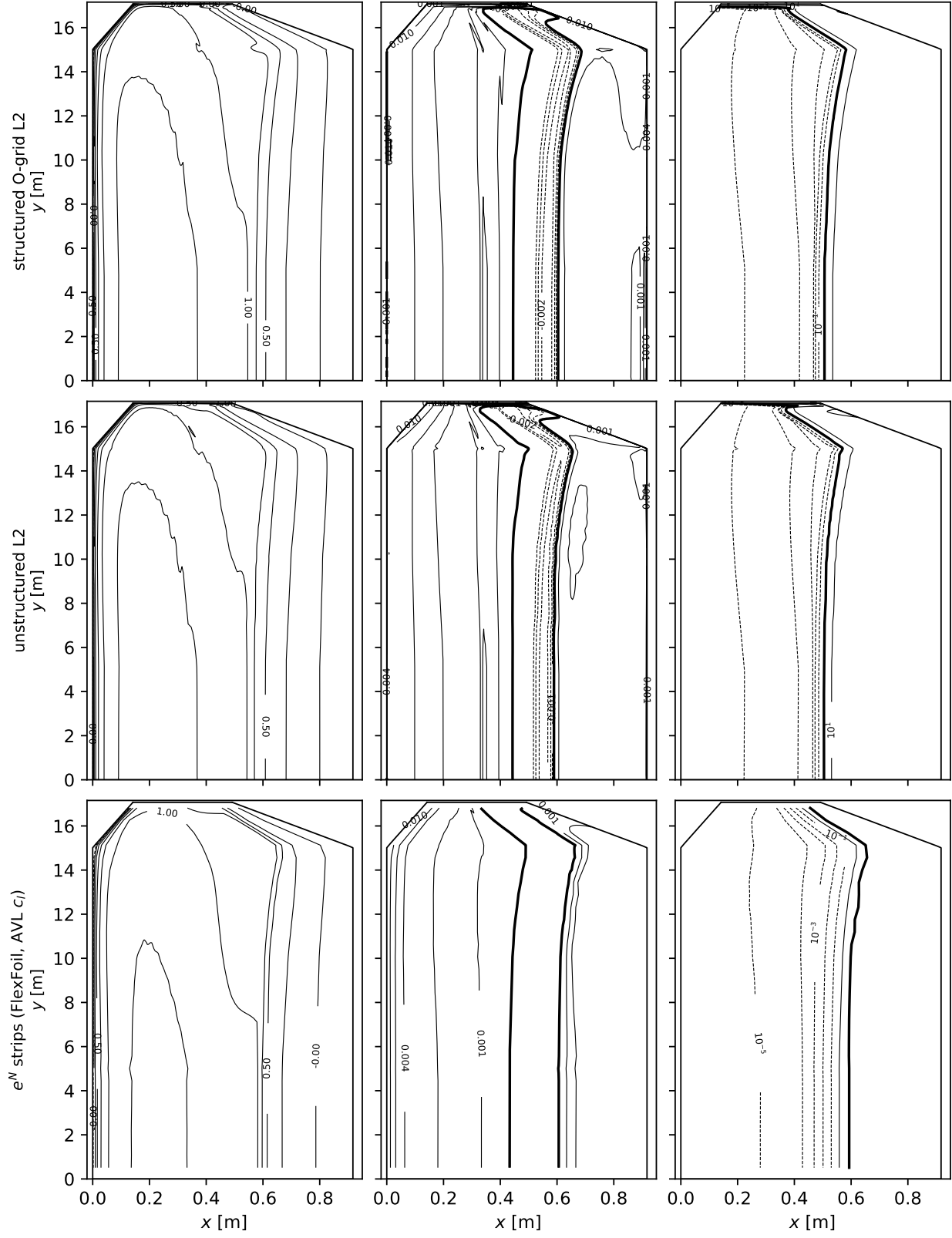
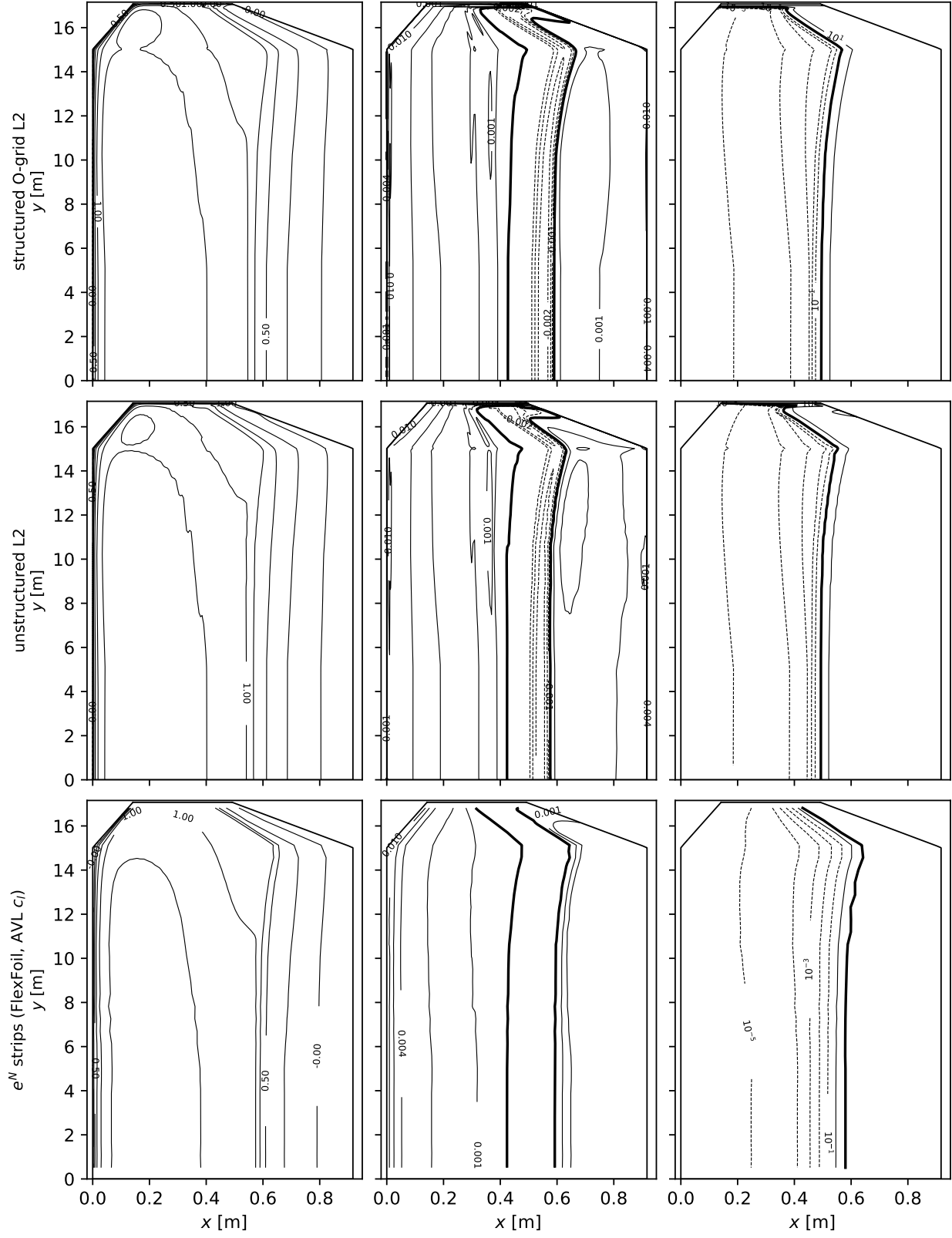
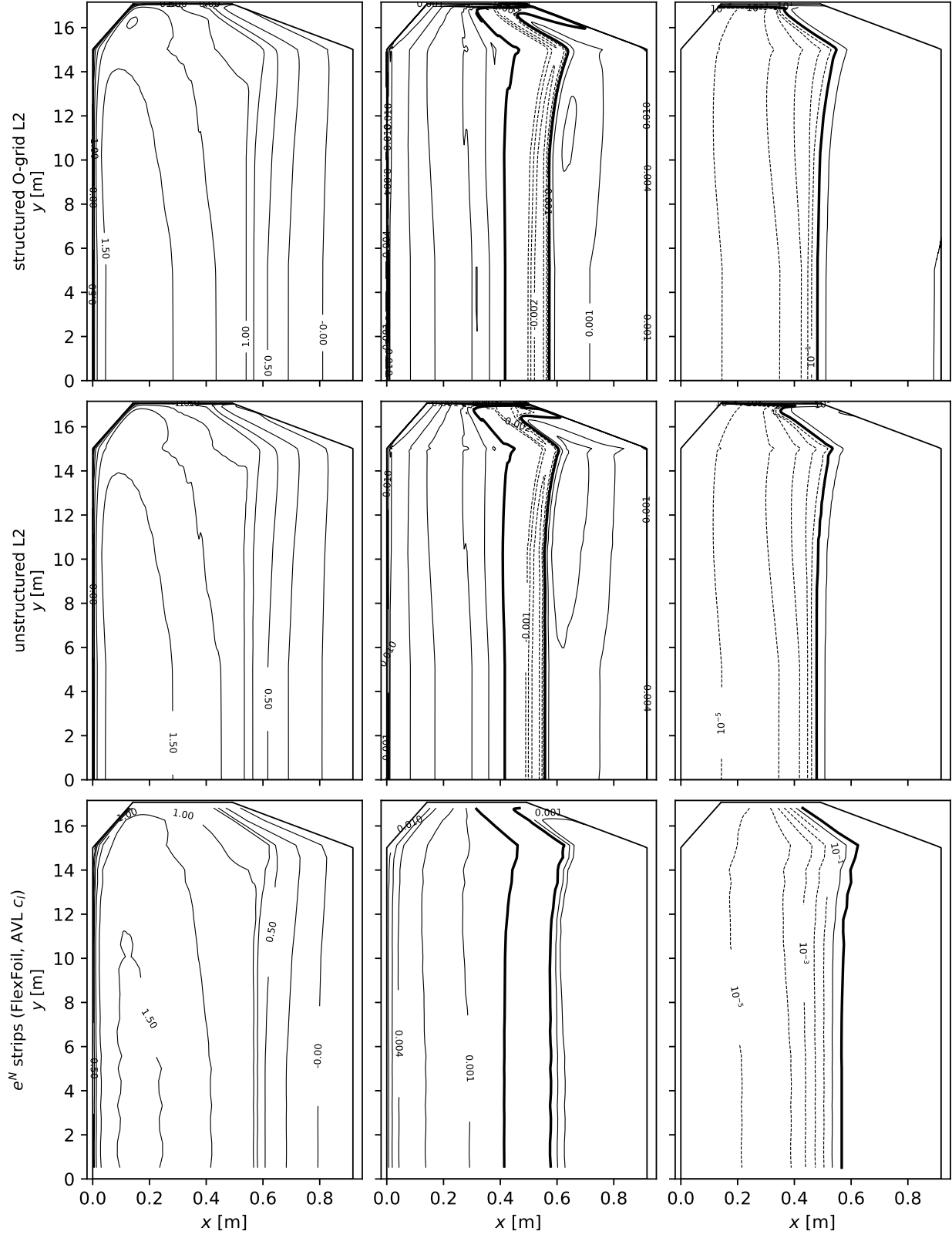


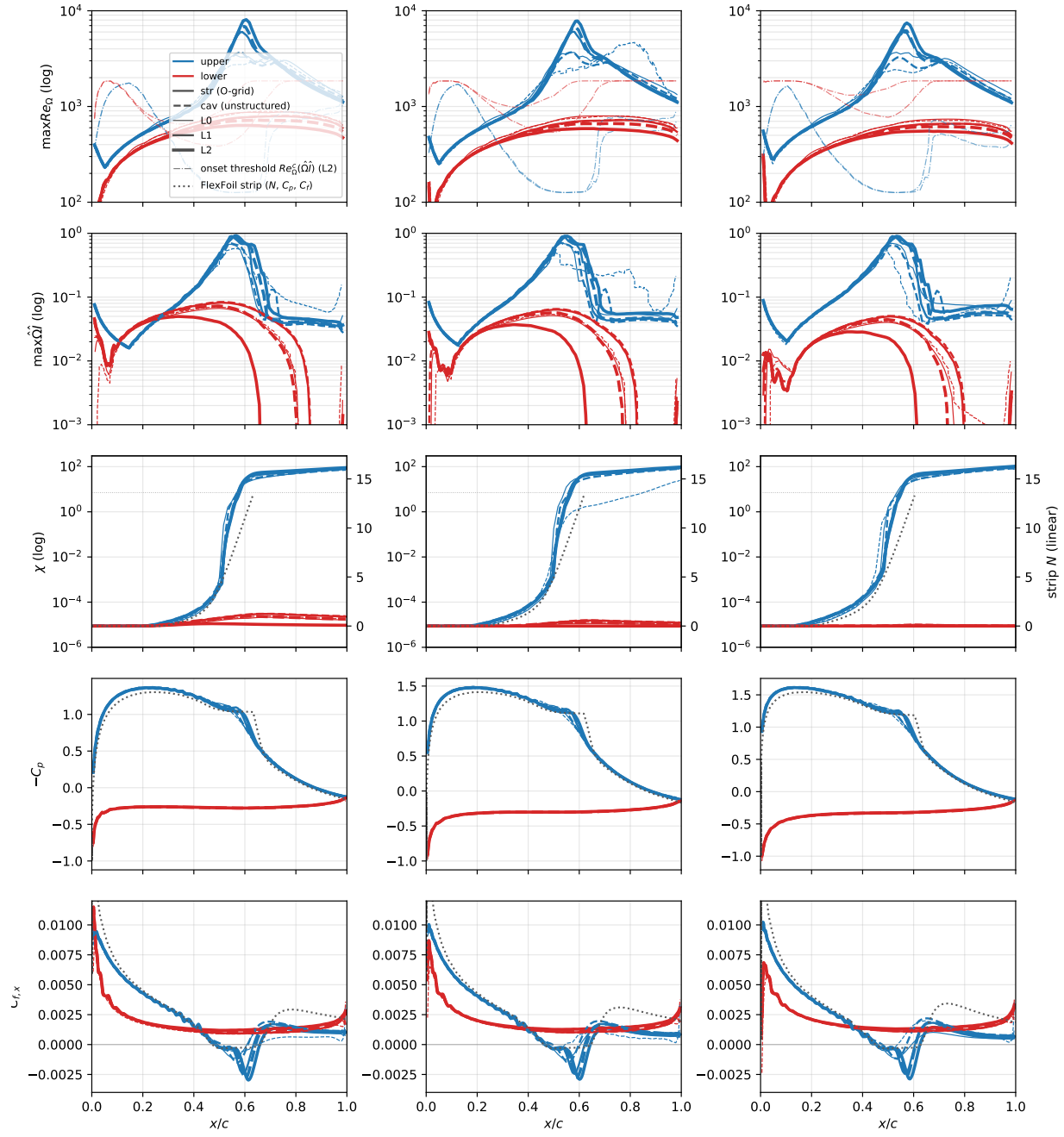
Figure 24: Daedalus surface maps, $\alpha = 4^\circ$, $Re_{\text{root}} = 5 \times 10^5$.

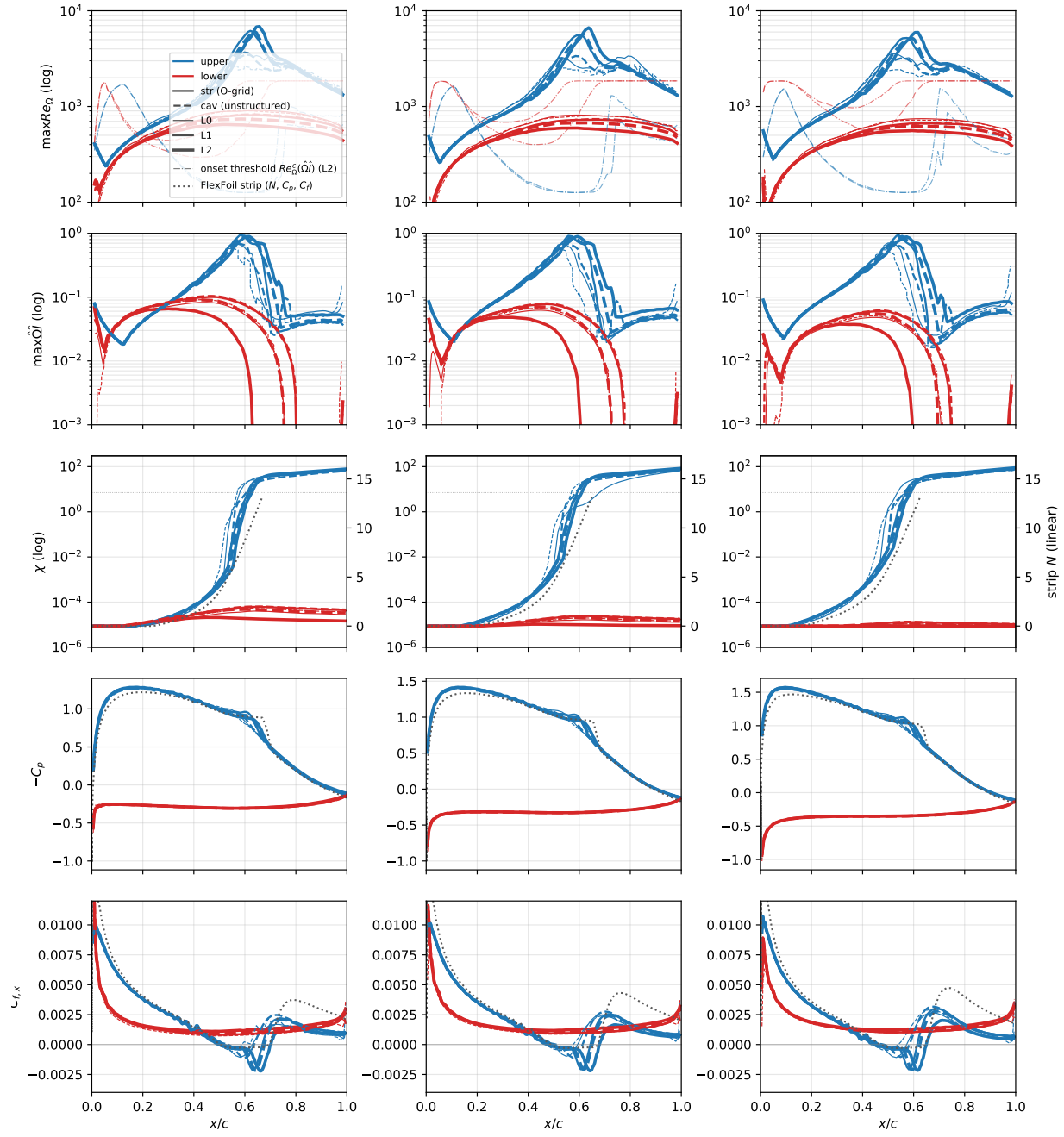
Table 10: Daedalus wing totals by grid level: SA-AI on both mesh families against the AVL+XFOIL reference trimmed to the RANS lift at each incidence.

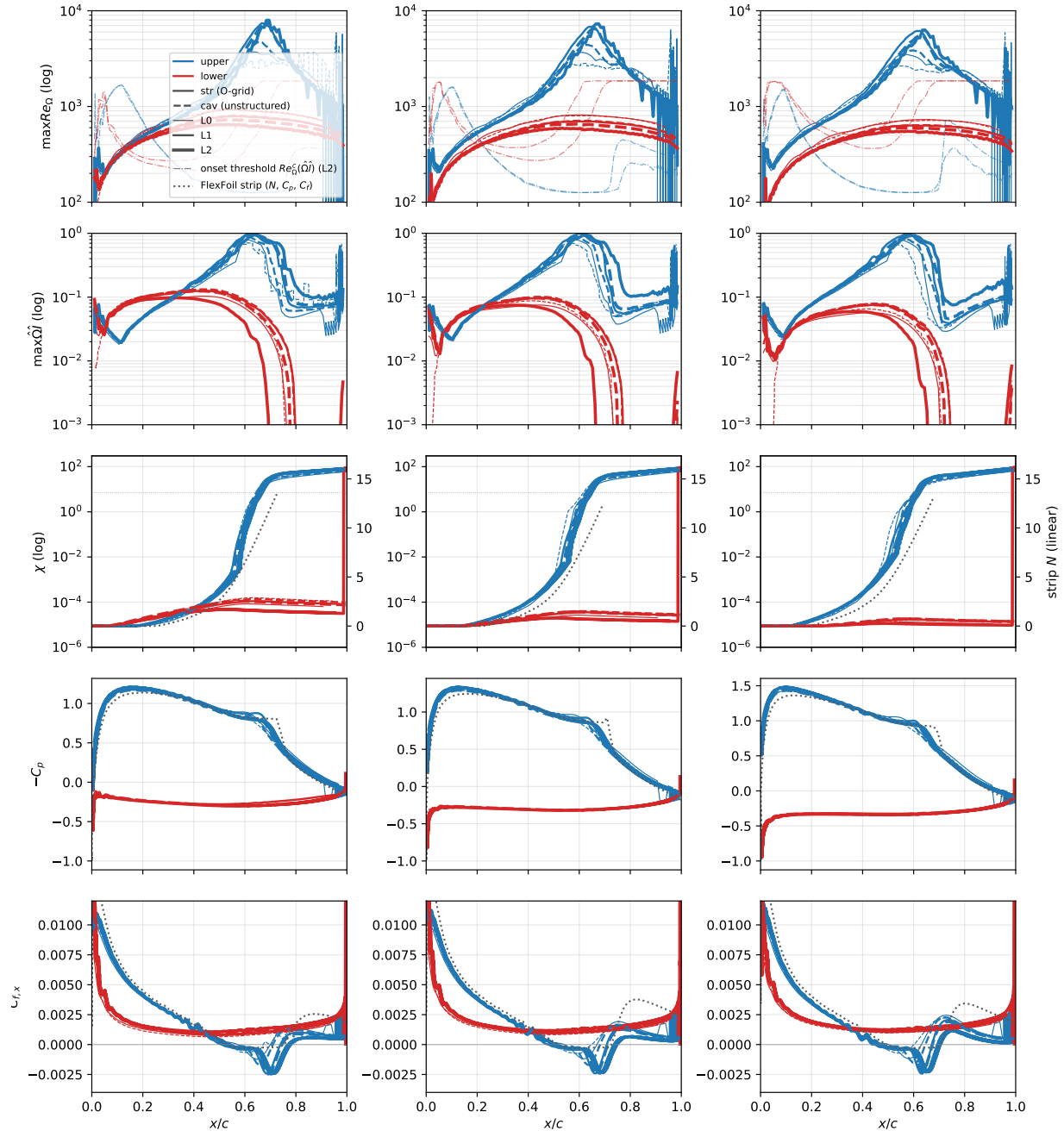
α	level	structured O-grid		unstructured	
		C_L	C_D	C_L	C_D
4°	L0	1.0244	0.01958	1.0156	0.02134
	L1	1.0279	0.01946	1.0232	0.01941
	L2	1.0284	0.01969	1.0219	0.01976
	AVL+XFOIL	1.0284	0.02125		
5°	L0	1.1254	0.02219	1.1145	0.02427
	L1	1.1307	0.02197	1.1248	0.02201
	L2	1.1290	0.02212	1.1236	0.02223
	AVL+XFOIL	1.1290	0.02315		
6°	L0	1.2262	0.02505	1.2171	0.02713
	L1	1.2321	0.02465	1.2256	0.02481
	L2	1.2282	0.02474	1.2238	0.02491
	AVL+XFOIL	1.2282	0.02530		











A Proposed two-branch kernel (for discussion)

Rate $a = \text{clip}(\text{softmax}_2(a_{\text{inv}}\langle\hat{\Omega}\hat{I}\rangle_+, a_{\text{visc}}\langle\hat{\Omega}\hat{Z}\rangle_+), 0, a_{\text{inv}})$, $\text{softmax}_2(x, y) = \sqrt{x^2 + y^2}$. Gate $Re_{\Omega}^c = \text{softmin}(A + B/P_I^2, A + B_c/P_{\text{curv}}^2)$, $P_I = \langle\hat{\Omega}\hat{I}\rangle_+$, $P_{\text{curv}} = \langle\hat{\Omega}\hat{Z}\rangle_+$, $\hat{Z} = -Z/R$, $\langle x \rangle_+ = \max(x, 0)$. Constants: $a_{\text{inv}} = 0.19$, $a_{\text{visc}} = 0.0276$, $A = 124.6$, $B = 1.424$, $B_c = 130$. (env-gated *AI_A_VISC*, *AI_REOMC_BC*; default 0 = canon.)

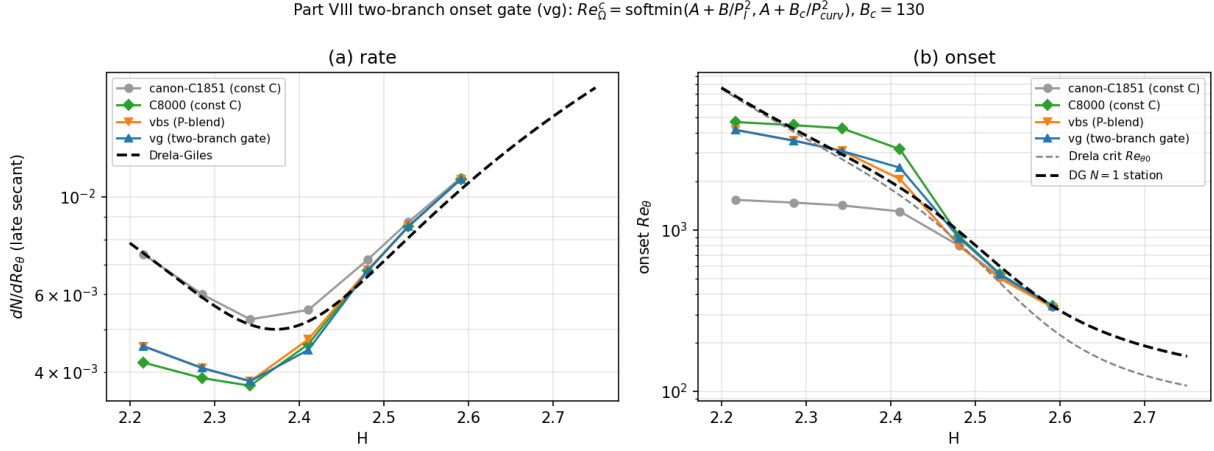


Figure 25: Curves: canon ($C=1851$), $C=8000$, vbs, vg. vg both-panel max $1.63\times$ DG; (a) rate; (b) onset on the DG $N=1$ station to $H=2.2$; low- H rate $0.56\times$ DG.

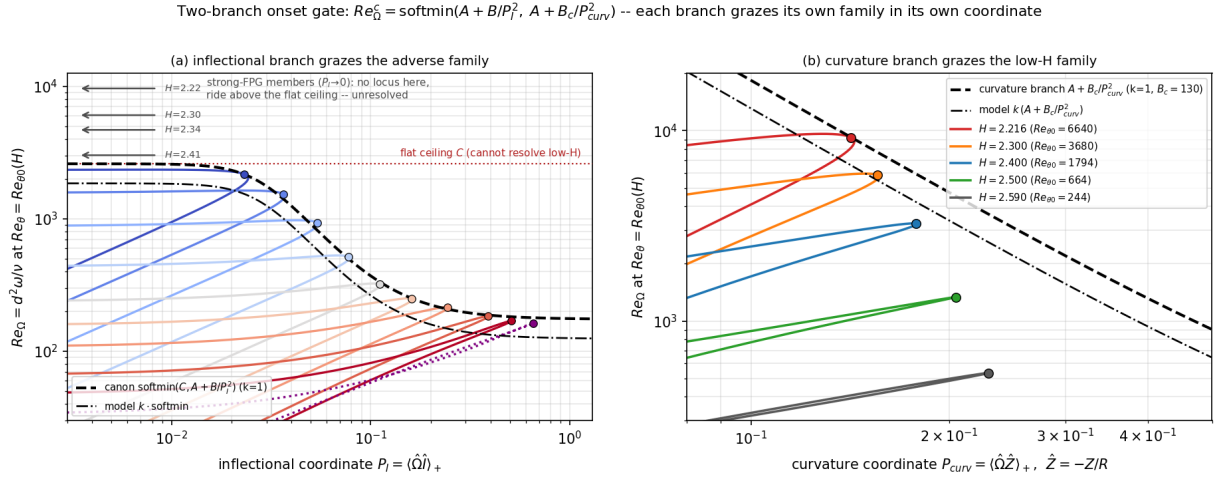


Figure 26: Left P_I (inflectional/adverse family); right P_{curv} (low- H family). Curvature-branch graze ratio by H : $2.216 \rightarrow 1.00$, $2.30 \rightarrow 0.76$, $2.40 \rightarrow 0.55$, $2.50 \rightarrow 0.29$, $2.59 \rightarrow 0.15$. Drela $Re_{\theta 0}$: $6640/3680/1794/664/244$.

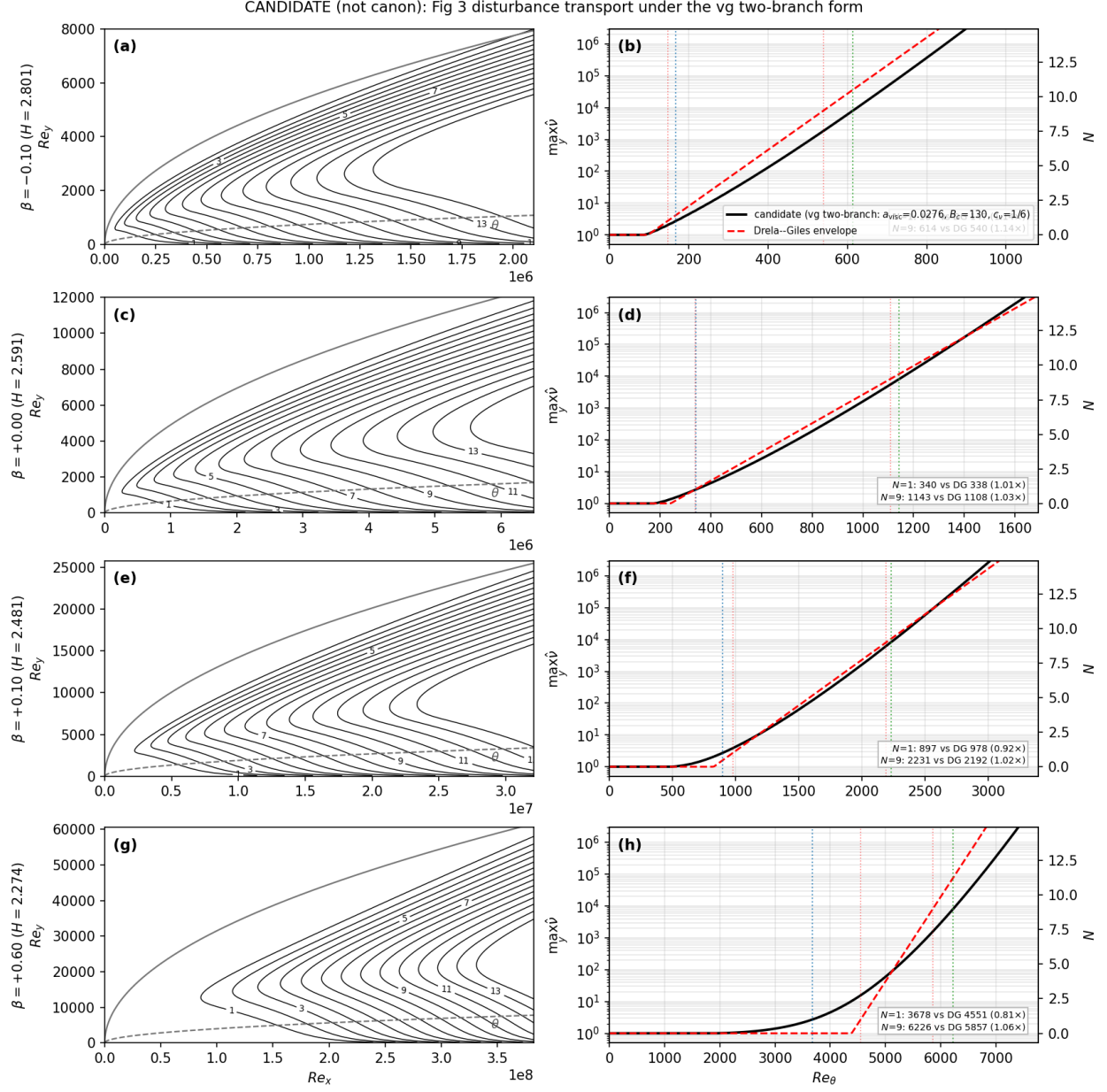


Figure 27: Rows $(\beta, H, \max \hat{\Omega} \hat{I})$: $(-0.10, 2.80, 0.162)$, $(0, 2.59, 0.078)$, $(+0.10, 2.48, 0.037)$, $(+0.60, 2.27, 6.5 \times 10^{-4})$. $N=1/N=9$ vs DG: 1.14/1.14; 1.01/1.03; 0.92/1.02; 0.81/1.06.

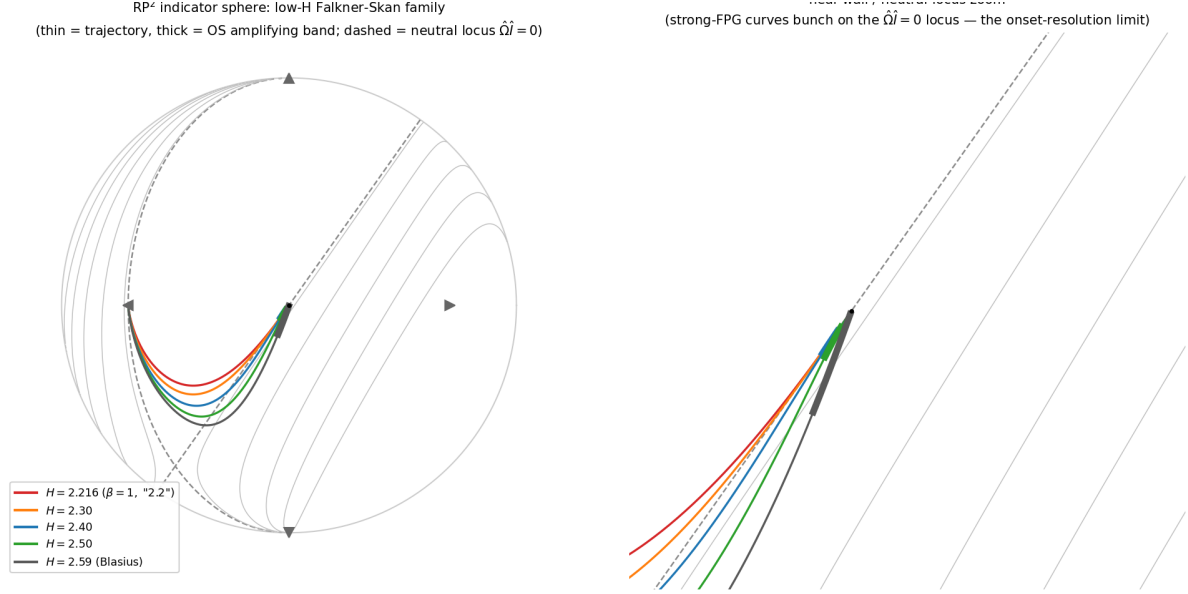


Figure 28: Low- H Falkner-Skan on $\mathbb{RP}^2$. $H = 2.216$ and 2.30 : $\max \hat{\Omega}\hat{I} = 1\text{--}3 \times 10^{-4}$ ($\Delta = 1.6 \times 10^{-4}$), Drela $Re_{\theta 0}$ 6640 vs 3680 ($1.81\times$).

In-solver (vg on, env-gated; canon untouched): $Re_D = 2 \times 10^6\text{--}2 \times 10^7$ vg vs canon $\Delta\theta_{tr} \leq 2.5^\circ$, $\Delta C_d \leq 6\%$; $C_d(10^9) = 0.209$ (canon 0.206), $C_d(10^{10}) = 0.199$ (canon 0.189); flat-plate $Re_{x,tr}$ -7.5% (Tu 0.16%) / -9% (Tu 0.08%); Eppler-387 $Re = 2 \times 10^5$, $\alpha = 2^\circ$ rooftop unchanged.

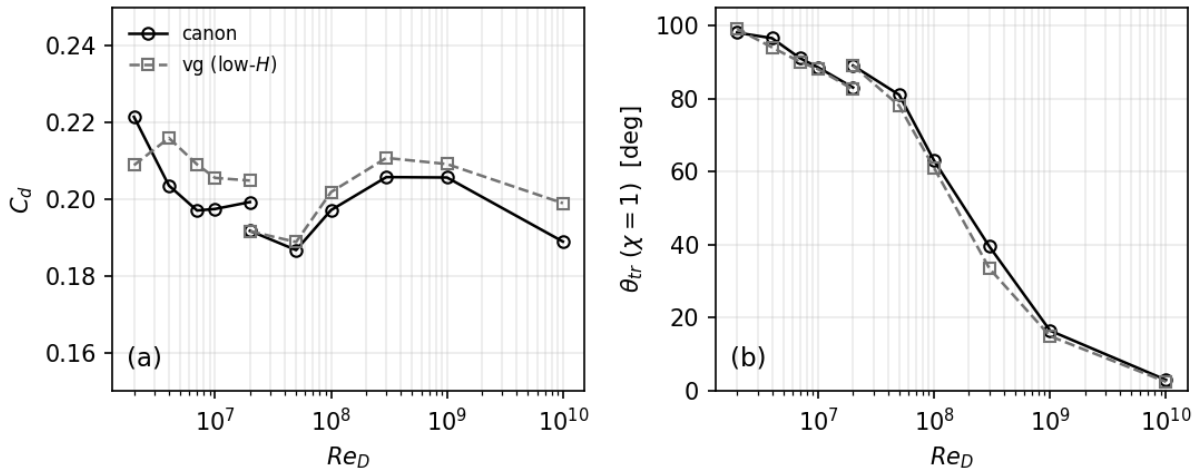


Figure 29: vg (low- H) vs canon, $Tu = 0.2\%$ up-ladder, $Re_D = 2 \times 10^6\text{--}10^{10}$ (highre mesh $\leq 2 \times 10^7$, ultra $\geq 2 \times 10^7$). (a) C_d : both 0.19–0.22, vg +3–6%; neither reaches the transcritical 0.5–0.7. (b) θ_{tr} ($\chi = 1$): front advances $98^\circ \rightarrow 2.5^\circ$; vg tracks canon within 6° .